

MAGNETOM

Installation Instructions System

Installation Instructions Aera Mobile

Document Version

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1.1 Introduction

This document describes the installation of a MAGNETOM Aera system in a mobile trailer or coach.

The main difference between the Aera Mobile and the static system is the relocation of the components (such as the TX-BOX, RFCEL, etc.) at the magnet service side into a new cabinet named **RCR** (Rearranged Component Rack) behind the magnet. Any other change or additional component will be introduced in the following chapters.

1.1.1 Safety information and references to safety information



Installation personnel and system engineers should refer to the operating notes for the magnet, which are included in the bag in the magnet bore.



Superconductive magnets are highly sensitive components. Proceed with extreme caution during transport and delivery to avoid hazardous conditions for both the system and the personnel involved. Only specially trained personnel may install and service the magnet. Installation and service personnel must read the Safety Information before working with the magnet.



Chains, belts, measurement devices, tools, lifting devices, hydraulic lifts, and transport rollers must be in good condition to ensure the safety of personnel and the system.



Please note the general safety guidelines given under "MR-Specific Safety Information" in this manual when performing the work described in this chapter.



For the installation of the system into the trailer / coach, the Aera Installation and Start-up Instructions are required.

1.1.2 Conventions and Abbreviations

The abbreviations used in the Installation Instructions are explained below:

Abbreviation	Description
ACS	Air Conditioning System
BC	Body coil
EPC	Electronic Power Cabinet
ERDU	Emergency Run Down Unit
GC	Gradient Coil
GPA	Gradient Power Amplifier

Abbreviation	Description
MRC	MR Console
PAM	Power Adaption Module
PDS	Power Distribution System
PMU	Physiological Measurement Unit
RCR	Rearranged Component Rack
RFPA	Radio Frequency Power Amplifier
SEP	SEParator
UPS	Uninterruptible Power Supply

1.1.3 Installation Certificate



Only trained personnel are authorized to install the system.

1.2 Items Included in the Shipment

The system comprises the following components:

- Magnet type OR 98 MAGNETOM Aera, including adapted covers
- MR electronics cabinet (comprises GPA / EPC / ACS)
- Patient table (fixed or mobile)
- RF filter plate
- Alarm box
- Helium compressor and SEP cabinet
- Host computer PC and console components
- RCR and generic mobile conversion kit

2.1 General Information



The following chapter describes the installation procedure for the Aera Mobile system.



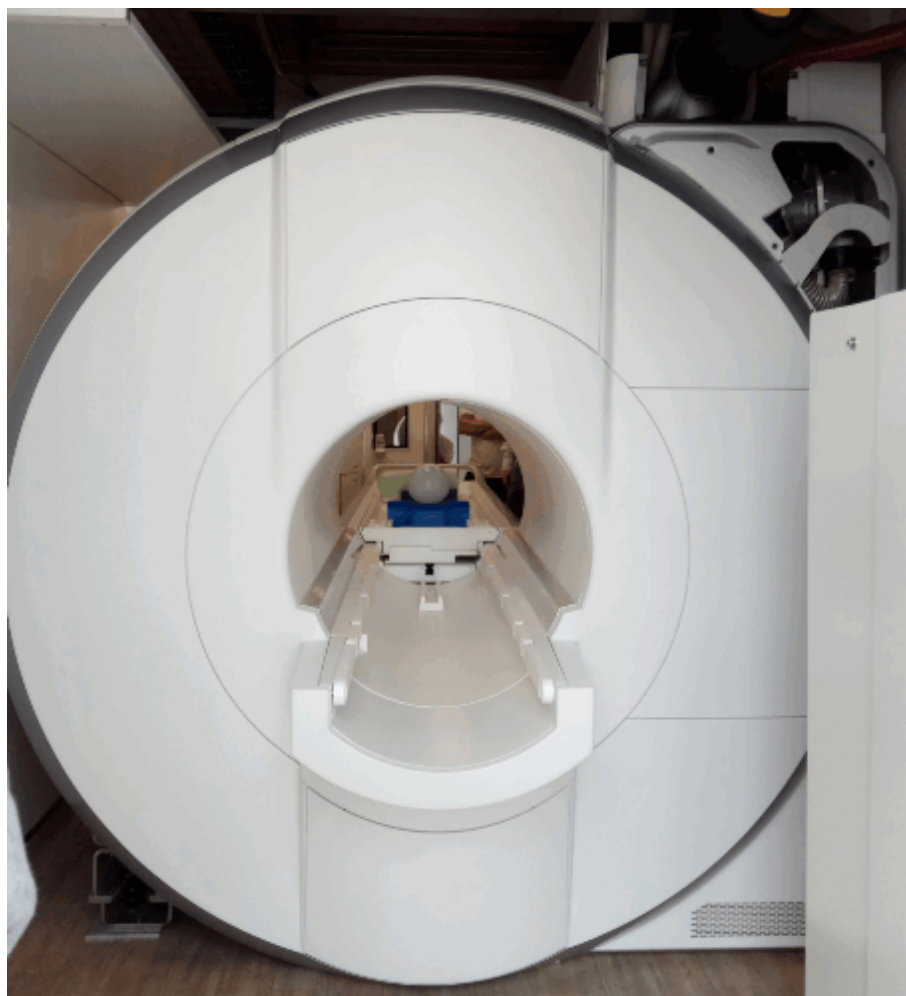
For the installation of the Aera Mobile system, reference to the existing Static system documentation is required.

2.1.1 Overview of the components

Fig. 1: System, front view (patient end)

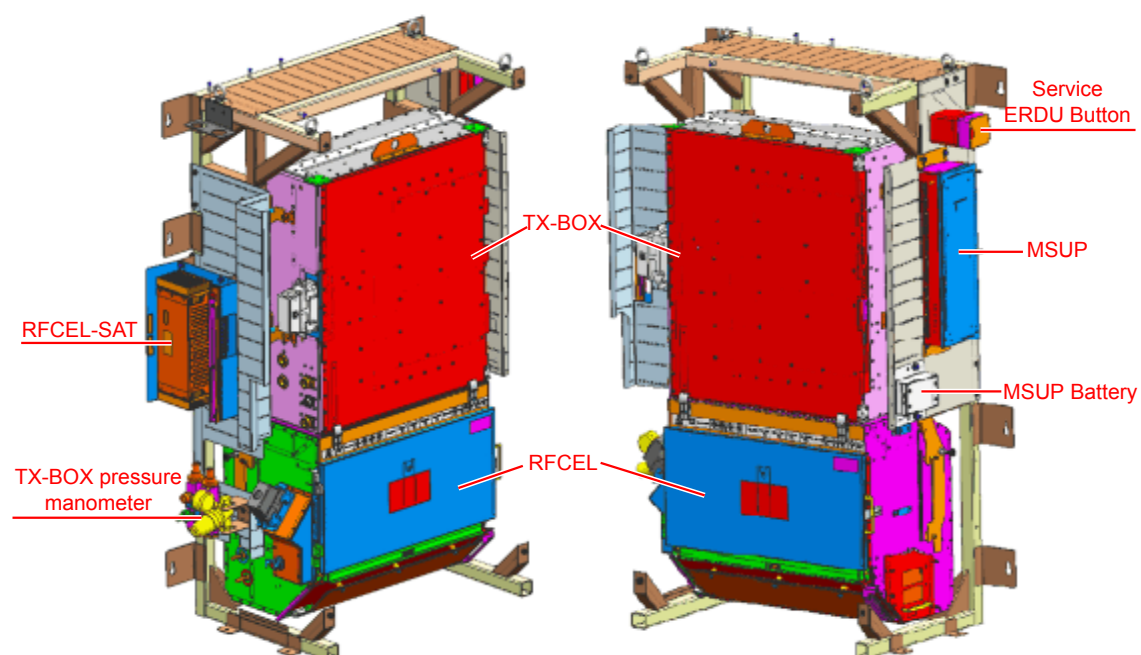


Fig. 2: System, back view (service end)



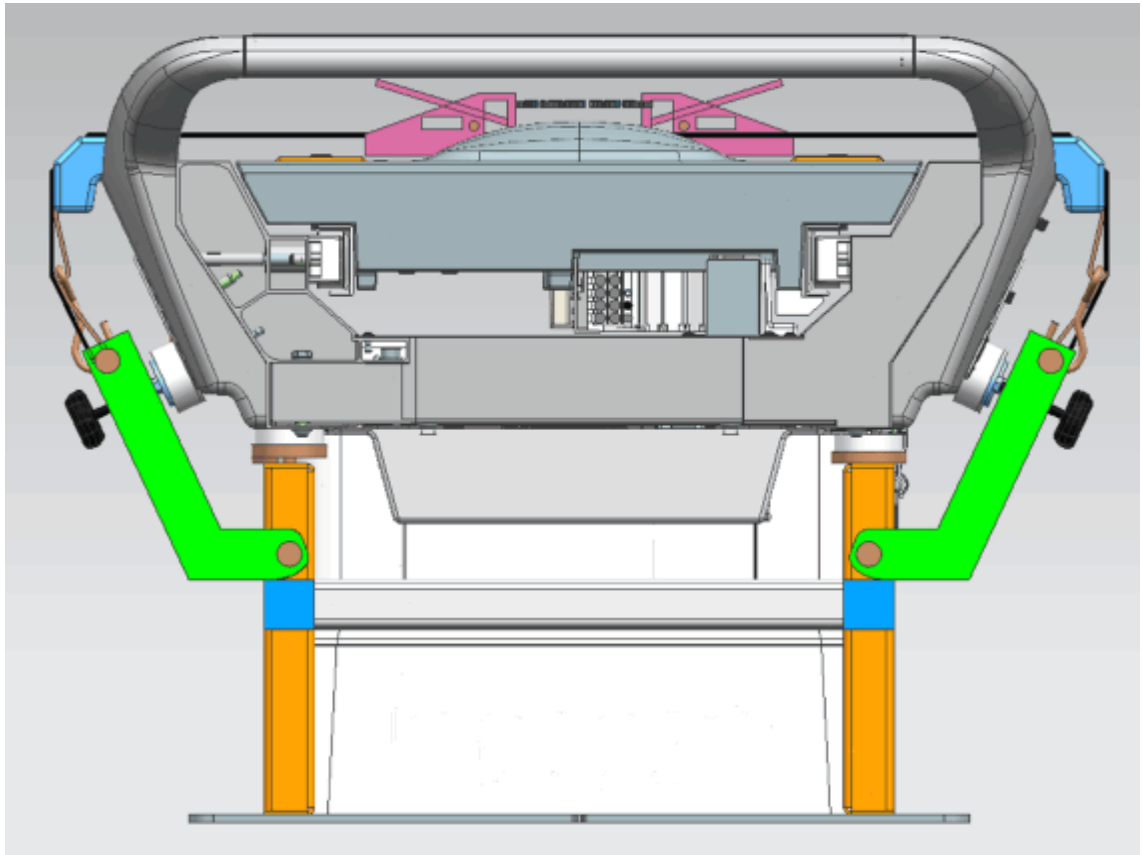
2.1.1.1 RCR (Rearranged Component Rack)

Fig. 3: RCR overview



2.1.1.2 Patient table support bracket

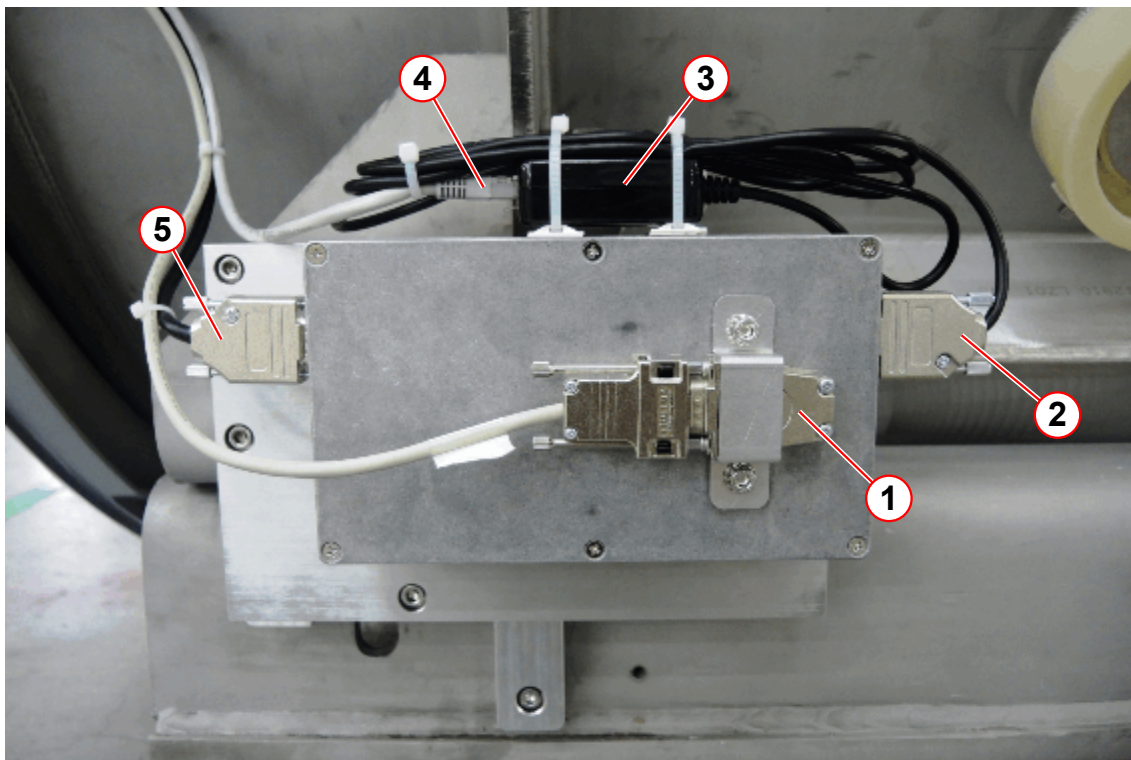
Fig. 4: PTAB support bracket



2.1.1.3 Shock-logger and external field sensor

- The installation of the RD298 shock-logger and the external field sensor is performed in the factory.

Fig. 5: Shock-logger and external field sensor



- (1) External field sensor
- (2) USB hybrid cable, 40121M
- (3) USB extender
- (4) RJ45 ethernet cable
- (5) Shock-logger cable, 40123M

2.1.1.4 Air Bladder Pump

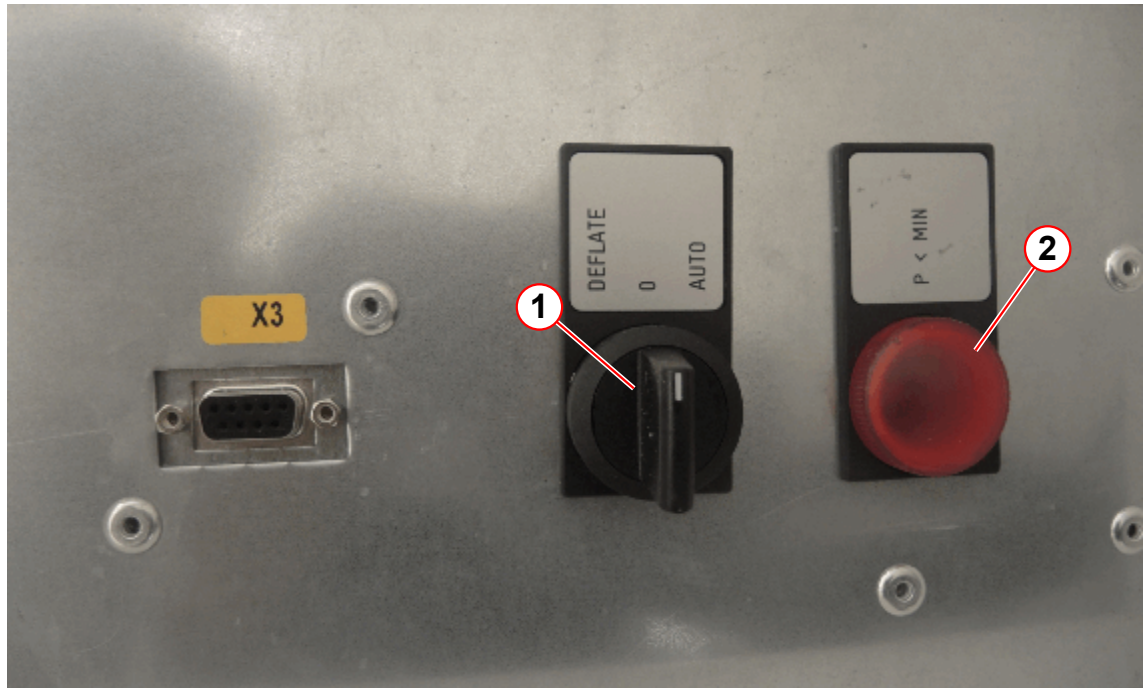
- The air bladder pump pressure device (PN: 10746803) is fitted in the Aera Mobile system. The air bladder pump will be located in the operator room.

Fig. 6: Air bladder pump pressure device



- Prior to any trailer transport, the bladder around the GC should be inflated to eliminate the possibility of GC movement.
 - Turn the bladder pump to the “**AUTO**” position, the bladder will be inflated automatically.
- During normal operation, the function selector should be in the “**0**” position.

Fig. 7: Air pump controls and indicator



- (1) Function selector
(2) Low-pressure indicator



The red indicator light will go off when the pump reaches the correct pressure.



It can take up to 2 minutes until the final pressure is reached.

2.1.2 Overview of the trailer

Fig. 8: Left side view of trailer



Fig. 9: Right side view of trailer



- (1) Patient lift
- (2) Entrance door of trailer

Fig. 10: Back view of trailer



- (1) RF door
- (2) Magnetic shielding door
- (3) Trailer door

2.2 Pre-positioning Steps

2.2.1 Moving the patient table into the trailer.



Move the patient table into the trailer before the magnet.

- Move the patient table into the trailer through the platform.

Fig. 11: Moving the patient table into the trailer.

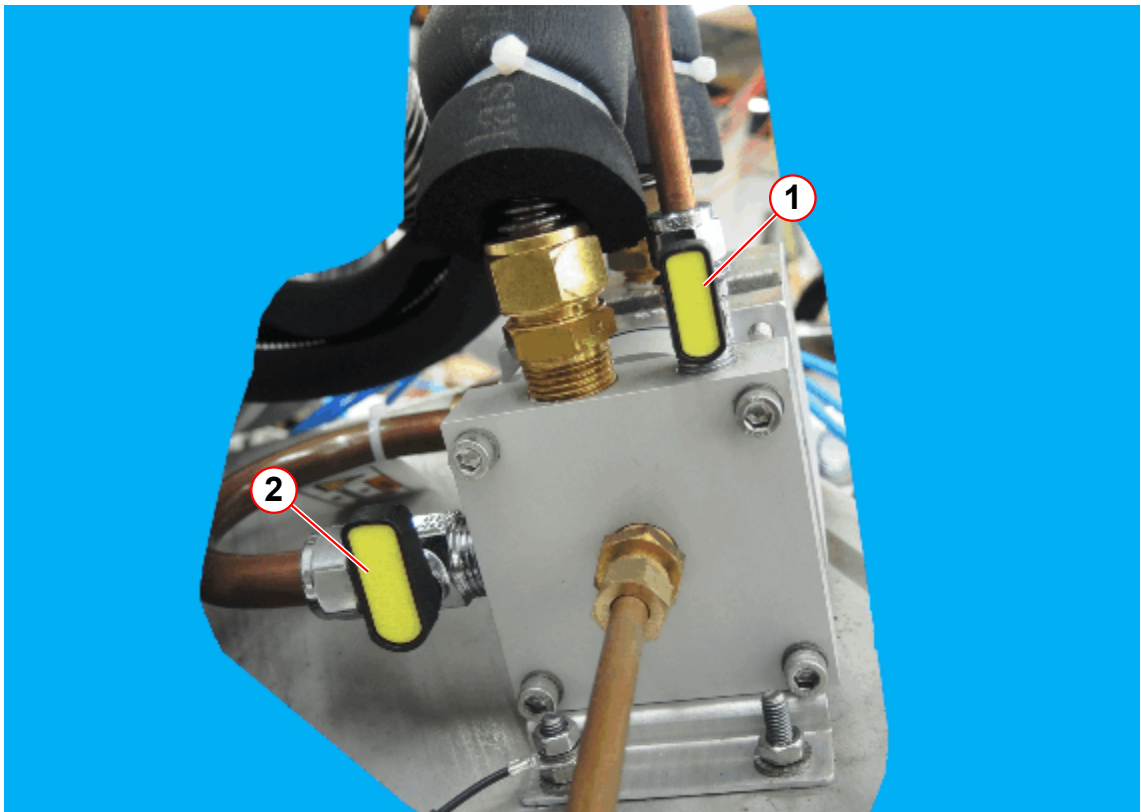


2.2.2 Checking the Cold-head Bypass Valve Position

Upon site arrival:

- The cold-head bypass valve [1] must be in the **OPEN** position.
- The magnet bypass valve [2] must be in the **CLOSED** position.

Fig. 12: Cold head bypass valve OPEN

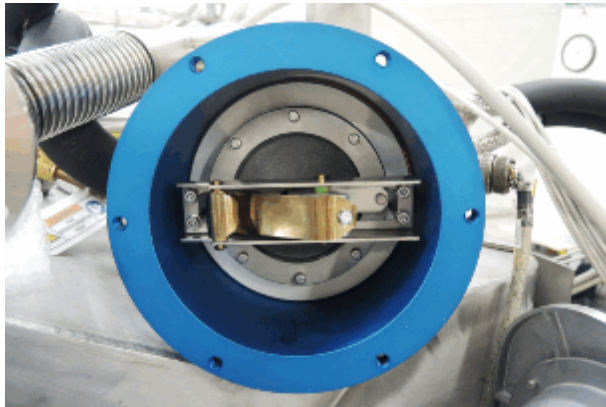


- (1) Cold head bypass valve
- (2) Magnet bypass valve

2.2.3 High-Altitude Quench Valve

- The Aera Mobile uses a high-altitude quench valve (PN: 10618505). The high-altitude quench valve is installed in the factory prior to delivery.
- During transport the high-altitude quench valve will reduce the risk of leaking and air ingress into the magnet at high altitudes above 3000 meters.
- The high-altitude quench valve is designed to remain closed during normal operation maintaining the internal magnet pressure at 15.5 psi (A). The valve will only open during a quench.

Fig. 13: High-altitude quench valve assembly



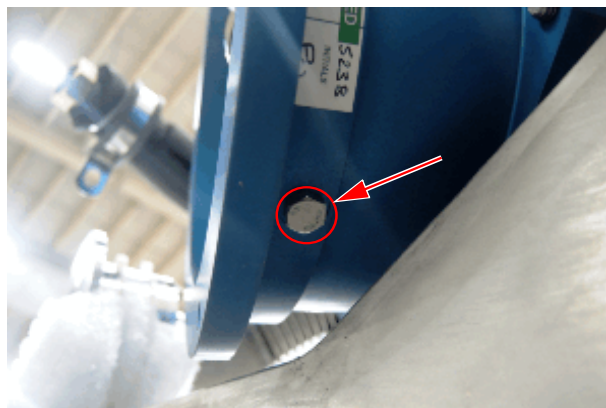
- For mobile systems, the high-altitude quench valve is pre-installed at the factory.
 - » The quench valve color coded blue to identify the high-altitude rating.

Fig. 14: High-altitude quench valve



- “HIGH ALTITUDE QV” is also engraved on the quench valve.

Fig. 15: High-altitude quench valve drain screw



Before the installation of the quench line:

- Locate the drain screw and remove it.
- Check to make sure the 1.5 mm drain hole is clear of debris.
 - » Confirm visibly that the hole is clear by using a cannula or suitably sized tool.



For Aera Mobile systems, the drain hole remains permanently open. This allows water / condensation to drip freely from the quench valve.

2.2.4 Installing the Extended Syphon

- The extended mobile syphon is not pre-installed at the factory. The extended syphon should be installed at the trailer manufacturer before the magnet is moved into the trailer.



WARNING

Pressurized helium vessel

Failure to observe the following safety information may result in physical injury.

- » **Do not disconnect any of the turret fittings without first depressurizing the helium vessel.**



CAUTION

Cryogenic liquids

Failure to observe the following safety warning may result in cold burns.

- » **Protective clothing, gloves, and eye protection must be worn.**

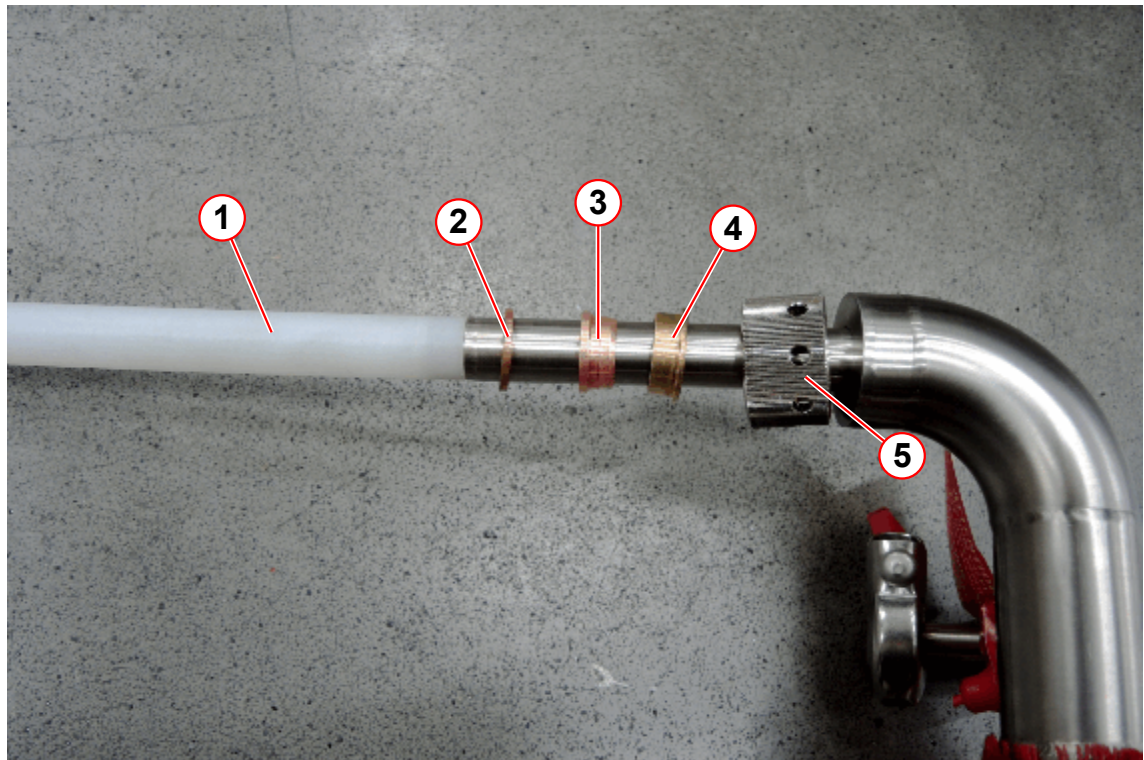
2.2.4.1 Tools Required

- Syphon kit mobile unit (PN: 10109298) (delivered with the system)
- Syphon support bracket (PN: 11282977) (delivered with the system)
- Pressure control tool (PN: 03861221)
- Syphon ring nut spanner
- Heat gun and paper towels
- Hand-held leak detector (PN: 10614850)
- Torque wrench suitable for 20 Nm
- 6 mm Allen (Hex) key socket
- Torque wrench suitable for 450 Nm
- 30 mm open-ended wrench
- Allen key (5 mm and 6 mm)
- 15 mm wrench / socket
- Cryogenic gloves, safety glasses and protective clothing and shoes

2.2.4.2 Extended Mobile Syphon Preparation

- Install the extended syphon fittings in the correct order as configured below:

Fig. 16: Syphon sealing configuration



- (1) Nylon diffuser leg
- (2) Copper sealing washer
- (3) Copper ferrule
- (4) Brass bush
- (5) Syphon ring nut

- Make sure the surface of the copper seals and syphon tube are clean and dry.

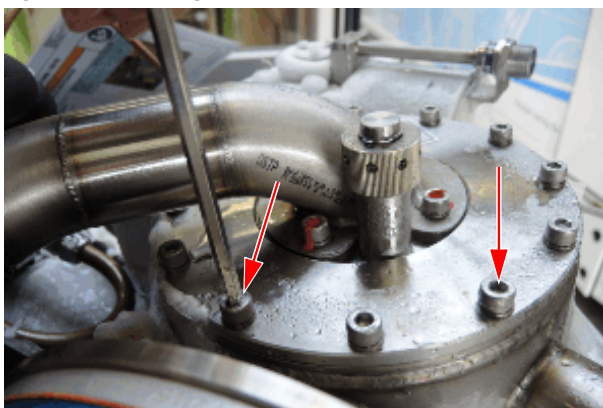
2.2.4.3 Installing the mobile syphon



The depressurization of the magnet must be performed in a well-ventilated area.

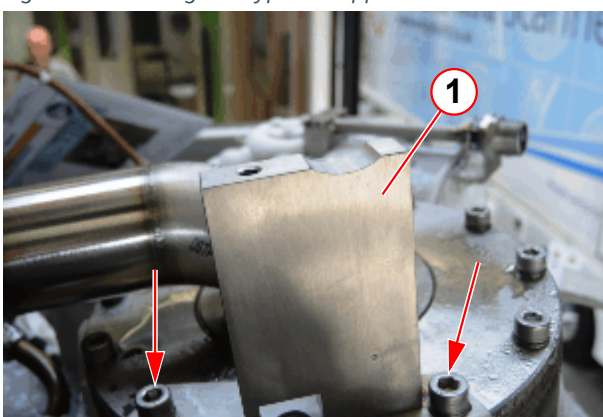
- **Fully open** the magnet bypass valve to depressurize the helium vessel.
- Wait until **zero psi** is displayed on the pressure gauge.
- **Close** the magnet bypass valve to prevent air ingress.
- **Warm up** the turret top plate and syphon port with heat gun.

Fig. 17: Removing the screws



- Remove the two screws (as shown) from the turret top plate with the 6 mm Allen (hex) key.

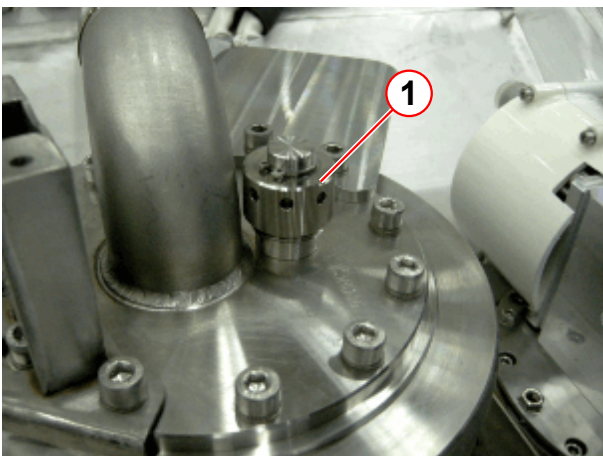
Fig. 18: Installing the syphon support bracket



(1) Syphon support bracket

- Install the syphon support bracket with the two screws.
- Torque the screws to 20 ± 1 Nm.

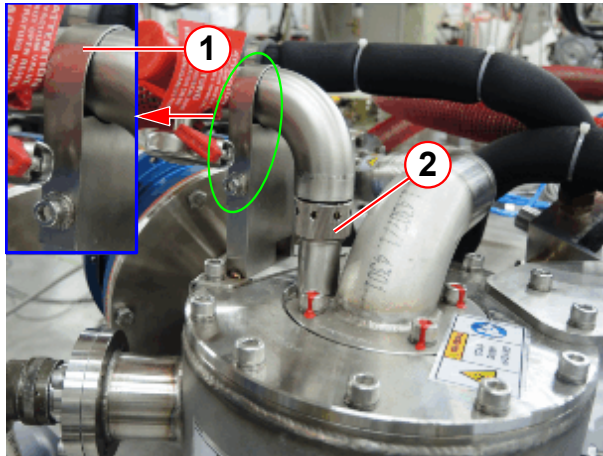
Fig. 19: Removing the turret ring nut



(1) Ring nut

- Using the supplied 'C' spanner, remove the syphon port ring nut.

Fig. 20: Installing the extended syphon



- (1) Clamp
- (2) Ring nut

- Install the extended syphon into the syphon port and tighten the ring nut.
- Warm the syphon ring nut with a heat gun and repeat tightening.
- Fix the clamp on top of the support bracket to secure the syphon in the bracket.

2.2.4.4 Final Tasks

- Use a heat gun to warm the turret and dry with paper toweling.
- **Leak check** the syphon entry port, turret top plate and syphon end cap.
- If a leak is detected:
 - » Warm up the turret and tighten the screws.
 - » Repeat the leak check.

Fig. 21: Routing the extended syphon

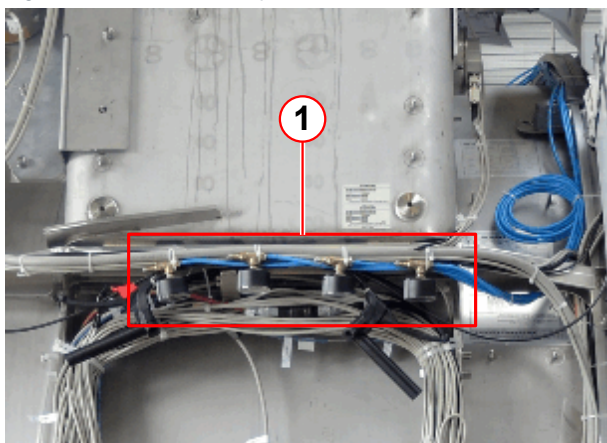


- Route the extended syphon on top of the magnet and fix in position with cable ties.

2.2.5 Installing the Air Bladder Pump Air Hose

- In Aera Mobile systems the air bladder pump is used to inflate the air cushion on top of the gradient coil.
- When the system arrives at the trailer manufacturer, install the air hoses connected to the air bladder pump before the magnet is moved into the trailer.

Fig. 22: Air cushion fill port and manometers



(1) Air cushion fill port and manometers

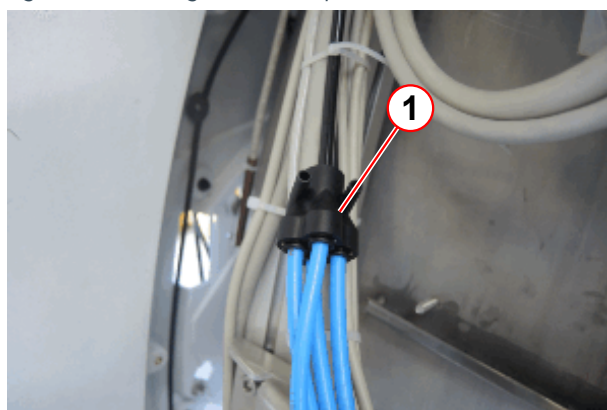
- Air cushion fill port and manometers as they arrive on site.

Fig. 23: Cutting the air hose



- Uncoil the air hose and cut it to the same length.
- The manometers and fill port are no longer required.

Fig. 24: Installing the 4-to-1 quick connector



(1) Quick connector

- Attach the 4-to-1 quick connector to the air hose.
- Join the air hose and cables using cable ties.
- The black air hose can be routed the same way as the cables to the filter plate.

2.3 Installing the Magnet in the Trailer

2.3.1 Moving the Magnet to the Platform

NOTICE

Exercise care, as magnet covers can be damaged during magnet installation.

- » Ensure that the magnet covers have sufficient clearance above the floor during installation.
- » Do not push or pull on the system cover or exert any force.



During transport, ensure that the lifting components do not detach from the box section.

Options for moving the magnet into the trailer manufacturer facility:

- Crane from the transport trailer onto a platform and roll into the factory.
- Roll from the transport trailer directly into the factory.

Fig. 25: Positioning the magnet



Rolling from transport trailer

- To move the magnet directly from the transport trailer to inside a building:
 - » Fit the 4 x swan necks and rollers.
 - » Lay steel spreader plates from the transport trailer to the building floor.
 - » Carefully push / pull the magnet into the building.

Fig. 26: Lifting the magnet by a crane

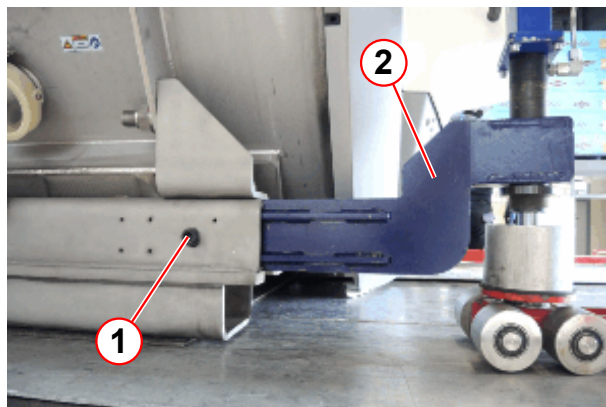


Craning from transport trailer

Use suitable straps or chains when lifting the magnet by crane.

- Fit straps / chains to the 4 x shackles attached to the magnet.
- Carefully lift the magnet from the transport trailer and move to the intermediate platform.

Fig. 27: Attaching the swan neck



- (1) Fixing screw
- (2) Swan neck

- Insert the 4 x swan neck tooling into the box section at each corner of the magnet.
- Secure the swan neck by the fixing screws.
- Make sure the swan neck will not detach from the box section when moving the magnet into the building.



The swan necks can remain fitted when lifting the magnet onto the platform for mobile trailer installation.

2.3.2 Positioning the Magnet in the Trailer

A suitable intermediate platform must be aligned with the rear trailer opening.

- Lift the magnet using a crane and position it onto the platform. Do not lower completely at this point.
- Carefully align the swan necks and lower the magnet onto the 4 x rollers.
- Lay spreader plates to distribute the load from the rollers on the trailer floor and to bridge any step between the intermediate platform and the trailer floor.

Fig. 28: Positioning the magnet on the platform



When moving and lowering the magnet, ensure that the covers do not touch the floor.

- Move the system on the rollers into the trailer.
- Position the magnet so that the box section holes are positioned over the mounting holes in the trailer floor.
- Continue to check that the magnet maintains the proper distance from the side walls.
- Fit 4 x hydraulic pumps to each swan neck and start lowering the magnet.
- Lower the magnet onto the spreader floor plates by slowly opening (clockwise) the set screws on the hydraulic pumps.



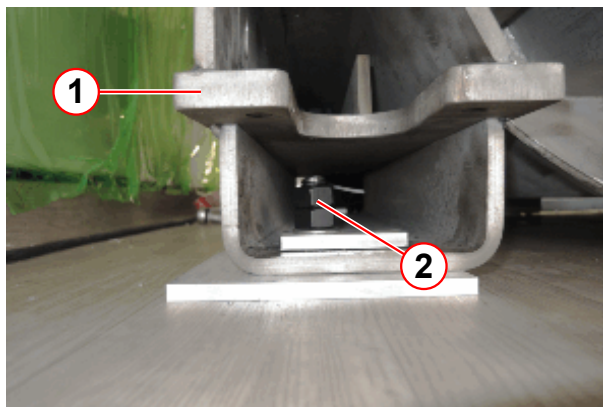
Ensure that the box sections are correctly aligned with the magnet base frame and trailer floor anchoring points.

If incorrect, raise the magnet using the correct lifting devices and realign accordingly.

- Remove the four swan necks from the box sections.
- Use a 30 mm open-ended (crows foot) adapter for the torque wrench to access the M20 nut.
- Tighten the screws that secure the box-section to trailer floor. Torque the screws to 430 ± 10 Nm.



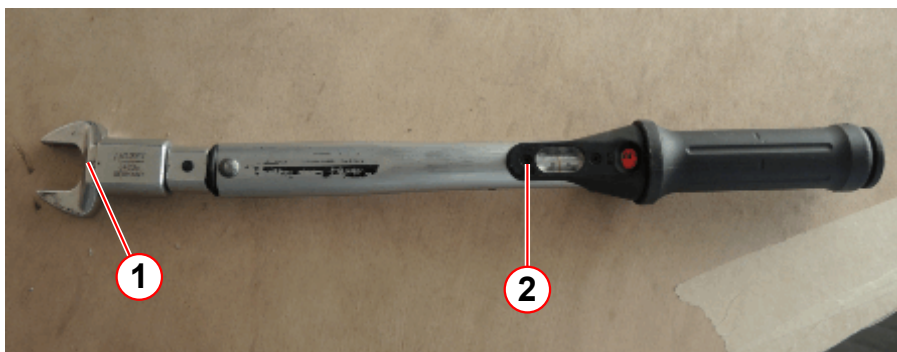
Fig. 29: Mount box section on the trailer floor



- (1) Box section
- (2) Securing nuts and bolt

- Attach the magnet to the floor, according to the trailer manufacturer's instructions.

Fig. 30: Torque wrench with an adapter



- (1) 30 mm 'crows foot' adapter wrench
- (2) Torque wrench

2.3.3 Adjusting the Magnet Level

- The magnet is leveled by trailer adjustment. No magnet action is required.

2.4 Post-positioning Steps

2.4.1 Moving the RCR into the Trailer

- After the magnet is moved into the trailer, lift the RCR to the intermediate platform by a crane.

Fig. 31: RCR on the platform



- Lift the RCR onto the platform.
- Then move the RCR into the trailer.

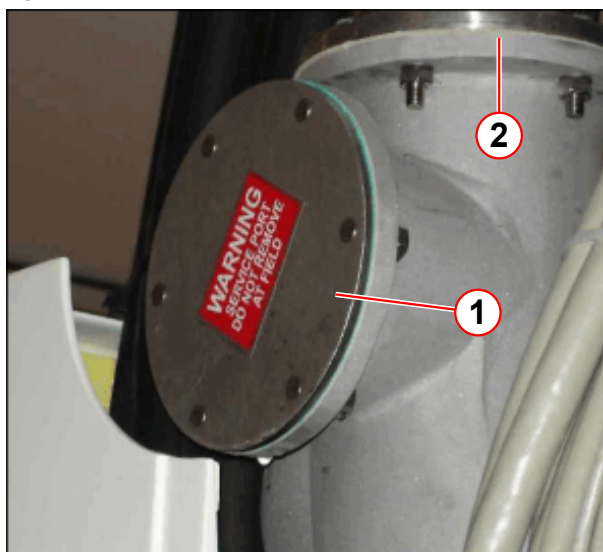
Fig. 32: Positioning and securing the RCR



- Move the RCR to the right position.
- Fix and secure the RCR on the floor and side wall.

2.4.2 Installing the Quench Line

Fig. 33: Quench elbow



- (1) Horizontal elbow outlet
(2) Vertical elbow outlet

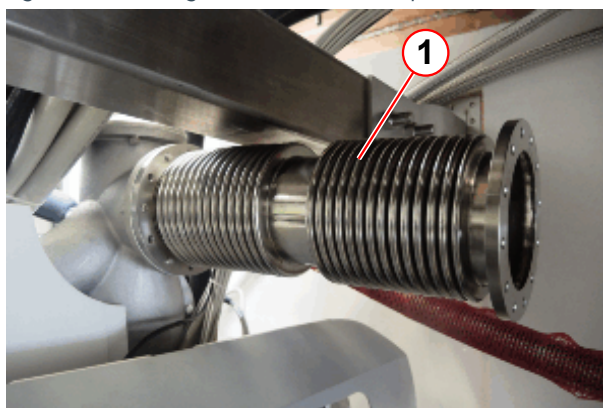
For a mobile magnet the elbow blanking plate position must be replaced.

- Remove the six M8 nuts and washers securing the blanking plate.
- Fit the blanking plate to the vertical outlet flange of the elbow.
- Secure with the M8 nuts and washers.



The de-coupler must be installed nominally straight, allowing for minor misalignment. Do not exceed the design limit of ± 20 mm.

Fig. 34: Installing the acoustic de-coupler



- (1) Acoustic decoupler

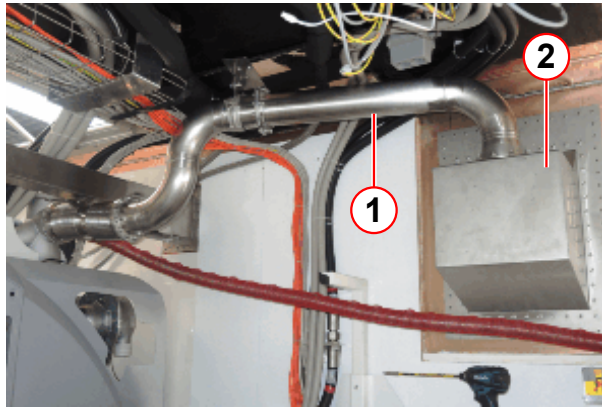
Locate the acoustic decoupler and the fixing kit supplied.

- Using the gasket seal, fit the acoustic de-coupler to the horizontal outlet of the elbow.
- Fit the 12 x isolation bushes (PN: 05761825) to the acoustic de-coupler flange.



The quench line and outlet must comply with the specifications in the Site Planning Guide.

Fig. 35: Connecting the quench line

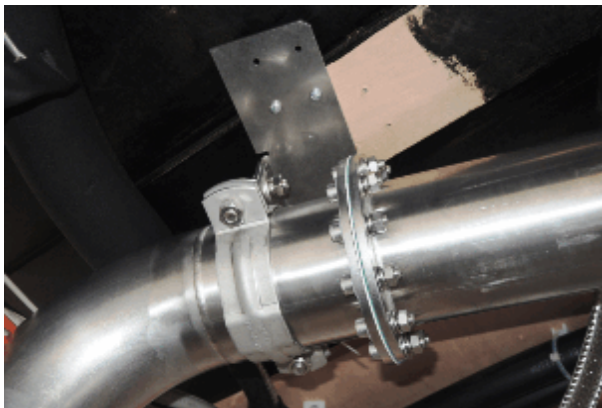


- (1) Quench line
- (2) Plenum box

- Connect the manufacturers quench line to the acoustic de-coupler.
- Connect the quench pipe from the acoustic de-coupler to the outlet.

The mobile quench line specification requires a 'Plenum' box to be fitted at the outlet.

Fig. 36: Securing the quench pipe to the ceiling



- Secure the quench pipe to the roof of the trailer with clamps to support the weight and reduce movement.



Ensure that the quench line is insulated along the entire length to avoid condensation dripping onto electronic components located directly beneath.

Fig. 37: Insulating the acoustic de-coupler and quench pipe



- Insulate the acoustic de-coupler and the complete length of the quench pipe.
- Fit warning signs to the insulation to identify the quench line.

Fig. 38: Louvre panel (Mobile example)



The quench line is covered at the outlet by a louvre panel.

- A wire mesh (size 10 +2 / -1 mm) must still be fitted behind the louvre panel.
- A warning sign must be attached next to the outlet.

2.4.3 Mounting the Extended Syphon

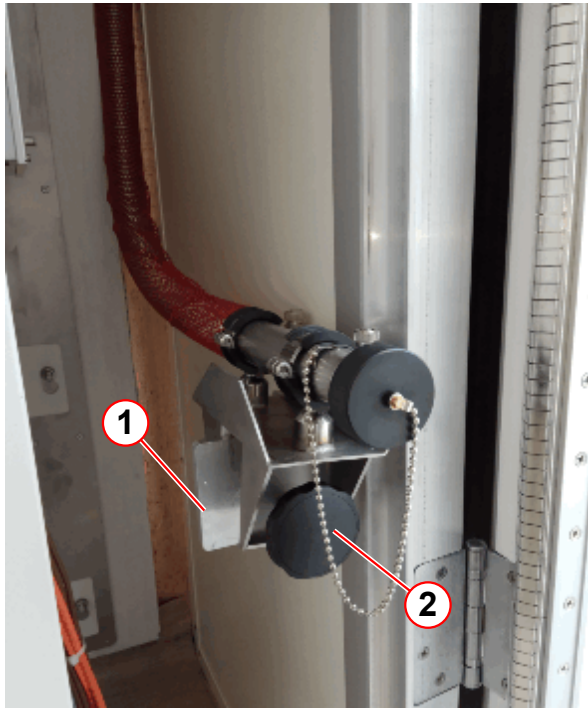
Fig. 39: Attaching the extended syphon



(1) Extended syphon

- Route the extended syphon along the quench pipe through the cover.
- Attach the extended syphon using the insulating clamps.
- The extended syphon should be above the RCR so that no related RCR service is affected.

Fig. 40: Syphon bracket and clamp



- (1) Syphon bracket
- (2) Syphon clamp

- Attach the syphon wall bracket in a suitable location. This should be at the rear of the trailer near the RF door.
- Fix the clamp to the syphon with the clamp fixings.

The syphon clamp can be removed from the clamp and moved with the syphon for helium filling.

2.5 Examination Room Installation

2.5.1 RCR Installation



Ensure that the cables are not damaged when unpacking and positioning the cabinet.

Fig. 41: RCR overview

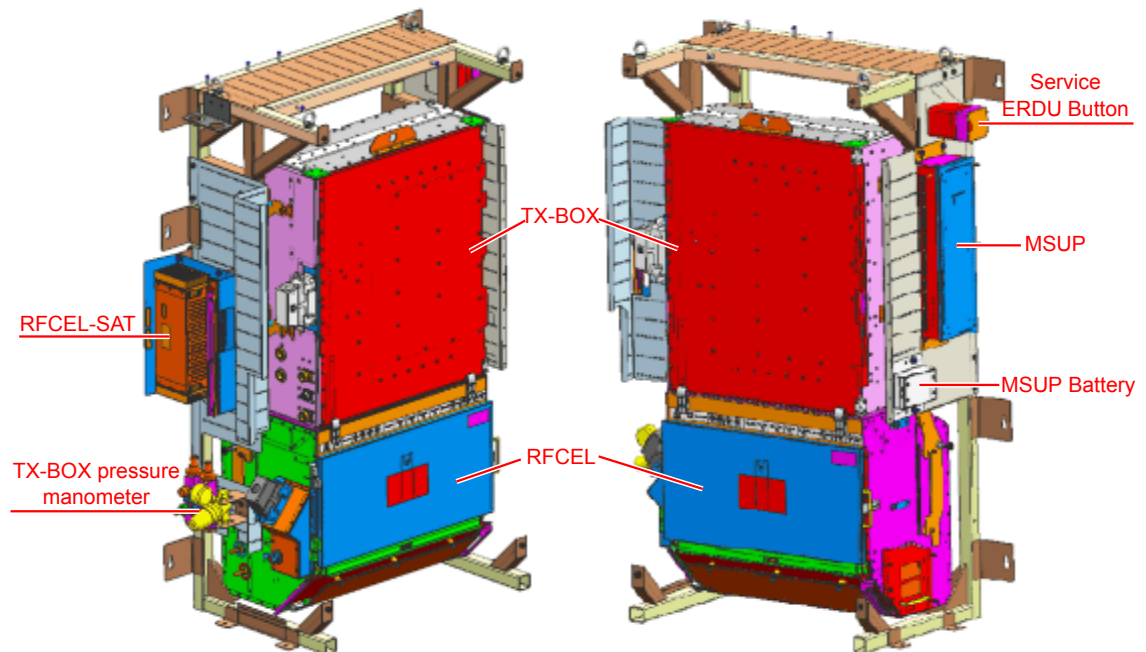
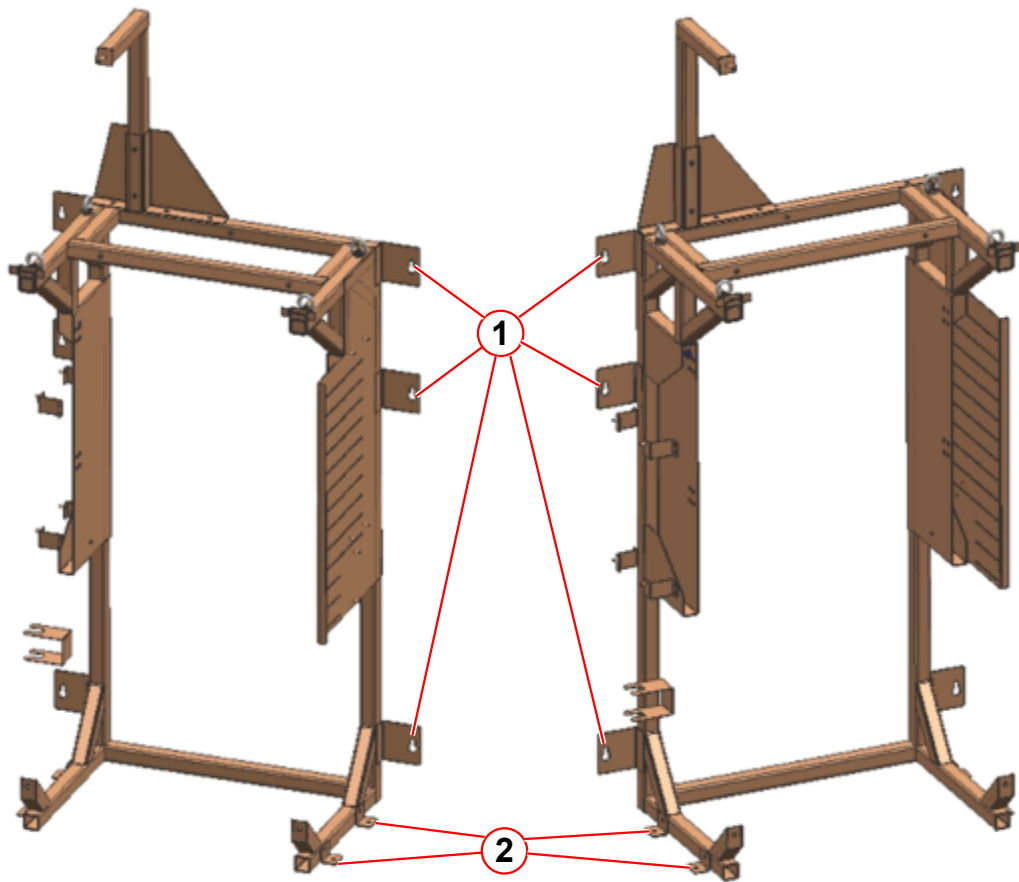


Fig. 42: RCR fixation points



- (1) Fixation points on trailer wall
- (2) Fixation points on floor

Fig. 43: RCR fixation dimension on trailer wall

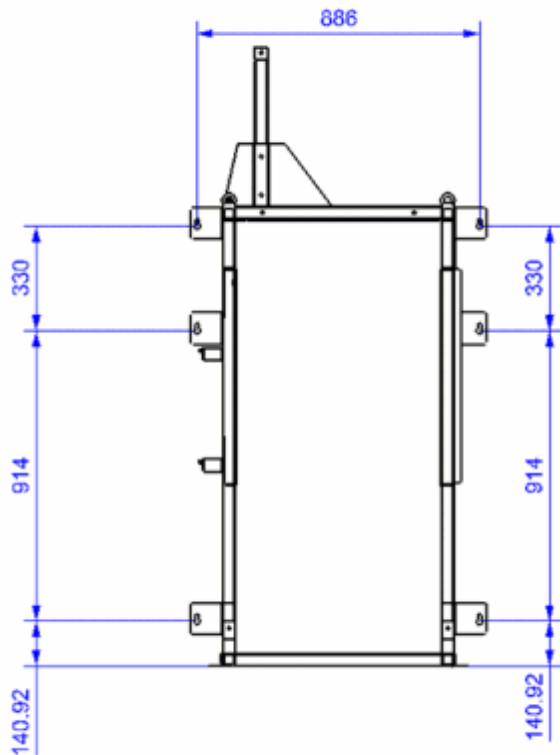
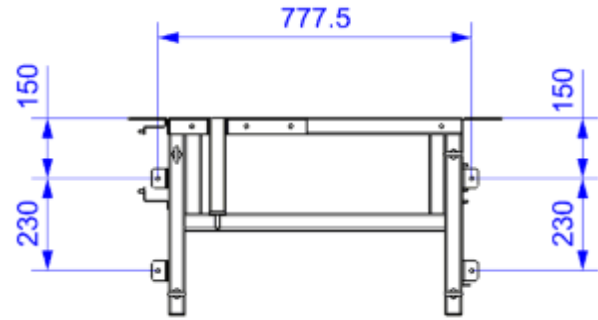


Fig. 44: RCR fixation dimension on floor



- There are six M10 bolts on the trailer wall and four M10 bolts on the floor for RCR fixation.
- After the position adjustment, secure the RCR by the nuts.

Fig. 45: Installing the RCR

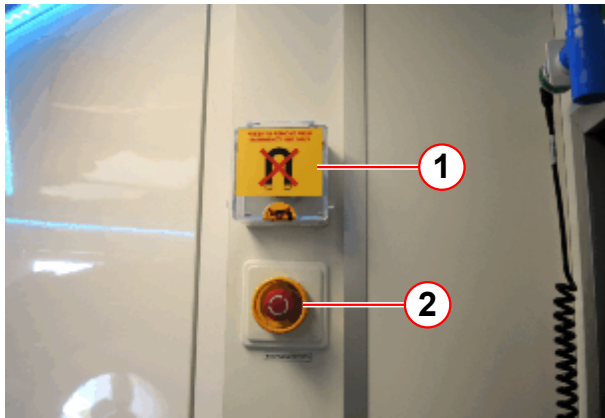


- Fix and secure the RCR on the floor and side wall.

2.5.2 ERDU and EPO Buttons

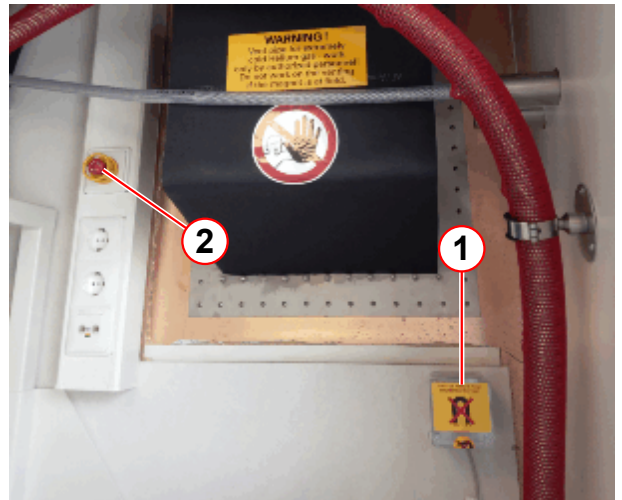
- Install and connect the following parts:
 - ERDU button
 - Emergency power-off button

Fig. 46: ERDU and EPO button at the patient end



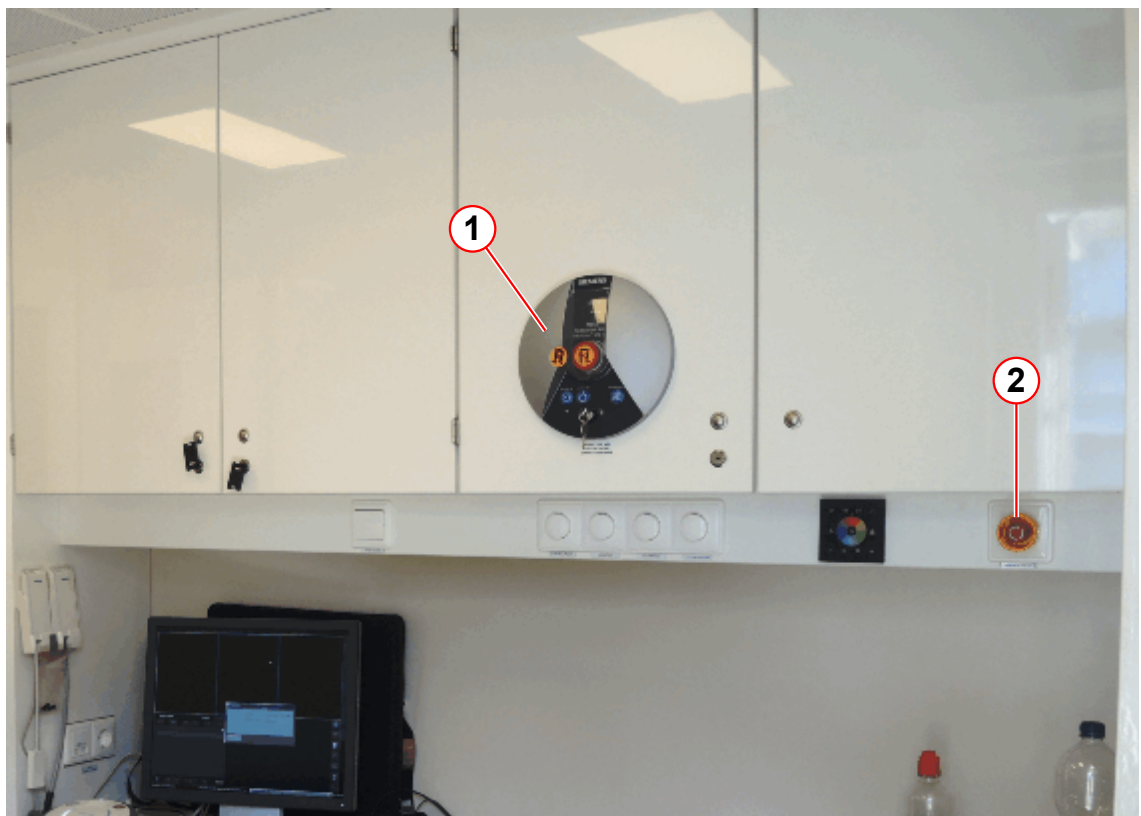
- (1) ERDU button
- (2) Emergency power-off button

Fig. 47: ERDU and EPO button at service end



- (1) ERDU button
- (2) Emergency power-off button

Fig. 48: Alarm box and EPO button in operator room



- (1) Alarm box
- (2) Emergency power-off button

2.5.3 Installing the Mobile 298 Shock-logger

2.5.3.1 General Information

This procedure covers the installation of the 298 shock-logger system that must be fitted to the OR98 when installed in mobile trailers. It is broken down into two installation stages:

- **Stage 1** - Installation onto the magnet (staging) and operational setup for shipment to the trailer manufacturer.
- **Stage 2** - Installation of the shock logger indicator and lead kit in the mobile trailer, with mains power to the shock-logger.

Safety



CAUTION

Electrical shock

Ensure all power is switched off when working inside the electrical cabinets.

- » Only trained CSEs must perform this procedure.



CAUTION

Strong magnetic field

Damage to laptop PC and equipment.

- » When downloading data from the shock-logger, always use a suitable length USB cable. Keep the laptop PC away from the magnet.

Required tools

Equipment required for shock-logger installation:

- 4 mm and 5 mm Allen (hex) keys
- Torque wrench (material number **08395829**)
- Service laptop and USB converter
- Standard tool set

Required parts

- 298 shock-logger kit - (**10621017**).

Spare parts

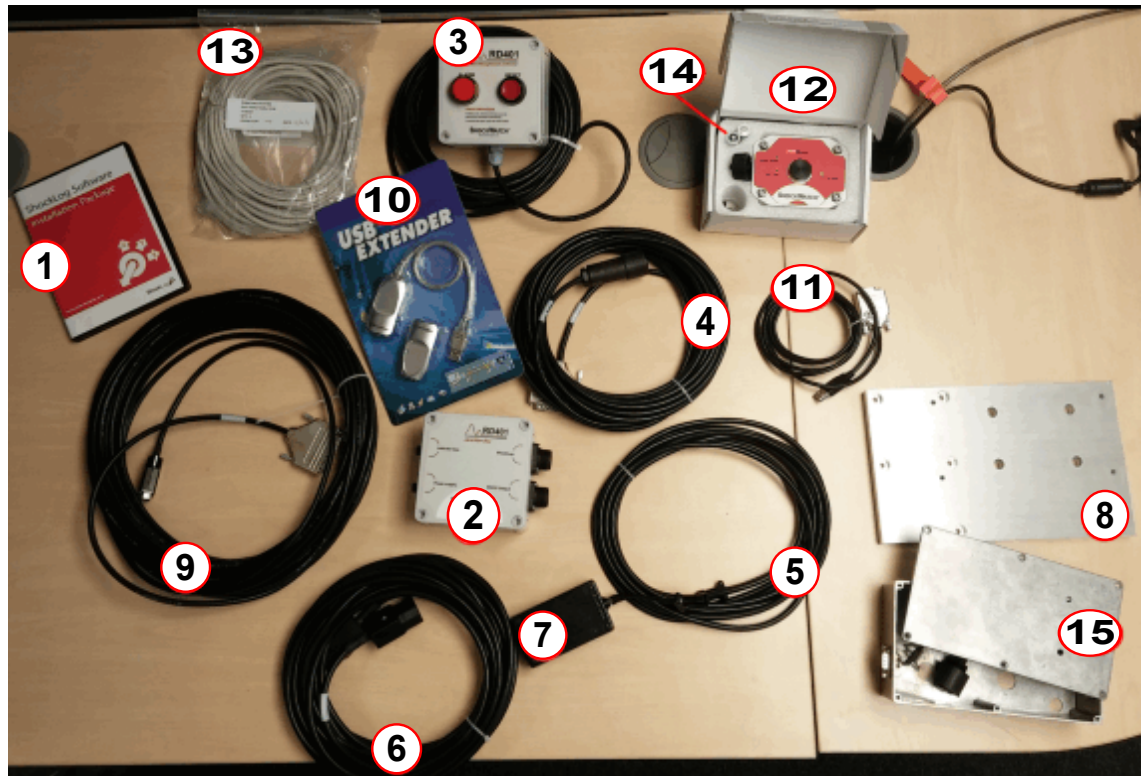
- 2 x 3.6 V lithium battery type AA (material number **10622033**).
- D15 RF filter - (material number **04753203**)



A D15 filter is required. If there is not a spare available on the RF-filter plate, order and fit a new one.

Shock-logger kit

Fig. 49: Shock-logger components



Tab. 1 Shock-logger components

Item	Description	Material number	Qty
1	Shock-logger software CD	298CDSW	1
2	Junction box	40101M	1
3	Indicator box	40102M	1
4	Cable - filter plate to junction box	40103M	1
5	Cable - power supply to junction box	40105M	1
6	Cable - UPS to power supply	40106M	1
7	Power supply	40108M	1
8	Shock-logger mounting plate - 40110M-RETRO	11256047	1
9	Cable - shock-logger to filter plate - 9 - 25 Pin	40123M	1
10	USB extender cable LAN - 0101	41135M	1

Item	Description	Material number	Qty
11	Cable - hybrid USB - 9 pin skt to USB	40121M	1
12	298 shock-logger unit	298 SH0Z0	1
13	Cable - RJ45 ethernet - 15 m	V192M	1
14	3.6 V lithium battery AA style	10622033	2
15	Shielding box aluminium - N507-M	11256048	1



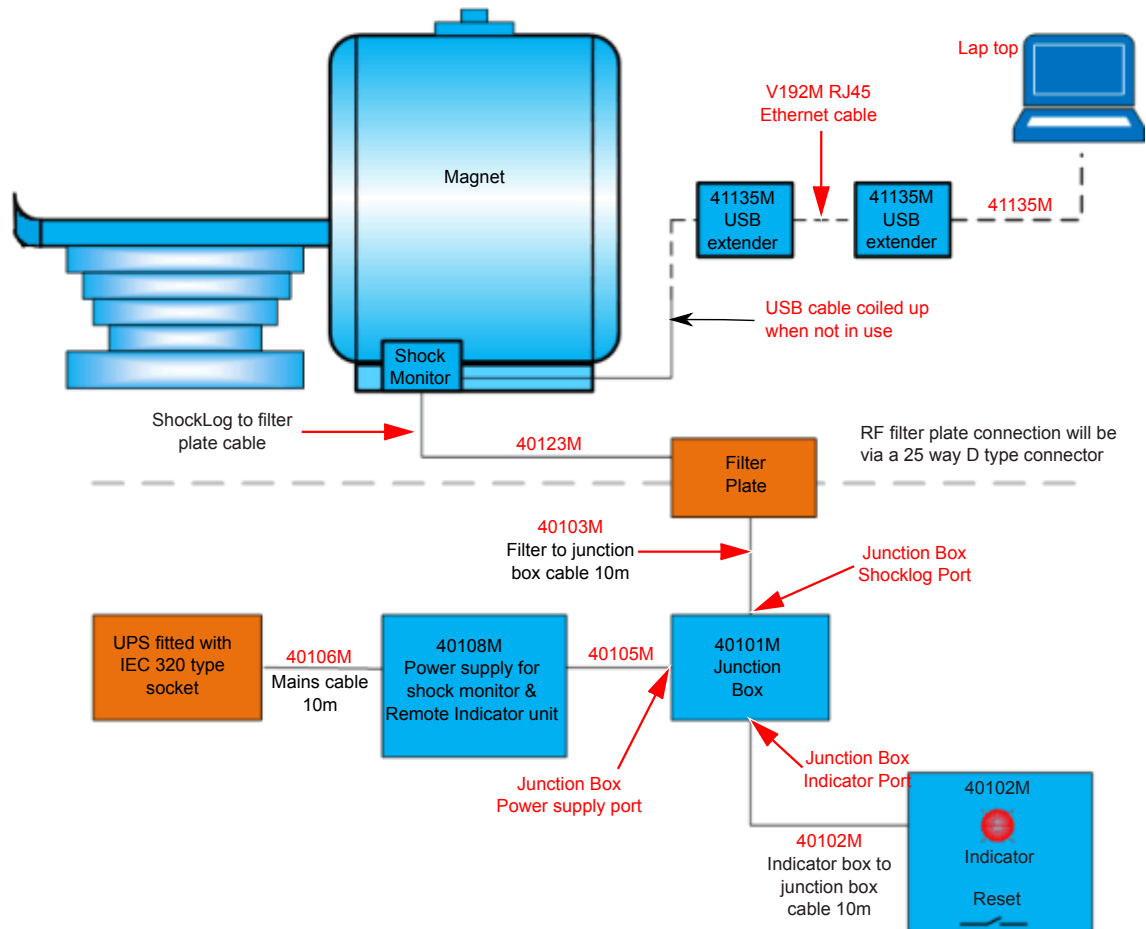
The inner box cables 40119M and 40120M are already fitted to the aluminium shielding box assembly.

Tab. 2 Attachment parts provided in shipment

Description	Material number	Qty
Screw M6 x 20 skt hd S.S.	60080	3
Screw M5 x 45 skt hd S.S.	60085	4
Screw M5 x 10 skt hd S.S.	S238	4
M5 washer shakeproof S.S.	61311	8
M6 washer shakeproof S.S.	61310	3
Screws S.S. - (shielding box)	n/a	6
Flex-Loc fastener M5	62244	4

Installation diagram

Fig. 50: Installation diagram

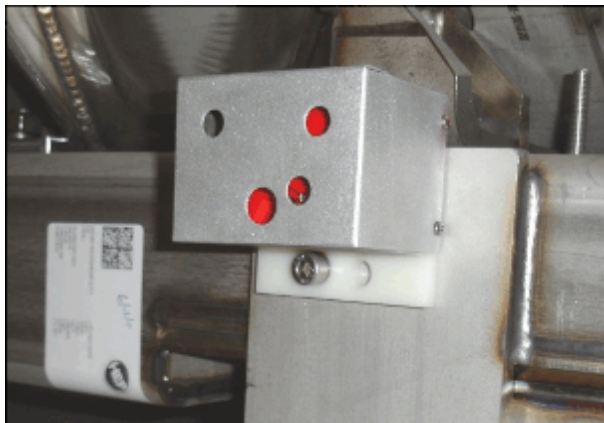


2.5.3.2 Stage 1 - Factory Installation (Staging) Procedure

Pre-requisites

The initial fitting and set-up of the 298 shock-logger is performed at the MR factory.

Fig. 51: ShockBug

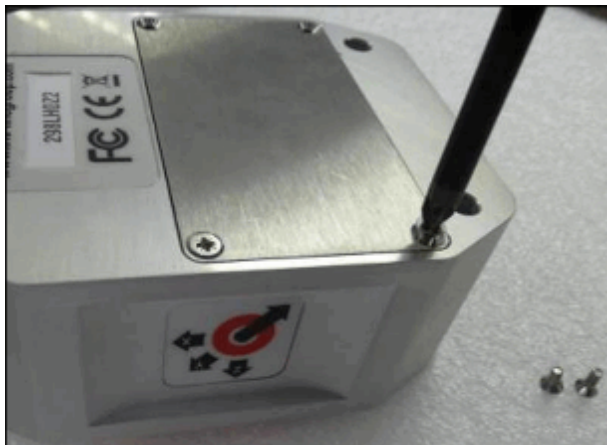


- Before fitting the 298 shock-logger, check that the shock bug indicator is not flashing **red**.
- Escalate all incidents of shock indicators found triggered to the HSC at - magnet.wschsc.team@siemens-healthineers.com

Installing new batteries

- Prior to fitting the shock logger, ensure that new batteries are fitted.

Fig. 52: Battery cover



- The battery compartment is located under the plate on the bottom face of the shock logger.
- Remove four screws to access the battery compartment.
- Remove the shock-logger battery cover.



The batteries discharge very quickly if installed incorrectly. Check the polarity before installing.

Fig. 53: Battery fitting

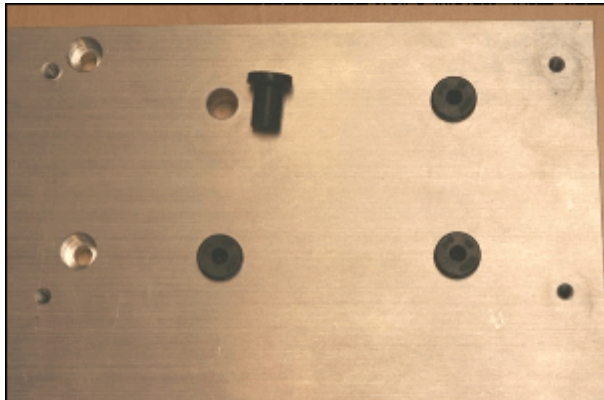


- Install two new 3.6 V lithium thionyl chloride batteries.
- Check the correct polarity.
- Reinstall the cover and tighten the four screws.
- Check that the shock-logger is running – "SLOT ALARM" LED and blinking every 10 seconds.

Fitting the shielding box and shock-logger

The initial assembly of the shielding box and shock logger can be completed before attaching to the magnet.

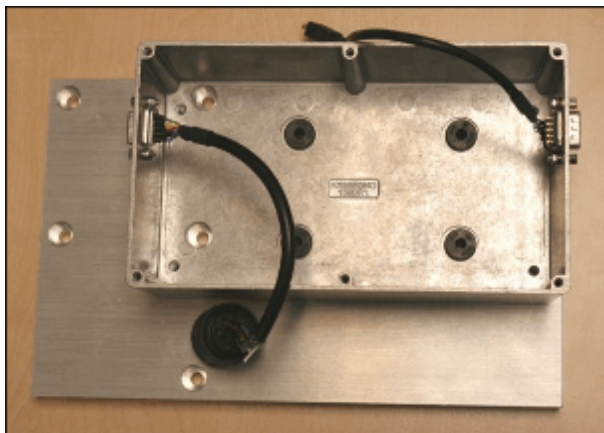
Fig. 54: Fitting the rubber bushes



Attach the shock logger to the aluminium plate:

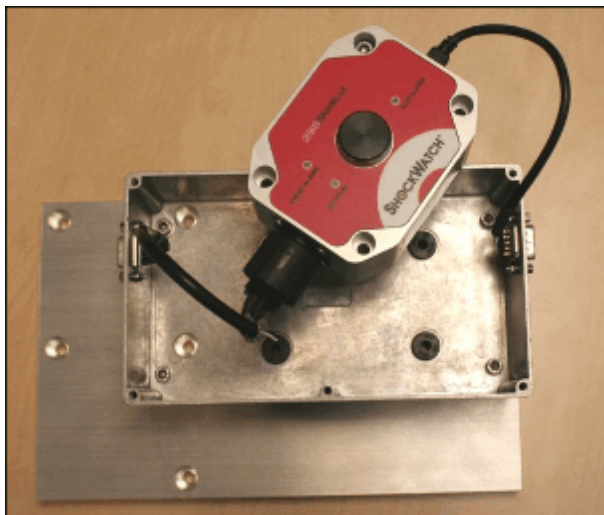
- Place the four rubber bushes into the holes provided in the mounting plate.

Fig. 55: Shielding box



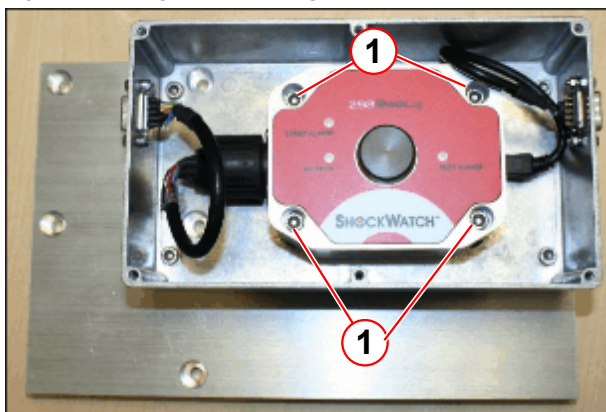
- Position the shielding box over the four rubber bushes.
- Fix the shielding box onto the mounting plate using the four M5 x 10 screws and washers.
- Tighten the M5 screws to 5 Nm.

Fig. 56: Fitting the inner cables



- Connect the two inner box cables to the shock logger ends.
- Check that the USB cable is fully engaged in the shock logger USB socket.
- If necessary, push the cable end further into the threaded bush to ensure a correct engagement.

Fig. 57: Fitting the Shock-log



(1) 4 x M5 x 45 mm screws

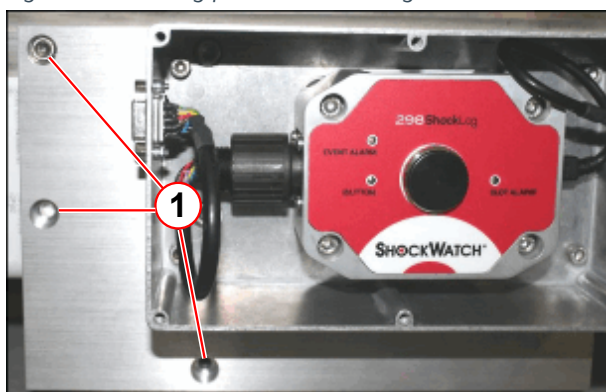
Attach the shock logger to the aluminium plate:

- Fit the four M5 x 45 S.S. screws and tighten into the rubber bushes in the mounting plate.
- Tighten the M5 screws to 4 Nm.
- Check that the shock logger is oriented (as shown) with the SL USB end oriented towards the magnet service end.

Attaching the shock logger assembly to the magnet

- Attach the aluminium mounting plate (40110M-RETRO) to the short magnet box section (on the right-hand side, as viewed from patient table end).
- Use the plate holes as shown, appropriate for the magnet type.

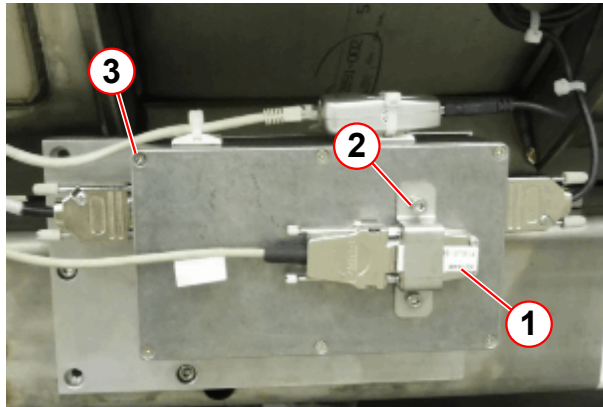
Fig. 58: Mounting plate onto the magnet



(1) Aera fixing holes

- Use the three M6 x 20 mm cap head screws and shake-proof washers provided.
- Tighten the M6 screws to 9 Nm.

Fig. 59: Shielding box lid



- (1) Magnetic field sensor
- (2) M5 fixing screws x 2
- (3) Lid fixing screws x 6

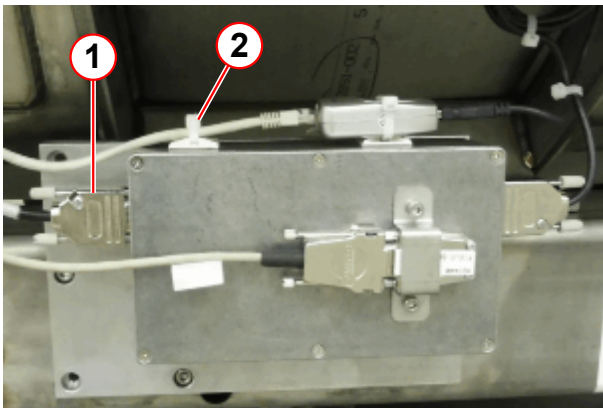
- Attach the magnetic field sensor to the shielding box cover using two M5 screws and washers. (Field sensor is not included in the kit).
- Secure the shielding box cover using the six screws.



The magnetic field sensor is not provided in the mobile kit.

Cable connections

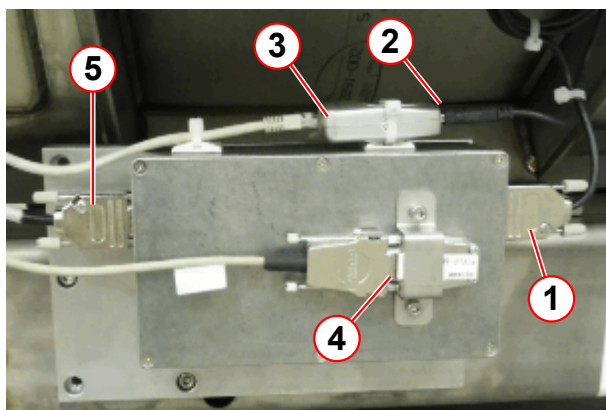
Fig. 60: ShockLog cable



- (1) Cable 40123M
- (2) Cable tie

- Connect the 40123M cable, with the end marked "shock log" to the left port of the aluminium box.
- Route the 40123M cable over the magnet to the point where the main cable trunking exits the magnet, so at the service end of the turret.
- Coil the cable and secure in place with cable ties.

Fig. 61: Cable connections



- (1) USB hybrid cable 10121M
- (2) USB extender cable 41135M
- (3) RJ45 cable
- (4) Magnetic field sensor
- (5) Cable 40123M

- Connect the hybrid USB cable 40121M to the shock-logger shielding box (N507-M).
- Connect the other end of the cable to the USB extender (41135).
- Connect the RJ45 LAN cable (V192M) to the USB extender (41135M) and coil up at the service end of the turret.
- Secure all cables in place with cable ties.

2.5.3.3 Setting up the 298 Shock-logger

To ready the shock-logger for service:

Required equipment

- Laptop PC + USB cable
- USB extenders (41135M)
- RJ45 LAN cable (V192M)
- Communications software for the 298 shock-logger provided by IMC Group Ltd.

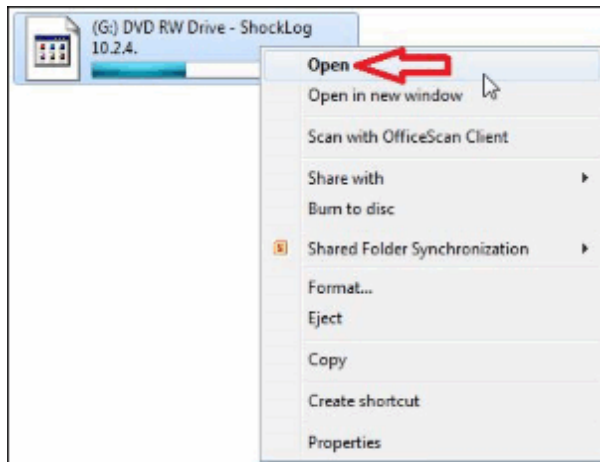
Installing the operating shock-logger software

- Log in as the 'administrator' and install the software as per instructions:



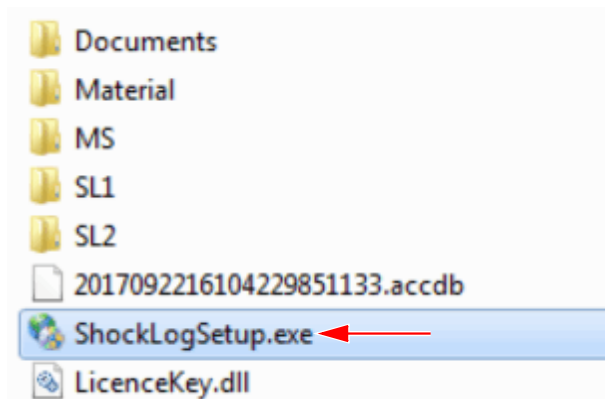
Skip this section for an initial installation.

Fig. 62: Open directory



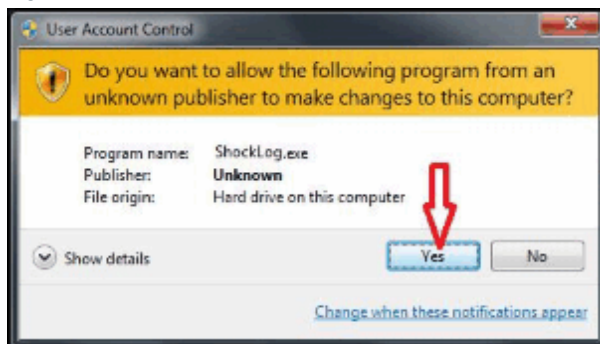
- Open the CD drive folder.

Fig. 63: ShockLogSetup file



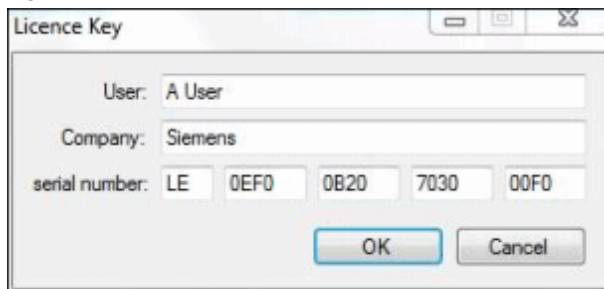
- Run "ShockLogSetup.exe" as the administrator.

Fig. 64: Select Yes



- If the following window appears, select "Yes".

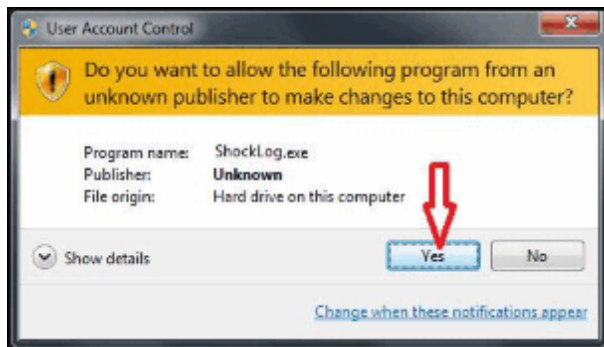
Fig. 65: Serial number



- Enter the following serial number when prompted:
 - » LE 0EF0 0B20 7030 00F0.

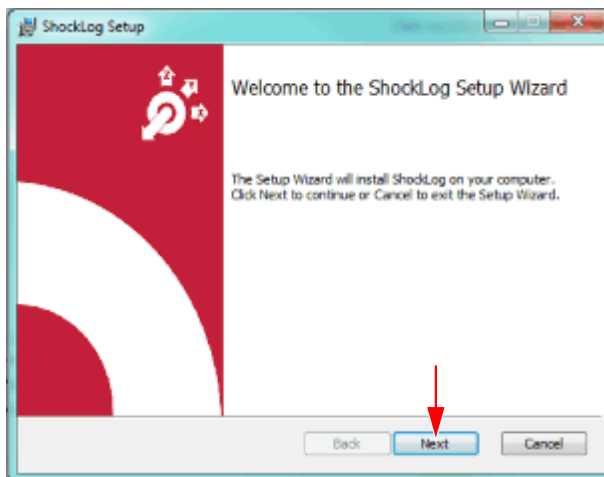
NOTE: 0 = zero

Fig. 66: Select Yes



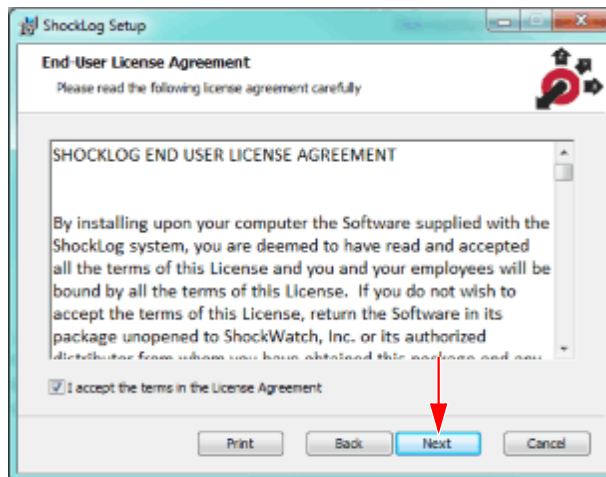
- If the following window appears, select "Yes".

Fig. 67: ShockLog setup



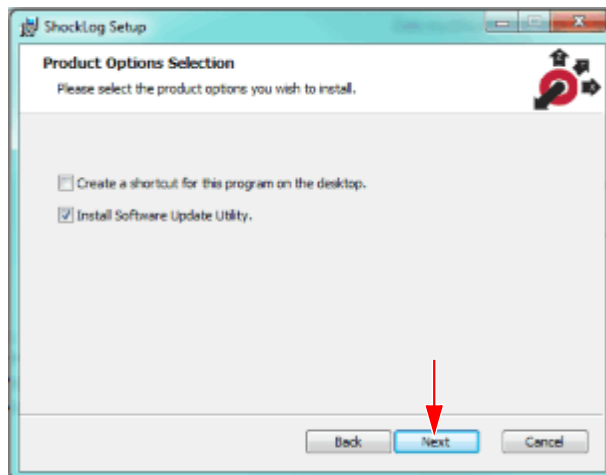
- Select 'Next'.

Fig. 68: Software agreement



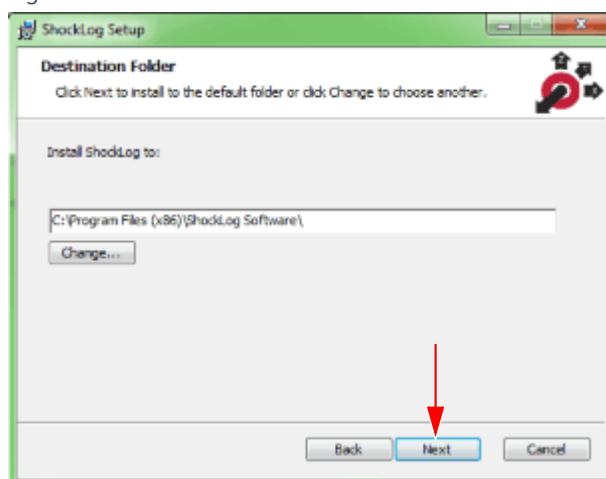
- Check the box "I accept the terms in the License Agreement"
- Select 'Next'.

Fig. 69: Product options



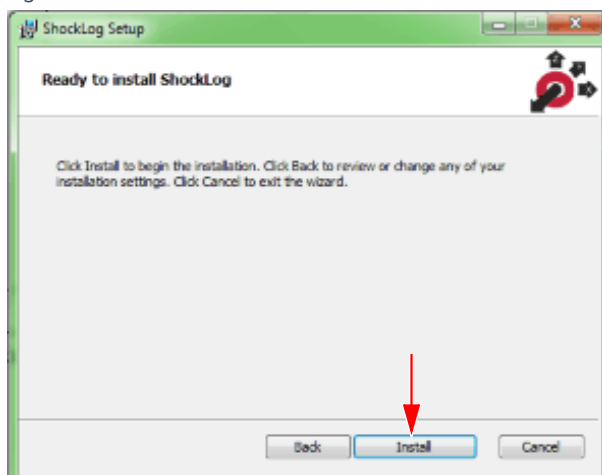
- Select 'Next'.

Fig. 70: Destination folder



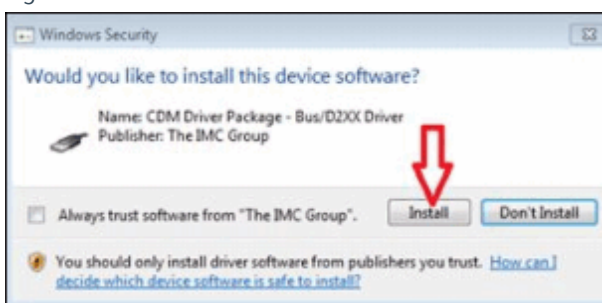
- Select 'Next'.

Fig. 71: Software installation



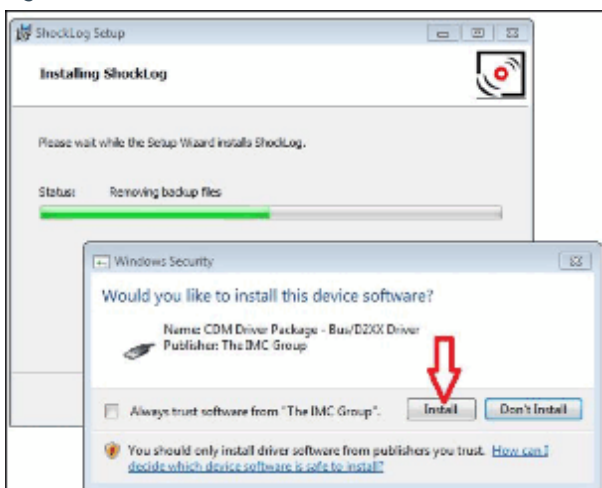
- Select 'Install'. Installation may take a few minutes.

Fig. 72: Software install



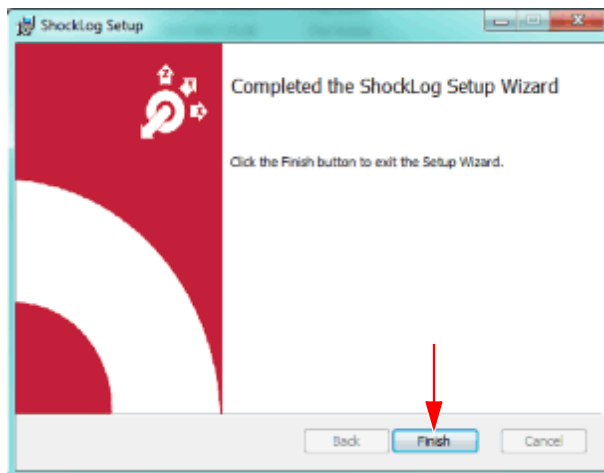
- If you get the following Windows Security windows, select "Install".

Fig. 73: Software install



- Repeat – press install for each new setup window

Fig. 74: Completing setup



- Once the installation is complete, press "Finish".

2.5.3.4 Downloading the 298 Shock-logger Data



CAUTION

Strong magnetic field

Damage to laptop PC and equipment.

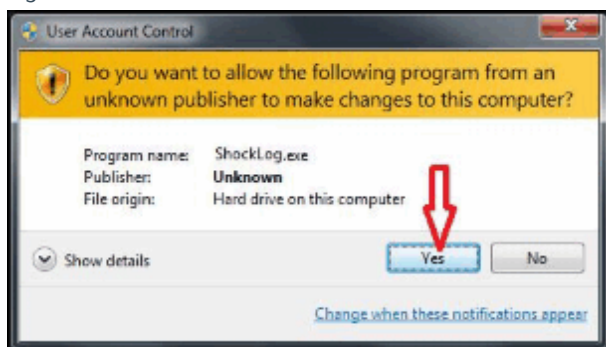
- » When downloading data from the shock-logger, always use a suitable length USB cable. Keep the laptop PC away from the magnet.

Fig. 75: ShockLog & Software



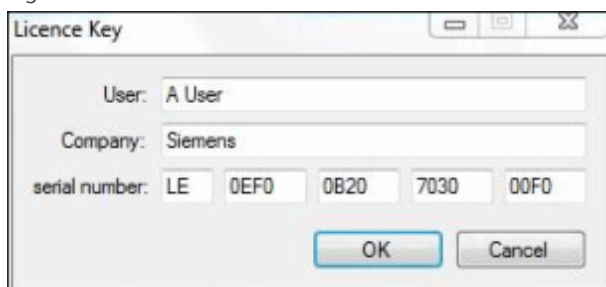
- Connect the shock-logger to your computer with the USB cable. If necessary, use the USB extender and the RJ45 patch cable.

Fig. 76: Select Yes



- Right-click the icon and select **“Run as administrator”**.
- If the following window appears, select **“Yes”**.

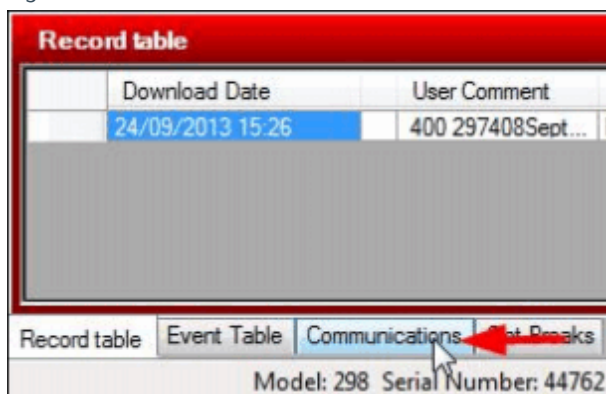
Fig. 77: Serial number



- Enter the following serial number when prompted:
 - » LE 0EF0 0B20 7030 00F0.

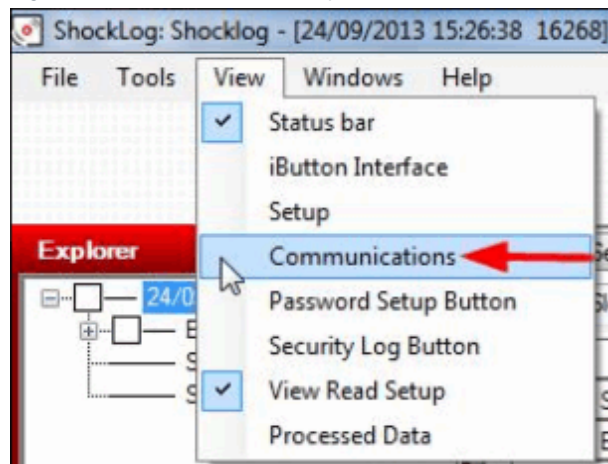
NOTE: 0 = zero

Fig. 78: Communication window



- Open the **“Communications”** window

Fig. 79: Communications drop down



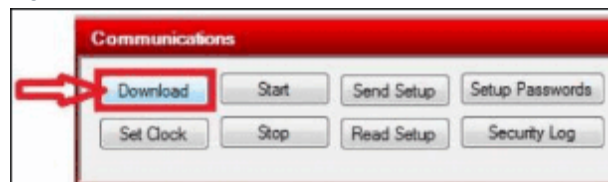
- The "Communications" window can also be accessed from the "View" drop-down menu.

Fig. 80: Communications



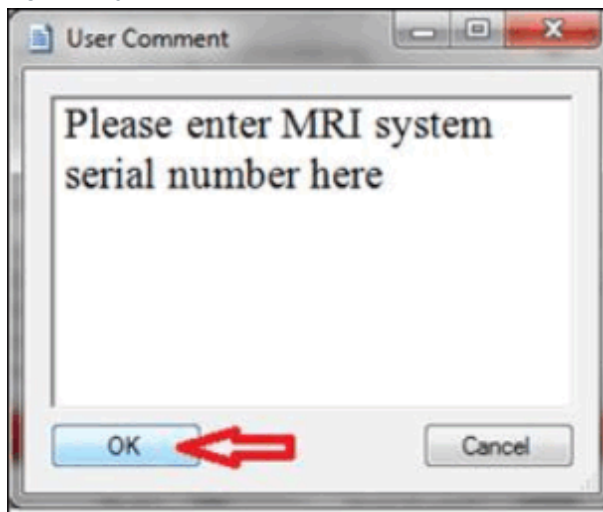
- In the Communication window select the "Stop" button.

Fig. 81: Communication 'Download'



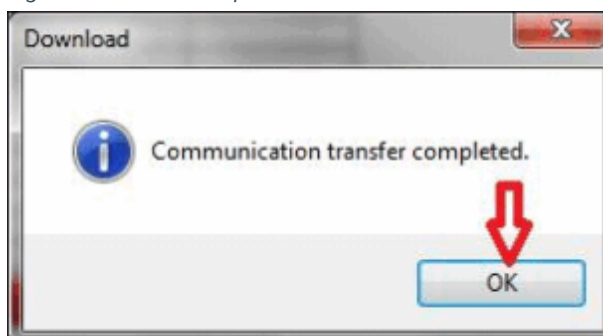
- After the state changes to "Stopped", select the "Download" button.

Fig. 82: System serial number



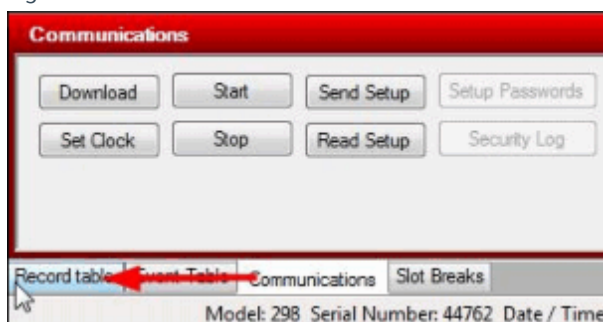
- Enter the MRI system serial number in the following window. Press "OK".

Fig. 83: Transfer completed



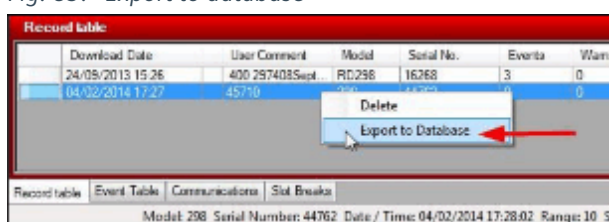
- After process download is complete, press "OK".

Fig. 84: Communications 'record table'



- Open the "Record table" tab.

Fig. 85: Export to database



- Find the downloaded file, right-click on it and select "Export to Database".

- Select "Desktop" and then press the "Save" button.

- Make sure that you have a new "*.accdb" file saved on your desktop.
- Close the shock-logger software and unplug the laptop.

2.5.3.5 Installing the 298 Shock-logger Setup Software



**Check that the "*.accdb" data files are saved on your desktop.
The following steps deletes all data from the shock-logger.**

Download the setup file from the Med CS intranet:

- Med CS intranet portal - MR Systems / Downloads.
- Or by emailing SMT HSC for a copy of the file.
- The shock-logger must be set up like the configuration shown below.

Fig. 86: Shock-log settings

The screenshot shows the 'Setup - 298' window with tabs for Firmware Options, Settings, Slots, Events, Slot Alarms, T&R Alarms, and Finish. The 'Settings' tab is active, displaying a table of configuration parameters:

Setup Name:	Mobile System		
Start Date:	10/02/2015 10:21:52		
Accelerometers:	Events:	Slots:	Summaries:
Range 10 g	Record Length 16K	Interval 30 Min	Interval 24 Hour
Wake (%) X:12 Y:20 Z:12	Max. Record Time 4 Seconds	Total run time 3246Days 5Hours	Total run time 1024Days 0Hours
Alarm (%) X:15 Y:25 Z:15	Max. No. Events 217	Max. No. Slots 155818	
Dropout (%) X:5 Y:5 Z:5	Filter Frequency 40Hz	No. Contents set: 3	
Hardware filter: 40 Hz	Always Max.	No. Alarms set: None Selected	
Standard Options	Factory Options	Peripherals	RD401
		RD401 Remote Indicator	On Alarms On Warnings Reset via External

Tab. 3 Alarm and warning levels

Acceleration wake	0.5 g
Acceleration alarm X (axial)*	1.5 g
Y (lateral)*	1.5 g
Z (vertical)*	2.5 g
* Axis dependent on shock-logger orientation	

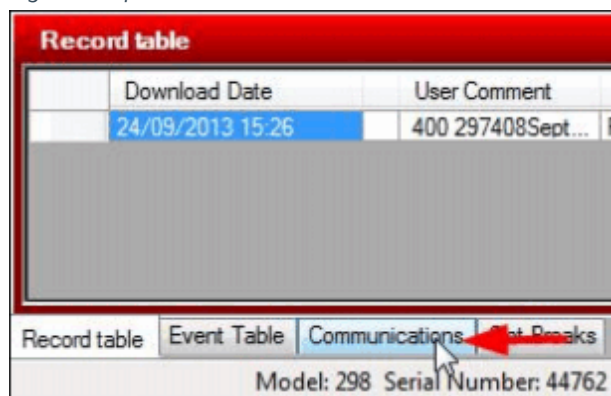


The configuration in the mobile setup program are the definitive settings. The table above is only for reference.

The steps for installing the setup file on a laptop are as follows:

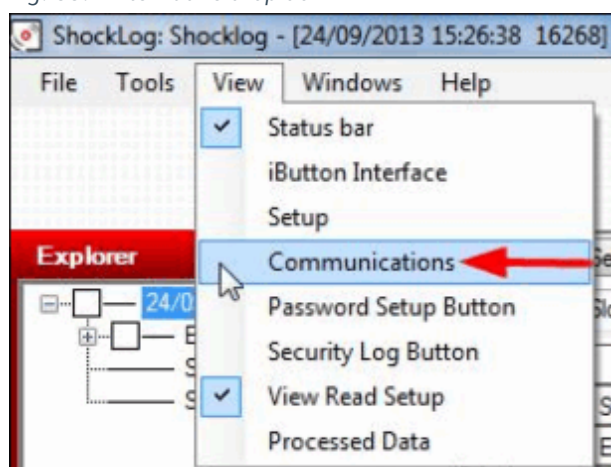
- Download the setup file "298 Mobile Setup V2.setup413" and save it on your desktop.
- Connect shock-logger to your laptop.
- Open the shock-logger software.

Fig. 87: Open communications



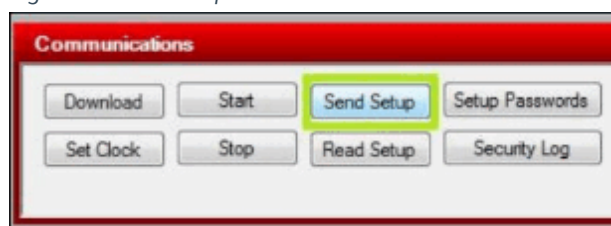
- Open the Communications window.

Fig. 88: Alternative drop down



- Alternatively, open the Communications window from the "View" drop-down menu.

Fig. 89: Send 'setup'



- In the Communications window select the "Send Setup" button.

- When the following window appears, go to "Desktop", select the relevant setup file and select "Open".

Fig. 90: Select 'Desktop'

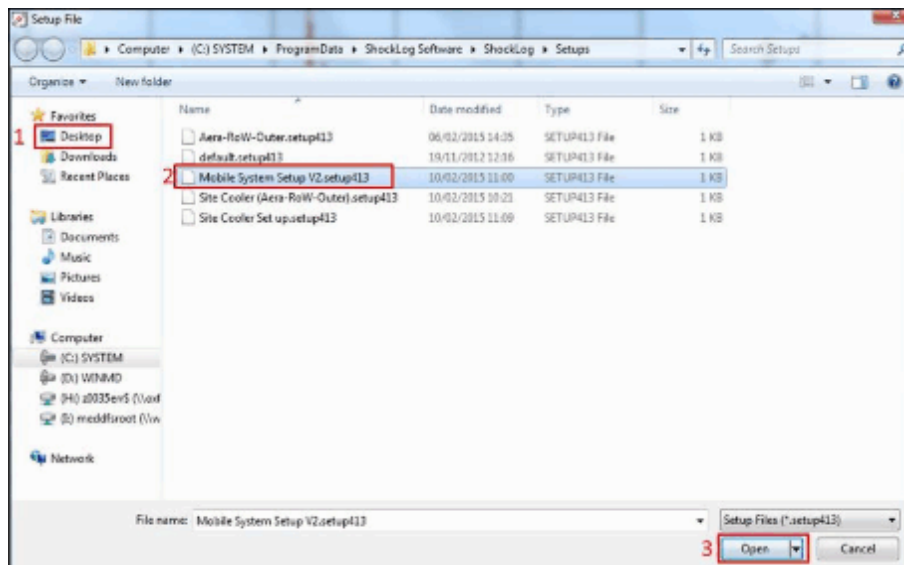
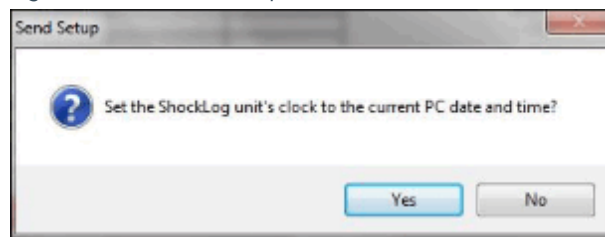
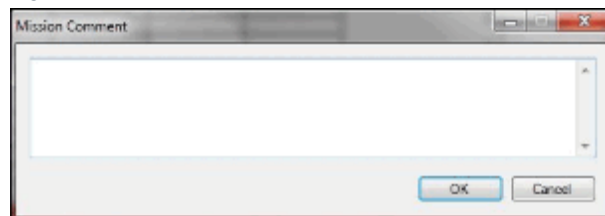


Fig. 91: Send clock setup



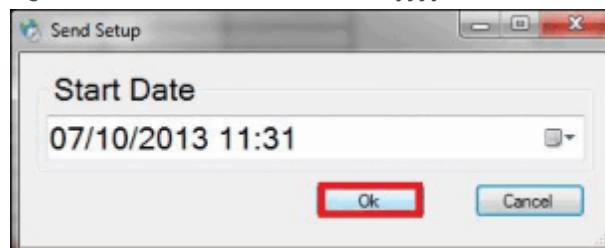
- When this window appears select "Yes".

Fig. 92: Select 'OK'



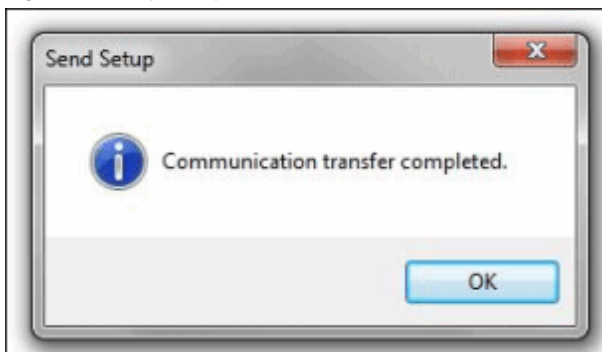
- Press "OK".

Fig. 93: Start date (Format: dd/mm/yyyy)



- Press "OK" again.

Fig. 94: Setup complete



- After setup is complete, the following window appears. Select "OK".

- After about a minute, open the Communications window again (see step 4) and make sure that the state is "Running".
- If it is in the "Waiting" or "Stopped" state, press the "Start" button.
- After 30 to 60 seconds, the state changes to "Running".

Fig. 95: Communications window

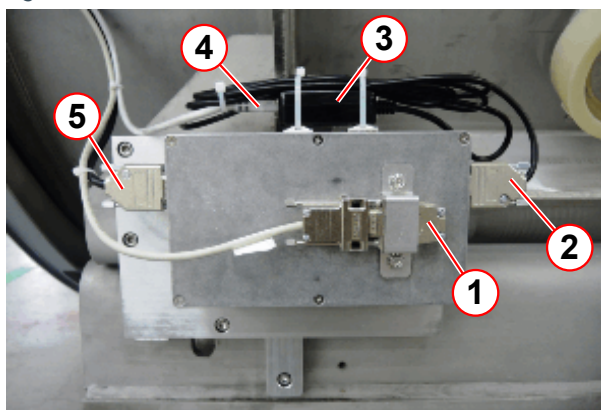


- When you are sure that shock-logger is running, unplug it from the laptop and coil up the cable.
- Secure the cable with ties.

2.5.3.6 Stage 2 - Trailer Manufacturer Installation Procedure

Follow these instructions when the MR system is being installed in the trailer at the mobile manufacturer premises.

Fig. 96: Cable connections



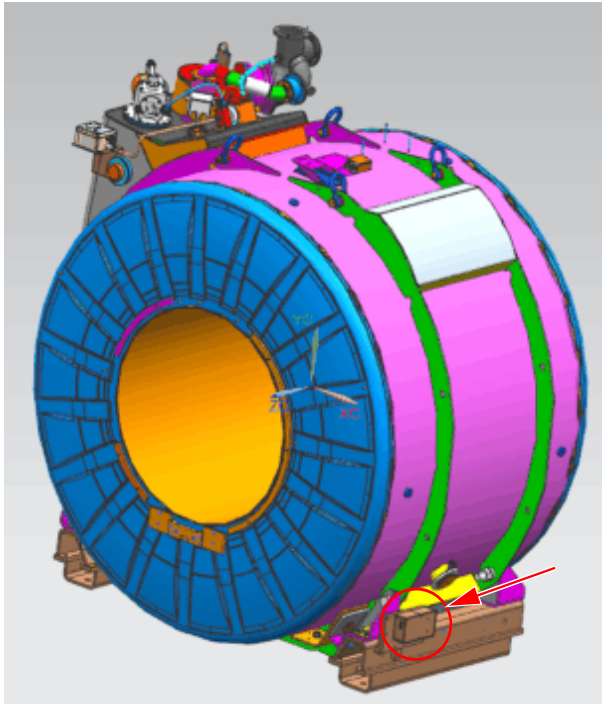
- (1) External field sensor
- (2) USB hybrid cable, 40121M
- (3) USB extender
- (4) RJ45 ethernet cable
- (5) ShockLog cable 40123M

The magnet is delivered with the cables connected to the shielding box:

- Shock-log™ cable (40123M)
- Hybrid USB cable (40121M)
- USB extender (41135M)
- RJ45, cable V192M
- External field sensor

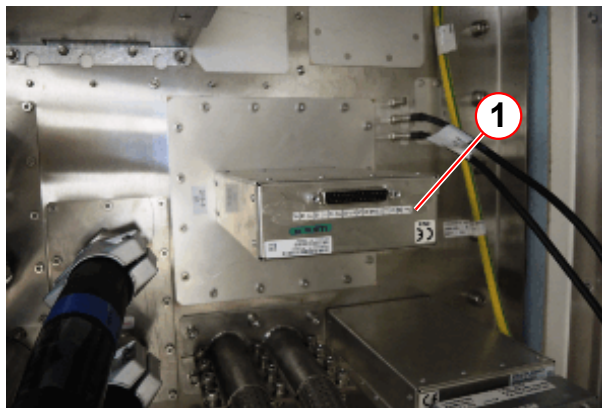
- The shock-logger plate assembly must be fixed to the service end left box section.

Fig. 97: Position of shock logger



- The position of the shock-logger and external field sensor is on the left box section of the service end.
- Use the original screws to fit the assembly.

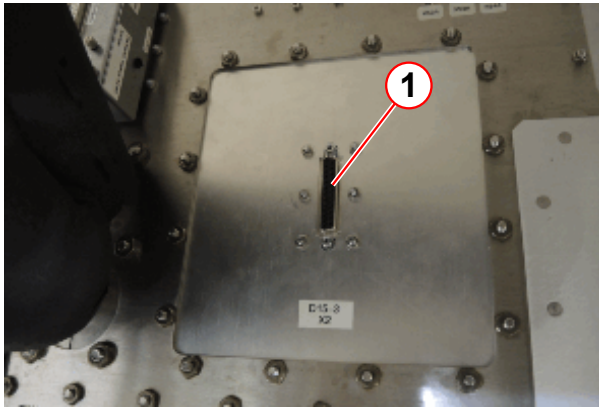
Fig. 98: Filter plate (operator room side)



- Uncoil the cable 40103M from the junction box and connect to D15-3 X1 connector.

(1) Shock logger filter (D15-3 X1)

Fig. 99: Filter connection (magnet room side)



(1) Connector D15-3 X2

- Connect the cable 40123M from shock logger to D15-3 X2 connector on the filter plate.

Fig. 100: RJ45 LAN cable



- Uncoil the RJ45 LAN cable (V192M) and route to the RF filter plate.
- Coil up the remaining part of RJ45 LAN cable (V192M) with cable ties in the filter plate cupboard. This cable is used for shock-logger data download and software shock-logger setup.



The RJ45 LAN cable is not used during normal operation.

Fig. 101: USB extender



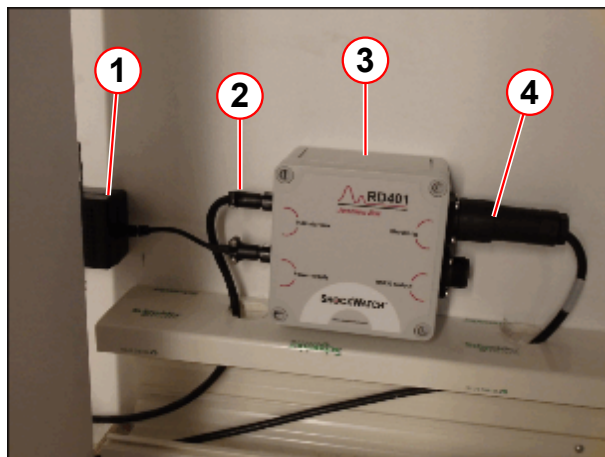
- When a service procedure is required connect the RJ45 cable to the other extender end.

- Uncoil the cable to the examination room and connect it to the laptop (with the shock-logger software installed).
- Follow the instructions for downloading and set up. When the data download is complete, store the cable in the filter plate cupboard in the examination room side.
- Install the junction box (40101M) securely in the control room using the screws and washers provided.



To stop RF noise, no cables must be left across the imaging room and control room during imaging operation.

Fig. 102: Junction box & battery



- (1) PSU - 40108M
- (2) Indicator box cable - 40102M
- (3) Junction box - 40101M
- (4) ShockLog cable - 40103M

- The junction box and the PSU are located in the control room.
 - Secure all cables appropriately with the "P" clips provided.
 - **NOTE:** The RS232 port is not used.
- Attach the cable (40103M) with the end marked 'Shock Logger Port' to the junction box and the end marked 'Filter Plate' to the RF plate.
 - When the installation is complete and the shock-logger is set up properly, store the USB laptop extender (41135M) with the Siemens Technical Documents.
 - Securely position the power supply unit (40108M).
 - Connect the cable (40105M) end marked "power supply unit Port" to the power supply unit (40108M).
 - Connect the cable end marked "Junction Box" to the junction box.
 - Secure the cable using the clips provided.



The cable 40105M is not used, as the PSU and the junction box are close to each other.

- Connect the cable (40106M) end marked "power supply unit Port" to the PSU (40108M).
- Connect the other cable end to the corresponding UPS in the trailer.

Fig. 103: Indicator box location



(1) Cabinet for indicator box location

- The indicator box is located at the cabinet on the left side of trailer.

Fig. 104: The indicator box



(1) Alarm indicator (shown as active)
 (2) Reset button

- Install the indicator box in the cabinet on the left side of trailer.
- Connect the free end of indicator box cable to the junction box and secure cable with the 'P' clips provided.

2.5.3.7 Checking the Shock-logger Function

Tooling

- Laptop PC and USB Extender (41135M laptop end).
- Communications software for the 298 shock-logger is provided by IMC Group Ltd.

Functional checks

- Install the operating software for the 298 shock-logger from the CD.
 - » Run the program as the Administrator.
- Check the set-up software for the 298 shock-logger.
- Check that the 298 shock-logger power is switched on.
- Check that the permanent power supply unit to the 298 shock-logger is connected.
- To be operational, the green LED on the power supply unit (40108M) is on.

Final Tasks

- When installation is complete and the 298 shock-logger is set up properly, store components with the Siemens technical documents and spare parts.
- IMC Group Ltd. shock-logger CD
- USB extenders, laptop side (41135M)

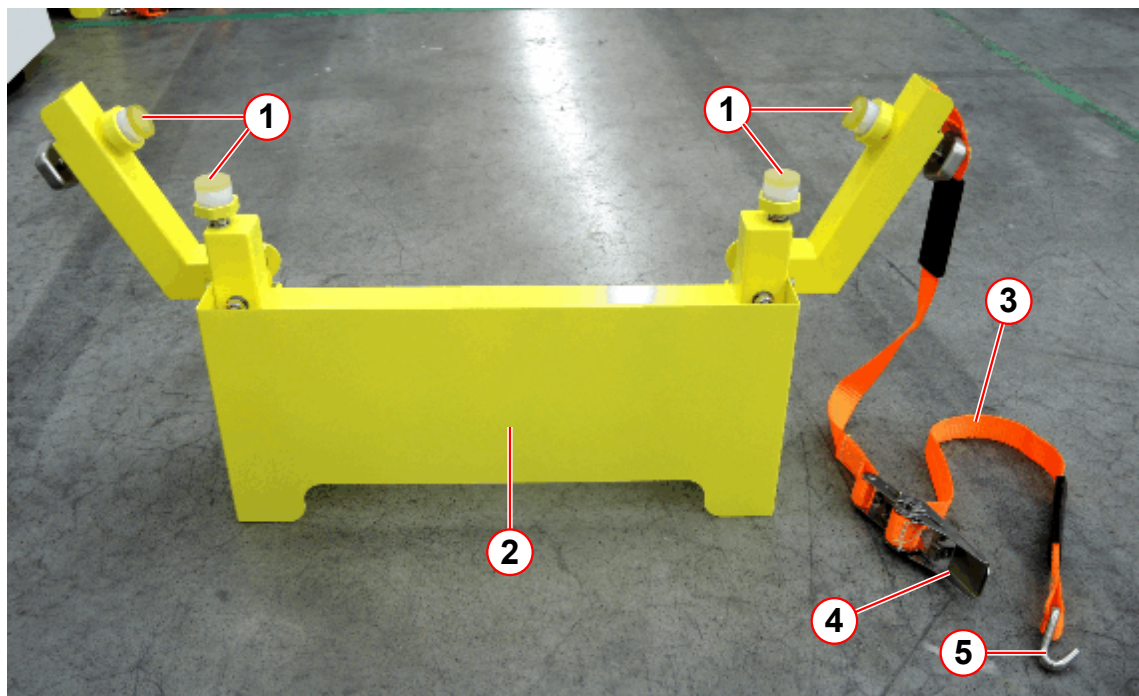
2.5.4 Patient table support bracket



It is the driver's responsibility to secure the patient table with the support bracket before transportation.

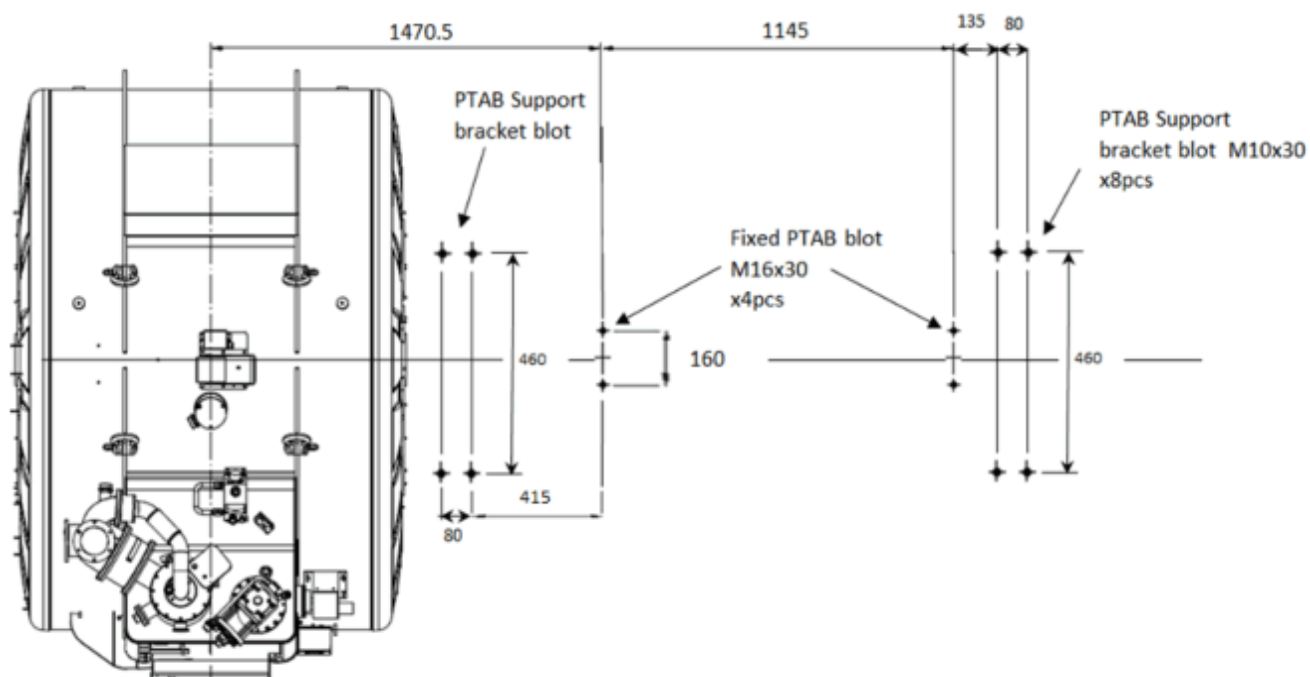
- The patient table support bracket is designed to secure the patient table during transportation. Before fastening the patient table with the belts, the patient table should be lowered down to touch the support points.
- During normal operation, the support beams are folded into the decorative panel cover.
- There are two support brackets for Aera Mobile systems, one fixed at the head side and the other one fixed at the feet side of the patient table.

Fig. 105: Patient table support bracket overview



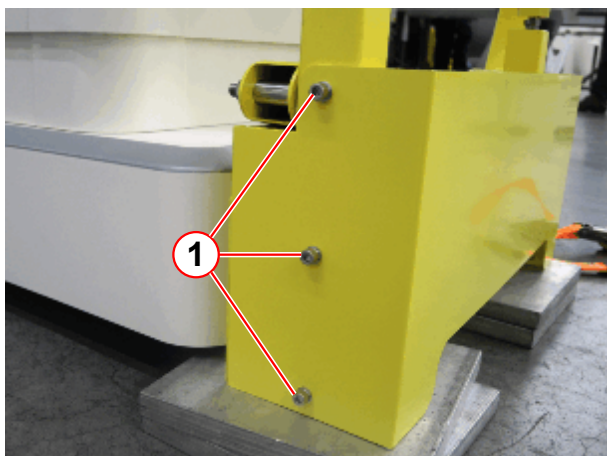
- (1) Support points
- (2) Decorative panel cover
- (3) Fastening belt
- (4) Lashing strap
- (5) Hook

Fig. 106: Fixation position of support bracket



2.5.4.1 Installation

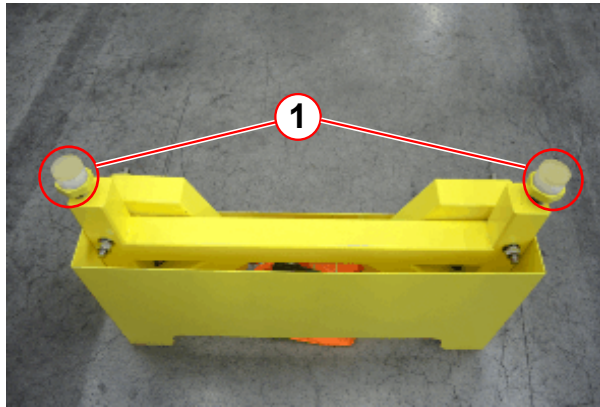
Fig. 107: Screws on decorative panel cover



(1) Screws (3x)

- Remove the three screws on both sides of the decorative panel cover.

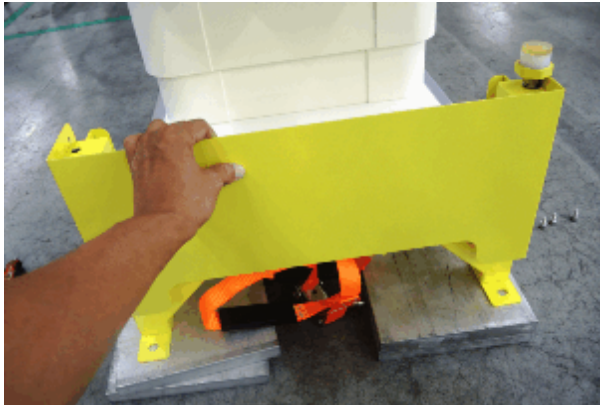
Fig. 108: Vertical support points



(1) Vertical support points

- Remove the two vertical support points.

Fig. 109: Decorative panel cover



- Lift and remove the decorative panel cover.

Fig. 110: Fixation holes of support bracket



- When the decorative panel cover is removed, the fixation holes can be seen.
- Fix the bracket on the bolts which are welded on the trailer floor and secure it.

Fig. 111: Reinstalling the decorative panel cover



- Recover the decorative panel cover.
- Install the two vertical support points.
- Install the three screws on both sides.

Fig. 112: Patient table support bracket



- The larger surface side of the decorative panel cover should face to the foot of the patient table to hide the support beams.

- Repeat the same procedure on the second of the patient table support brackets.

2.5.4.2 Unfolding the support beams

- When the trailer is prepared for transport, the support bracket should be unfolded and the patient table lowered until it touches the support beams. Subsequently secure the patient table with the belt.

Fig. 113: PTAB support bracket

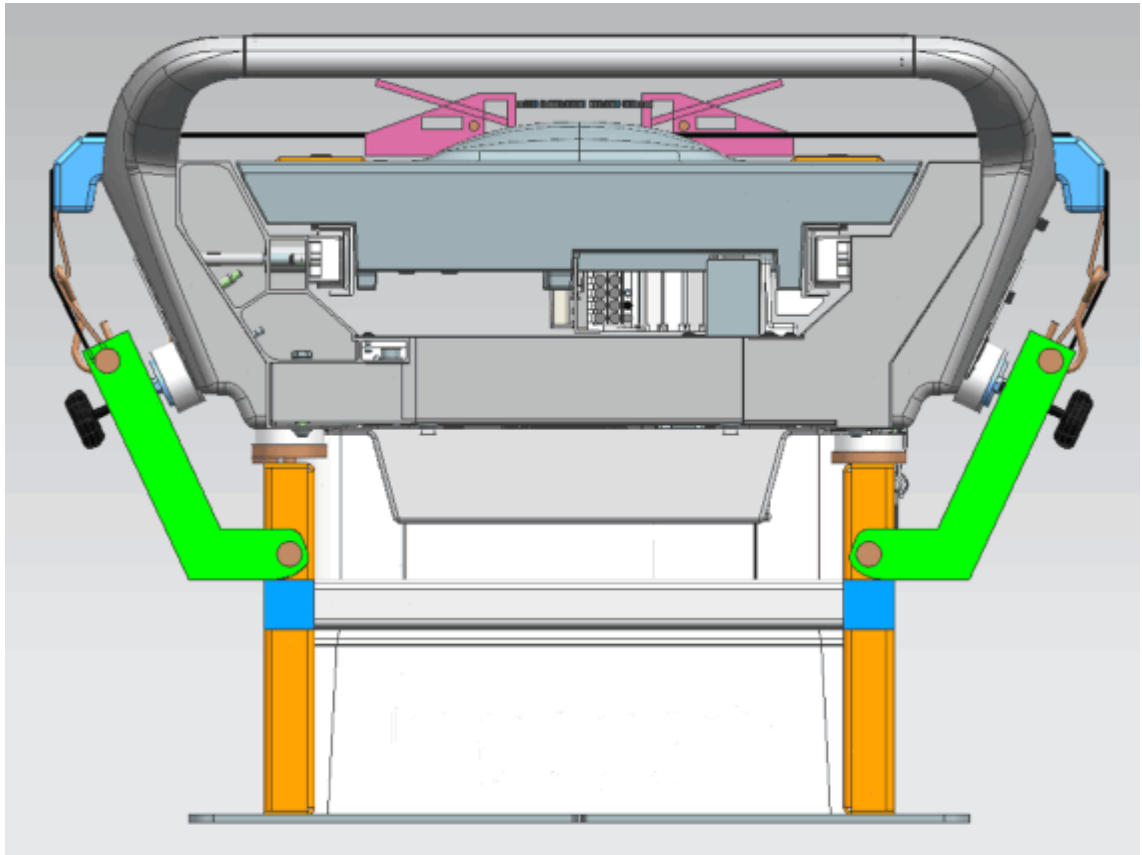
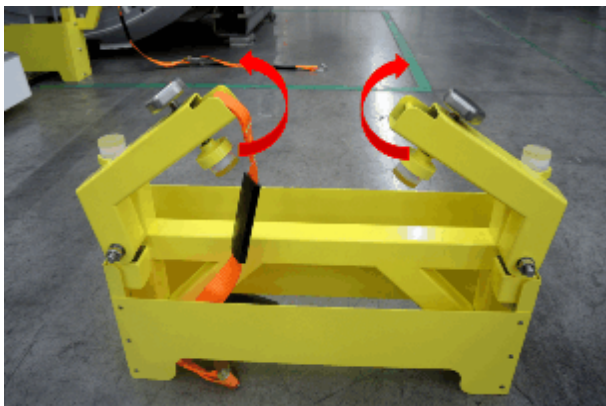


Fig. 114: Folded support beams



- During normal operation the support beams and the belts are stored in the decorative panel cover.

Fig. 115: Unfolding the support beams



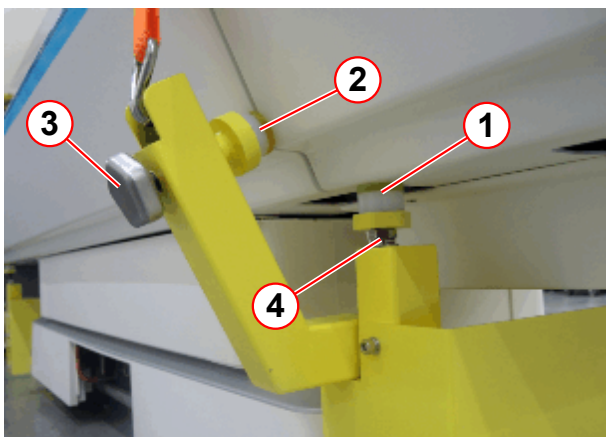
- If transportation is needed, swivel out the support beams.

Fig. 116: Final position of support beams



- The final position of support beams should be like in the picture.

Fig. 117: Support points



- Lower the patient table to make sure it touches the vertical and side support point sufficiently.
- Adjust the adjusting screws if necessary.
- Tighten the lock nut.

- (1) Vertical support point
- (2) Side support point
- (3) Adjustment screw
- (4) Lock nut

Fig. 118: Hook on the bolt



(1) Hook

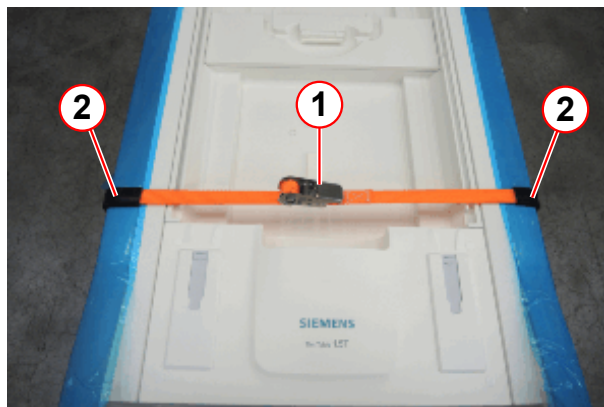
- Lay the belt across the top of the patient table.
- Hook the belt on the bolt.

Fig. 119: Lashing strap



- Turn the lashing strap left and right until the belt fastens the patient table tightly.

Fig. 120: Fastening belt on patient table



(1) Lashing strap
(2) Protecting rubber

- Tighten the belt in the same way as the picture shows.

- Repeat the procedure for the second support bracket of the patient table.

2.5.4.3 Folding the support beams

During normal operation, we could swivel and store the support bracket in the decorative panel cover.

Fig. 121: Folding bracket 1

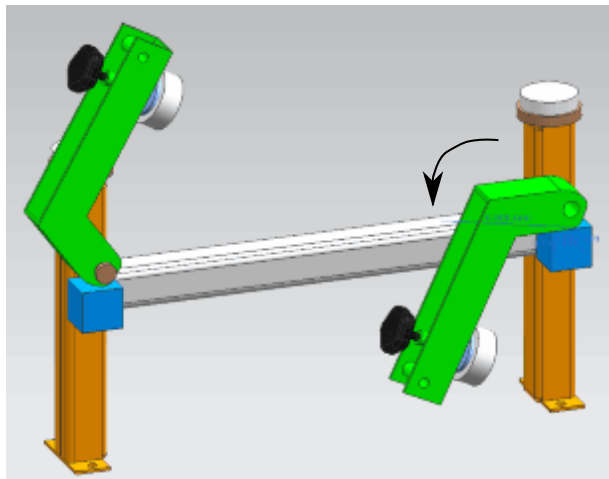


Fig. 122: Folding bracket 2

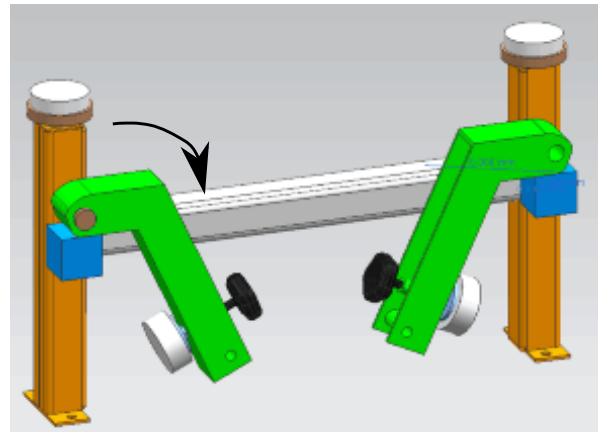
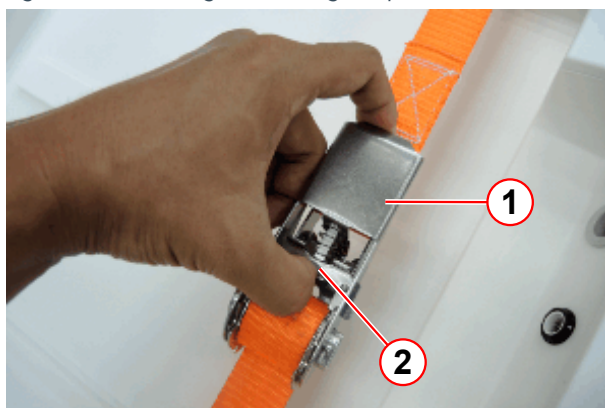


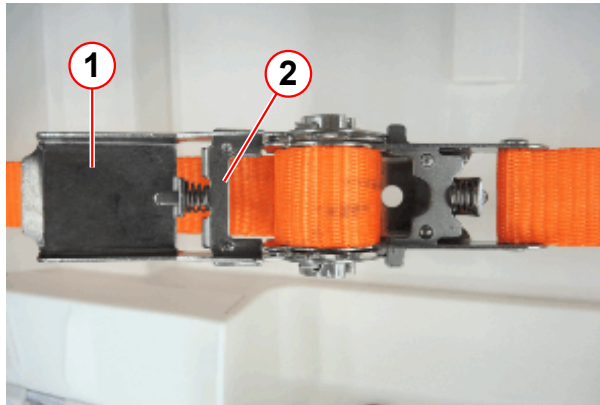
Fig. 123: Releasing the lashing strap



- (1) Lashing strap
- (2) Release pin

- Press the release pin and pull the lashing strap at the same time.

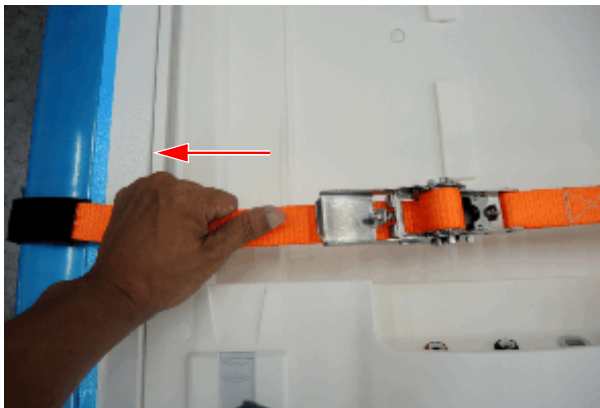
Fig. 124: Release position of the lashing strap



- (1) Lashing strap
- (2) Release pin

- Release the lashing strap as shown in the picture.

Fig. 125: Loosening the fasten belt



- Pull the belt near the hook side.
- Pull the belt until it is long enough to remove the hook.

Fig. 126: Hook on the bolt



- (1) Hook

- Remove the hook from the bolt.
- Raise the patient table to the home position.
 - » Before the patient table is raised, make sure the belts are removed on both sides.

Fig. 127: *Folded support bracket*



- Fold the support beam and fasten belt into the decorative panel cover.

- Repeat the procedure for the second support bracket of the patient tablet.

2.6 Equipment Room Installation

2.6.1 Connecting the Cooling Water Lines

2.6.1.1 Mobile water supply

Fig. 128: Mobile water supply

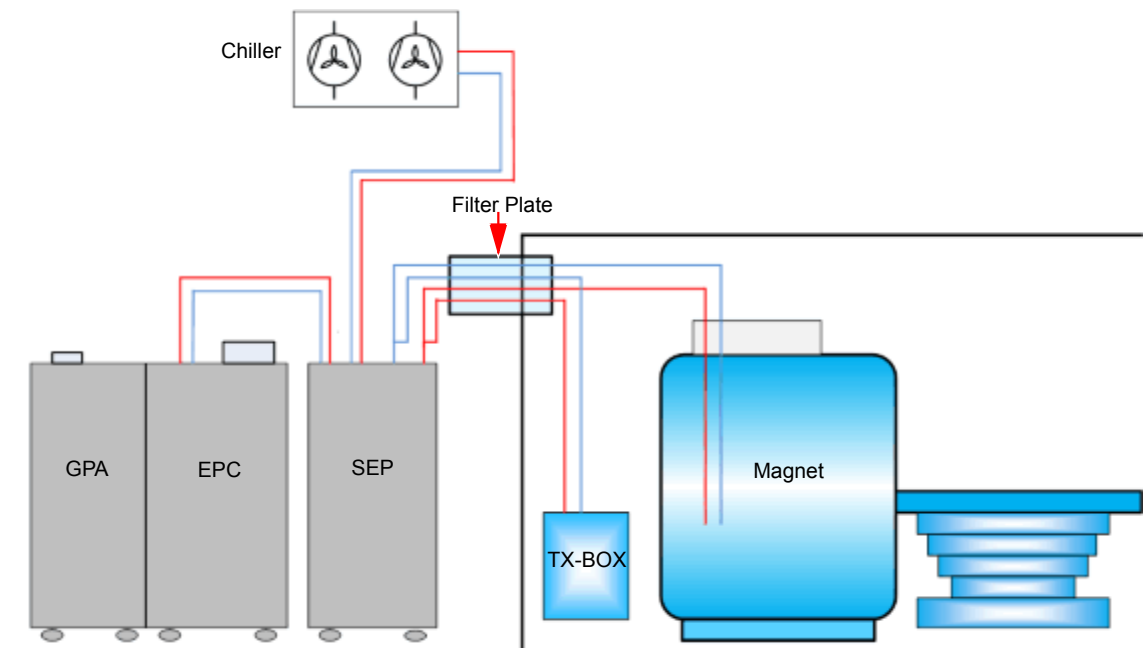
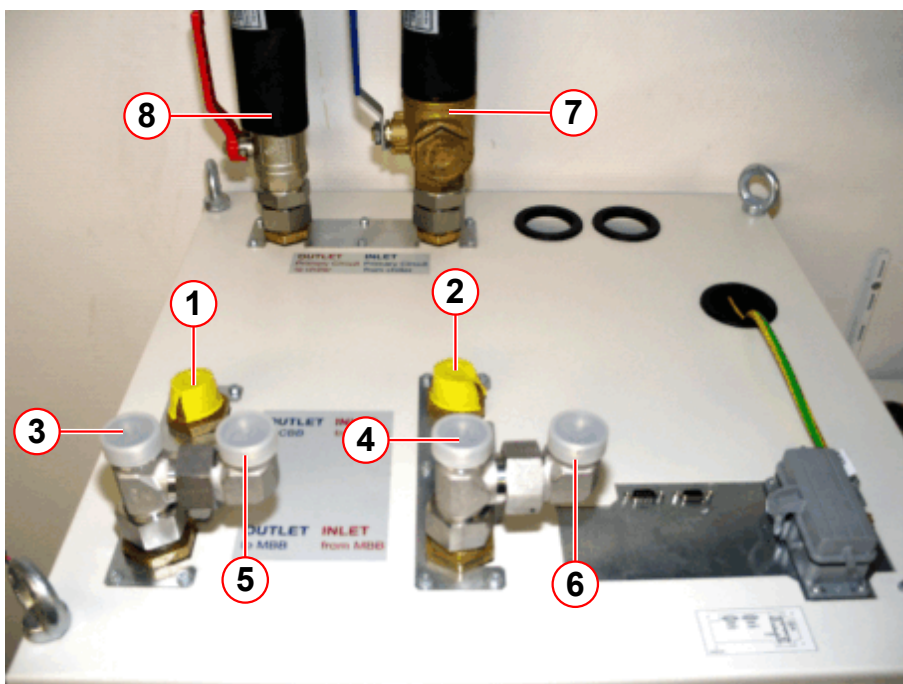
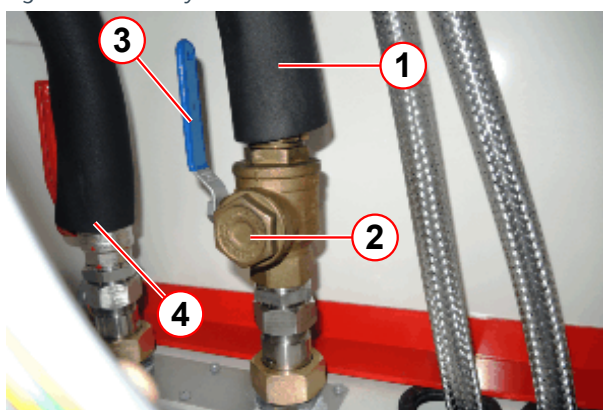


Fig. 129: SEP water connections



- (1) OUTLET to CBB (to EPC)
- (2) INLET from CBB (from EPC)
- (3) OUTLET to MBB (to filter plate - GC)
- (4) INLET from MBB (from GC - filter plate)
- (5) OUTLET to MBB (to filter plate - RFCEL/TX-box)
- (6) INLET from MBB (from RFCEL/TX-box - filter plate)
- (7) Supply (on-site)
- (8) Return (on-site)

Fig. 130: Primary water connection



- (1) Water supply
- (2) Filter
- (3) Valve
- (4) Water return

- Connecting the water hoses from the SEP to the EPC, valves on supply and return must be opened before use.
- The supply at the SEP is connected with a filter. Attach the filter (rotate it) so that the filter can be accessed from the front by Service.
- Connect the return to the SEP.
- Ensure all cooling hoses are not kinked.



Refer to the standard procedure for the static Aera system (Aera Installation Instructions > Establishing Water and Helium Connections > Water connections > Connecting the Water Hoses).

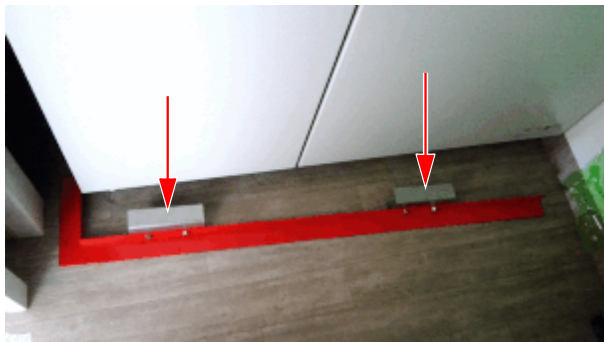
2.6.2 Blocking Cabinets in the Trailer

Fig. 131: Attaching the top rail of EPC and GPA



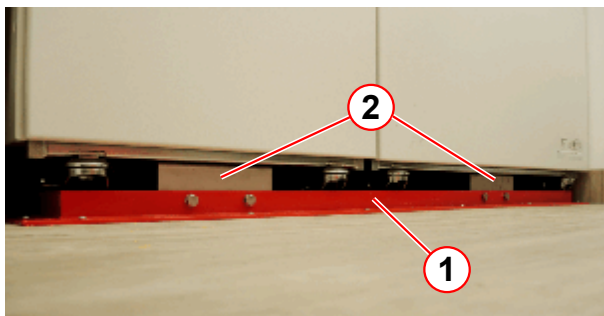
- Attach the top rail of the EPC and GPA to the side wall of the trailer.

Fig. 132: Installing the two vertical movement blocks



- Install and adjust the height of the two vertical movement blocks.

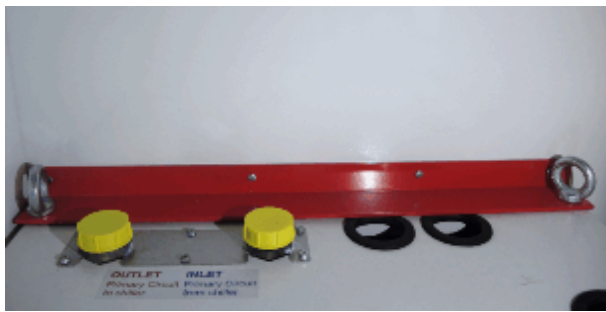
Fig. 133: Blocking the wheels of EPC and GPA



- (1) Horizontal movement block
(2) Vertical movement block

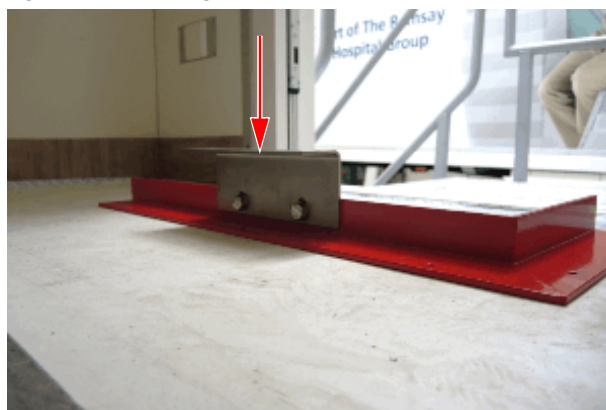
- Turn all four front wheels perpendicular to the cabinet door.
- Install and secure the red rail to the trailer floor. Make sure all four wheels contact the rail.
- Ensure that the two vertical movement blocks contact the bottom of the cabinets sufficiently.

Fig. 134: Securing the top rail of SEP



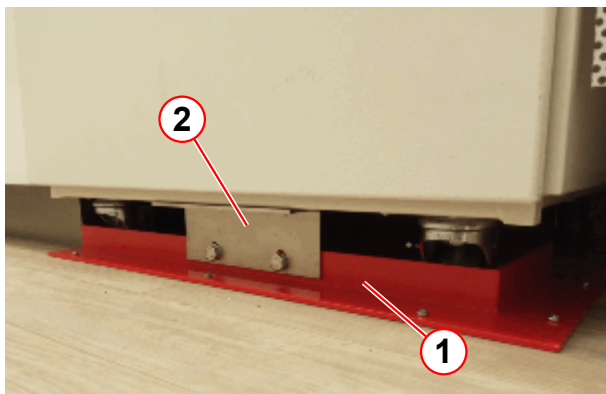
- Secure the top rail of the SEP to the side wall of the trailer.

Fig. 135: Installing the vertical movement block



- Install and adjust the height of the vertical movement block.

Fig. 136: Blocking the wheels of SEP



- Turn the two front wheels perpendicular to the SEP door.
- Install and secure the red rail to the trailer floor. Make sure the two wheels contact the rail sufficiently.
- Make sure the vertical movement block contact the bottom of the SEP sufficiently.

- (1) Horizontal movement block
(2) Vertical movement block

2.7 Operator Room Installation

Fig. 137: Overview of operator room



2.7.1 MRAWP Connection

- Install the host and secure it with a belt so it cannot move and adhere the router for external connections to the top of the host.

Fig. 138: Host and router



- (1) Host
- (2) Router
- (3) Fastening belt

2.7.2 WireLess Network Connection

Fig. 139: Wireless network connection diagram

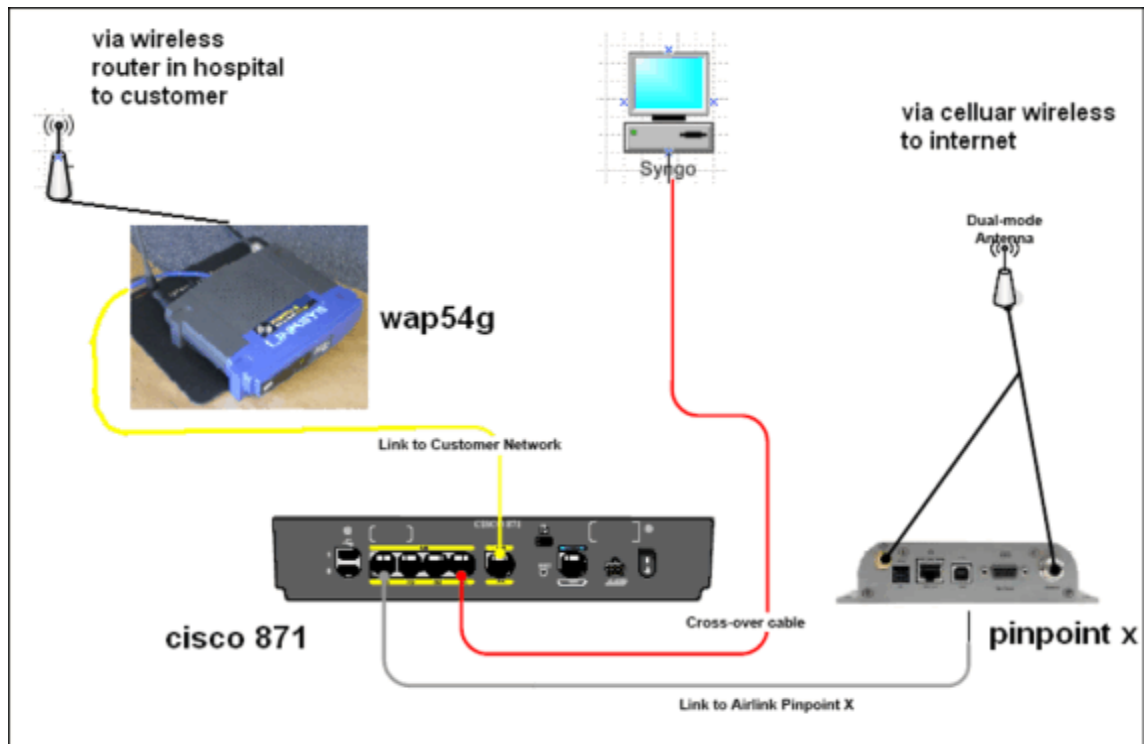


Fig. 140: Wireless network antenna diagram

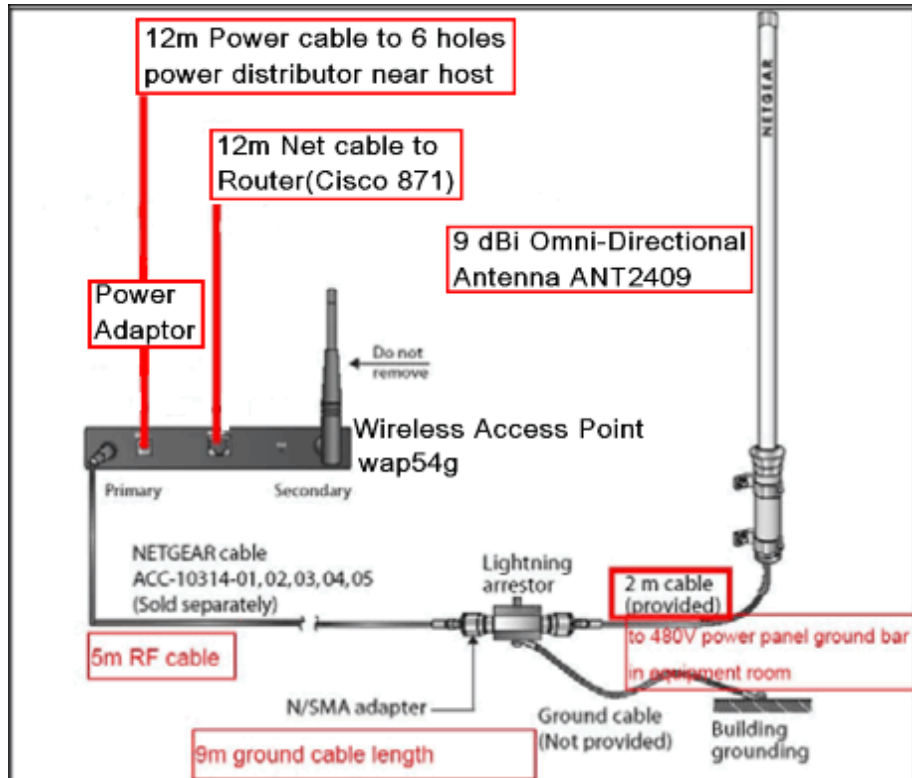


Fig. 141: Fixing the wap54g to the wall



(1) Wap54g

For parameter configuration, refer to A-Doc.

2.8 Cabling

2.8.1 Cabling on RCR

Fig. 142: Cables on the RCR, left side



Fig. 143: Cables on the RCR, right side



2.8.2 Cabling on the Filter Plate

Fig. 144: Cables on filter plate (Examination room)



Fig. 145: Cables on filter plate (Operator room)



2.8.3 Cabling on Cabinets

Fig. 146: Cables on top of EPC and GPA



2.9 Magnet Cooling

2.9.1 Chilled Water

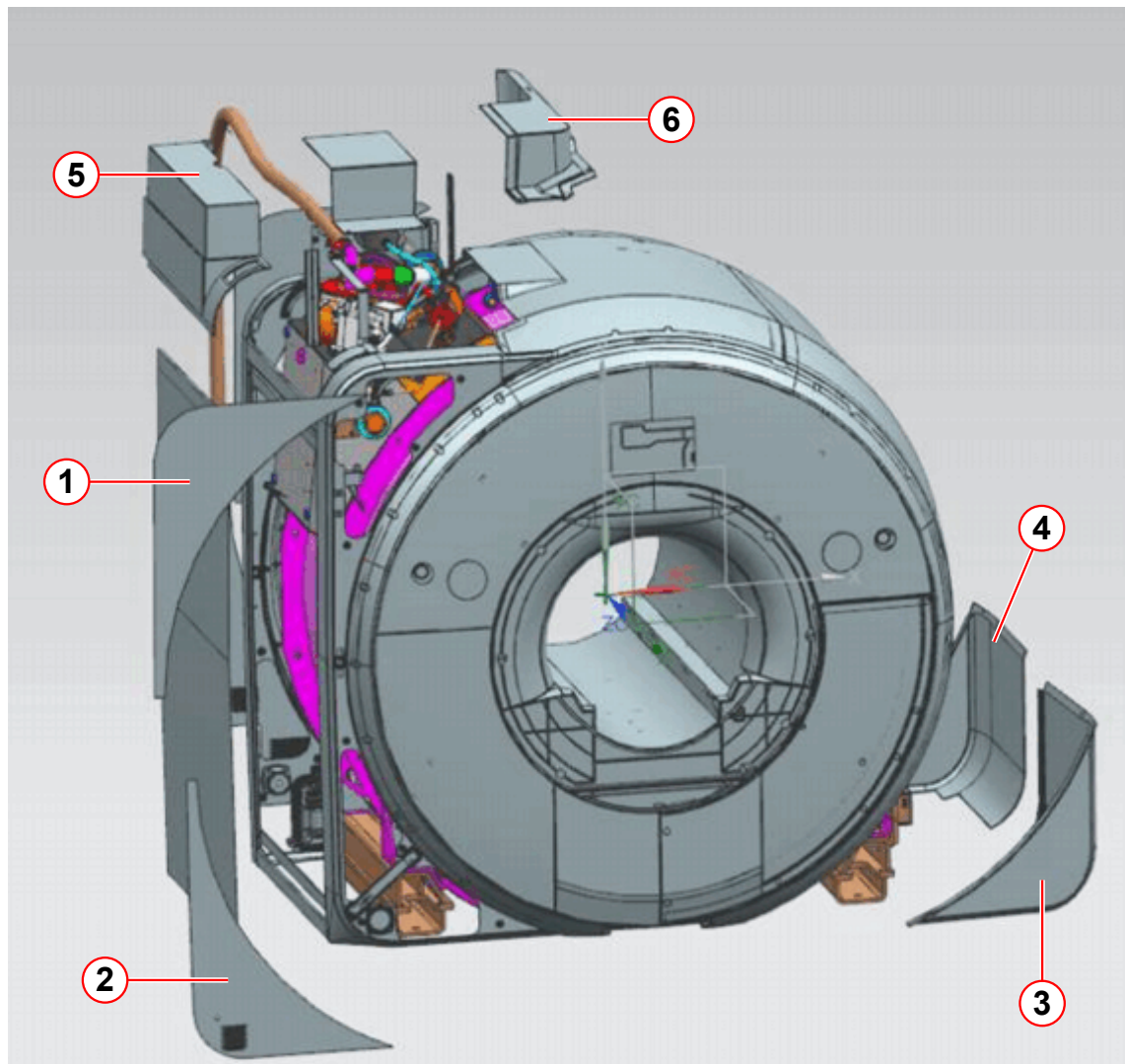


The water must consist of 35% to 38% ethylene glycol (corresponds to the scale on the frostbite meter from -21 °C to -23 °C).

Do not use automotive glycol. Use ethylene glycol only.

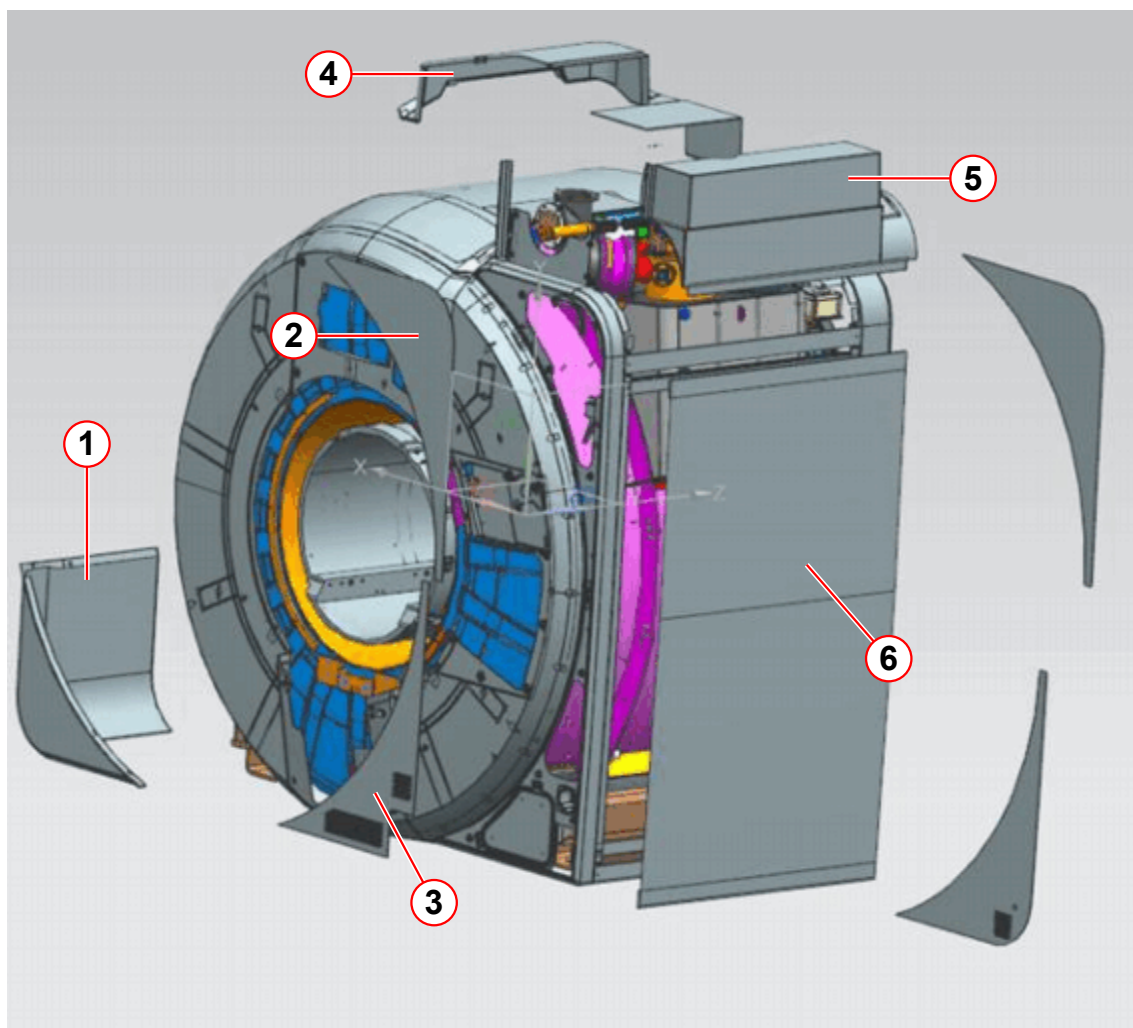
2.10 Installing the Aera Mobile Covers

Fig. 147: Mobile cover, patient side



- (1) Front design panel cover upper
- (2) Front design panel cover lower
- (3) Lower right cover front mobile
- (4) Lower right cover rear mobile
- (5) Turret cover left mobile
- (6) Turret cover right mobile

Fig. 148: Mobile cover, service side



- (1) Lower right cover rear mobile
- (2) Rear design panel cover upper
- (3) Rear design panel cover lower
- (4) Turret cover right mobile
- (5) Turret cover left mobile
- (6) Service cover left mobile



Refer to the standard procedure for the static Aera system (Installation Instructions (M7-010.812.01) > Final work steps > System covers > Installing the covers).



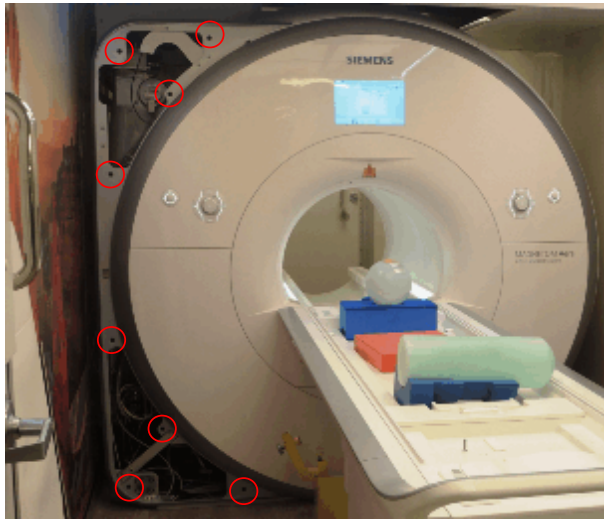
Covers that differ from the normal Aera system are described in the following chapters.

2.10.1 Front Design Panel Cover Upper / Lower

- For mobile systems, the decorative panel cover split into two parts, the front design panel cover upper and lower.

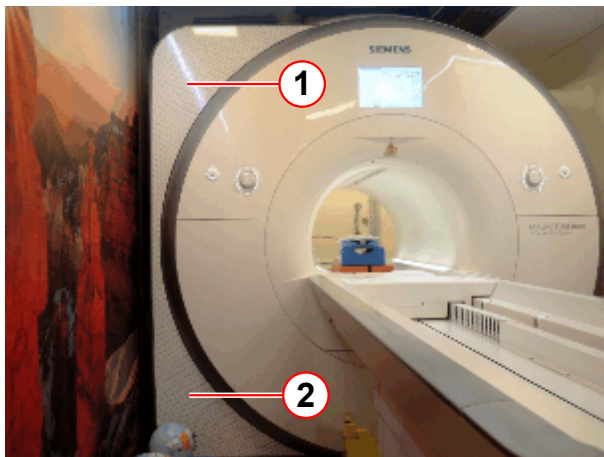
- The front cover and rear cover also make changes to fit the width of the trailer / coach.

Fig. 149: Snapping locks on the front cover



- The front design panel cover upper / lower is attached to the front cover by the snap locks.

Fig. 150: Installing the front design panel cover upper / lower

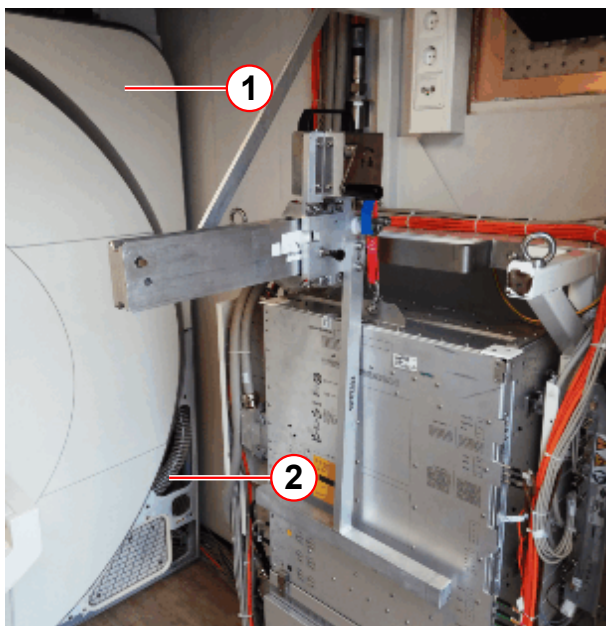


- (1) Front design panel cover upper
- (2) Front design panel cover lower

- Install the front design panel cover upper and lower.
- Push on the cover to lock the snap locks.

2.10.2 Rear Design Panel Cover Upper / Lower

Fig. 151: Installing the rear design panel cover upper / lower



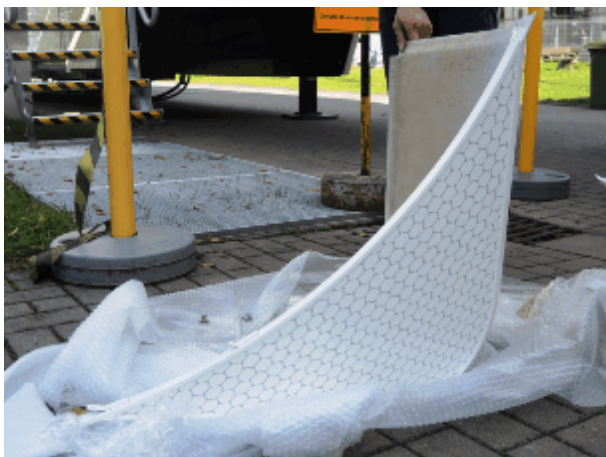
- (1) Rear design panel cover upper
(2) Rear design panel cover lower (shown as not installed)

- The rear design panel cover is same as the front one. Attach to the rear cover by snap locks.
- Install the rear design panel cover upper and lower.
- Push on the cover to lock the snap locks.

2.10.3 Lower right cover front / rear mobile

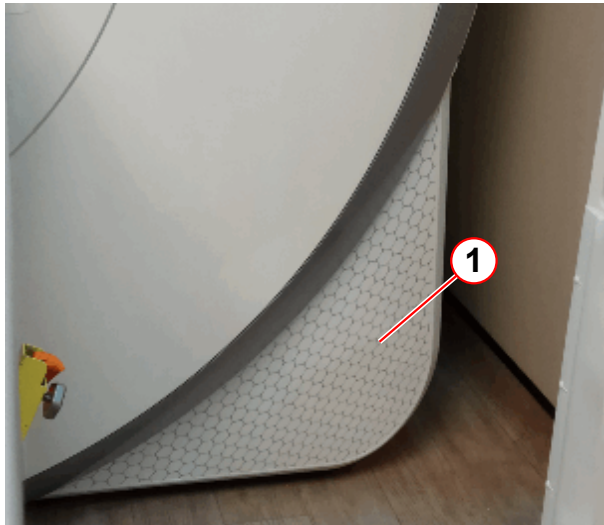
- Due to the narrow space between the magnet and the trailer wall, the lower right cover split into two parts, the lower right cover front and rear.

Fig. 152: Lower right cover front mobile



- The lower right cover front mobile.

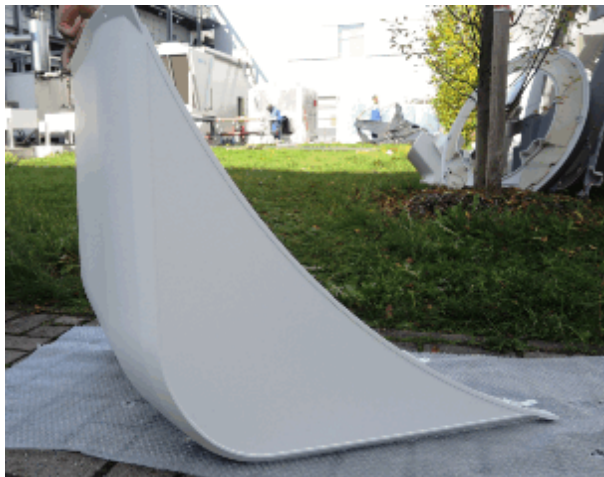
Fig. 153: Installing the lower right cover front mobile



(1) Lower right cover front mobile

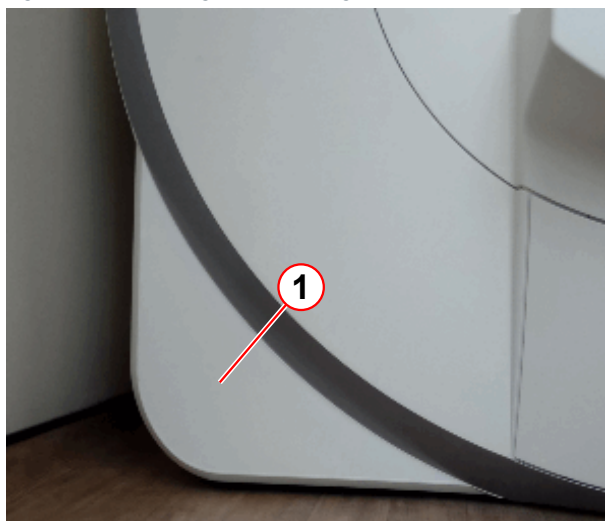
- Install the lower right cover front, using two countersunk screws and the 4 mm Allen key.

Fig. 154: Lower right cover rear mobile



- The lower right cover rear mobile.

Fig. 155: Installing the lower right cover rear mobile

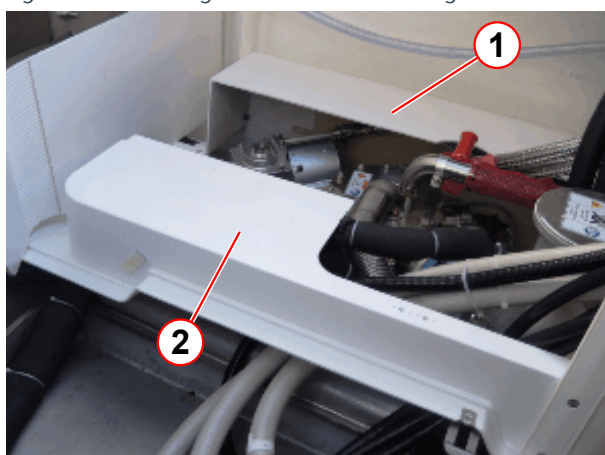


(1) Lower right cover rear mobile

- Install the lower right cover rear, using two countersunk screws and the 4 mm Allen key.

2.10.4 Turret Cover Left / Right mobile

Fig. 156: Installing the turret cover left/right mobile

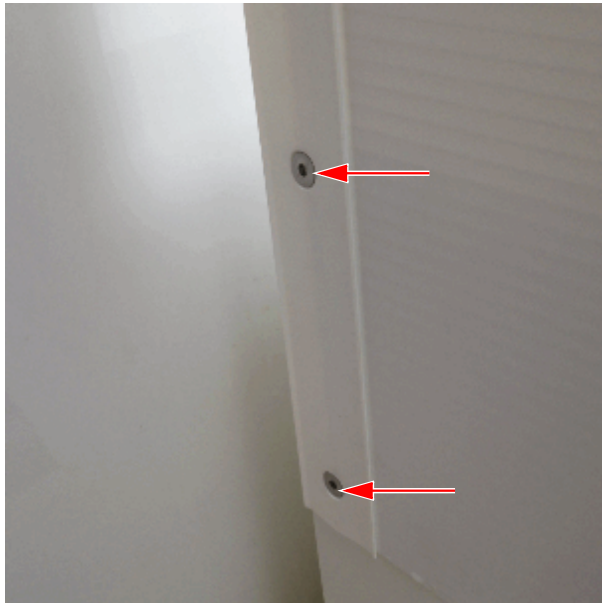


(1) Turret cover left mobile
(2) Turret cover right mobile

- Install the turret cover left / right mobile using the 4 mm Allen key.

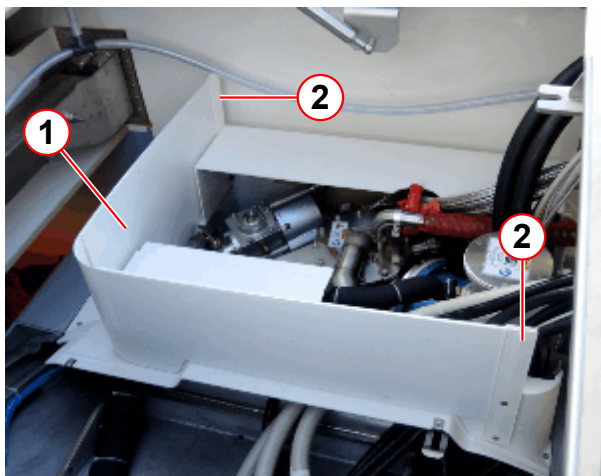
2.10.5 Turret extension mobile

Fig. 157: Installing the H-rail



- Install the two H-rails, one on the turret left cover, the other one on the turret right cover.

Fig. 158: Installing the turret extension mobile



- Attach the turret extension mobile to the H-rails.

- (1) Turret extension mobile
(2) H-rail

2.11 Final Work Steps

2.11.1 Mobile Warning Signs

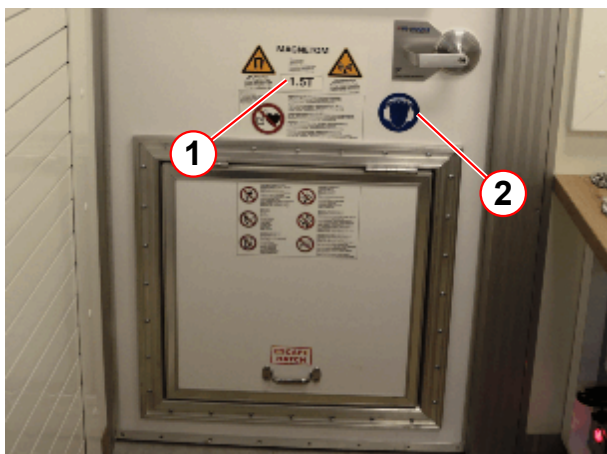
Fig. 159: Warning sign



(1) Field strength

- Attach the warning signs at eye level and so that they are clearly visible on the outer walls of the trailer.

Fig. 160: Warning sign and hearing protection sign



(1) Field strength
(2) Hearing protection sign

- Attach the warning signs on the examination room door.

2.11.2 Cleaning the System



The system must be cleaned before it is handed over to the customer.

Only use cleaning agents recommended by the manufacturer to clean the outer surfaces.

- **Never** use sprays. These could seep into assemblies, destroy electrical parts, change the structure of plastics, or form explosive combinations of air and solvent vapors.
- **Never** use cleaning agents that contain the following components or ingredients:
 - Abrasive or organic solvents such as aldehyde, acetone, spot remover, benzene, or alcohol. These can weaken surfaces or result in hairline fractures that will cause the material to break under any load. (Exceptions: alcohol may be used to clean monitors, and aldehyde-based compounds may be used to disinfect the gantry or hardened surfaces of the patient table).
 - Cleaning agents that can produce ammonia gas as a result of decomposition or evaporation. Ammonia gas causes stress corrosion.
 - Cleaning agents that contain silicone. Old silicone causes electric coupling problems.
 - Phenol-based or chlorine-releasing disinfectants, which can weaken materials.

Permitted and recommended cleaning agents

- The following cleaning agents may be used to clean the patient table, patient table pad, and patient positioning supports:
 - Tego, 10% vol.
 - Gigasept, 6% vol.
 - Somplex Perfekt AC, 2% vol.
 - Melsitt, 10% (slightly discolors surfaces)
 - Peform, 2% (slightly discolors surfaces)

Magnet and patient table:

- Clean these components using a damp cloth.

Computer, UPS, cabinet:

- Clean these components using a damp cloth.

Monitor:

- Monitors have sensitive anti-reflective layers.
- Clean the monitors using a soft cloth.
- You can use either pure alcohol or alcohol diluted in distilled water at a ratio of 1:3 or 2:3.
- **Never** use solvents.

2.11.3 Reporting

- Fill out the report (Service Provider Documentation Aera Mobile) in the Technical Documents.

3.1 General Information

The start-up procedure of the Aera Mobile is almost the same as for a static system, please refer to Start Up/System of Aera system. The differences are listed in the following chapters:

3.1.1 Prerequisites

- Room light has to be available in every room of the trailer.
- The system components have been positioned and installed according to the Planning Guide and Installation Instructions.
- The RF integrity of the RF-room was measured after the installation of the RF-room and its components.

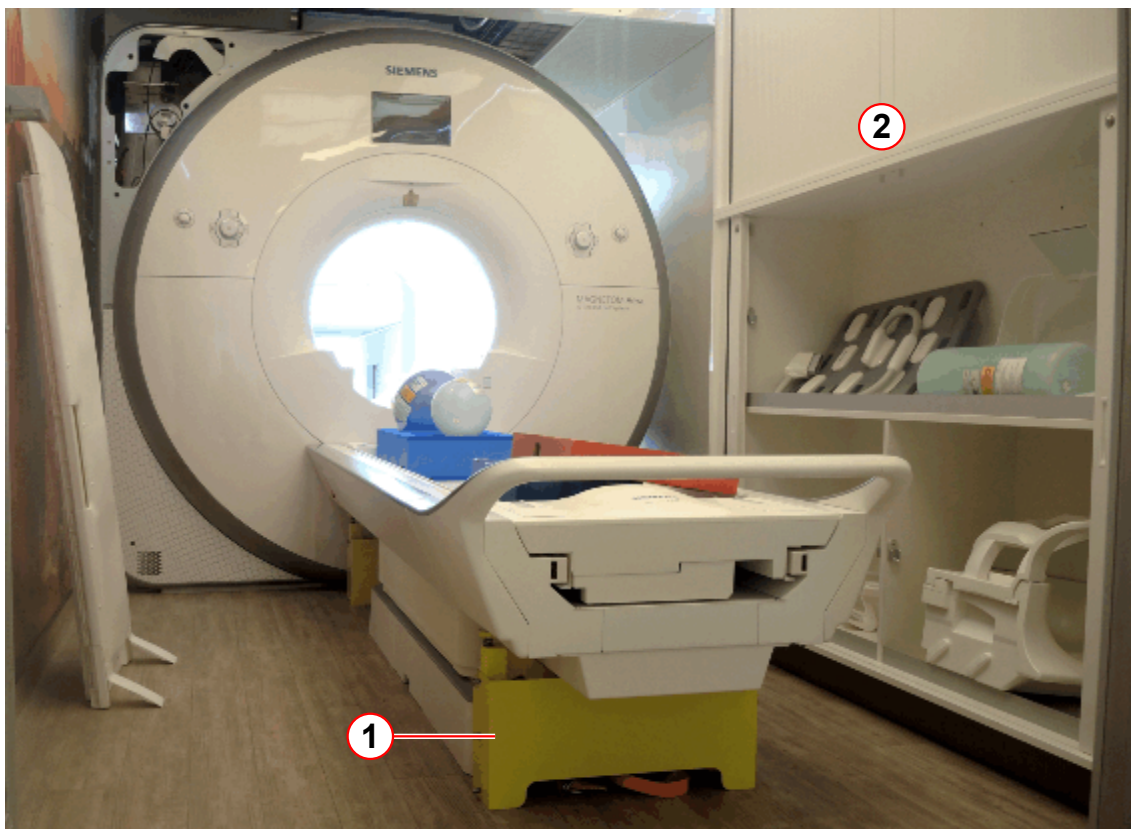


Only trained personnel, who have successfully completed the MAGNETOM Aera course, are allowed to start-up the system. The warranty is void if problems occur in the system due to work performed by untrained personnel or third parties.

3.2 Examination Room

- Before system start up, make sure that the support beams of both of the PTAB support brackets are folded inside the decorative panel covers. All coils and phantoms shall be positioned and secured in the cabinets.

Fig. 161: Examination room status



- (1) PTAB support bracket (beam folded)
(2) Cabinet for local coil and phantom

3.3 Equipment Room

- Make sure the gradient supervision status is OK, no FAULT or FIRE is displayed. The VLM jumper setting is the same as for normal Aera systems.

Fig. 162: Equipment room setup



- (1) VESDA device
- (2) EPC
- (3) GPA

- Close all the doors of EPC, GPA and SEP.

Fig. 163: SEP and trailer cabinets in equipment room

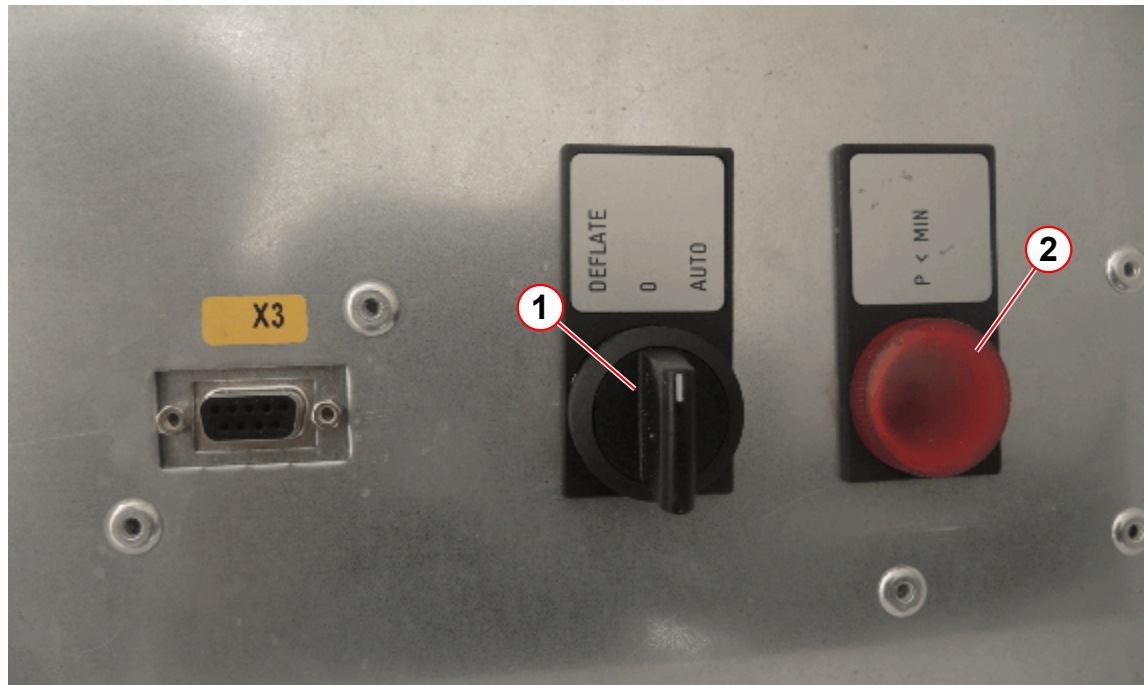


- (1) SEP
- (2) Trailer cabinets

3.4 Operator Room

- Before system start up, make sure the control button for the air bladder pump is **not** in the **AUTO** position.

Fig. 164: Air pump controls and indicator



- (1) Function selector
- (2) Low-pressure indicator

- Make sure all the doors of the cabinets are closed and locked.

Fig. 165: Operator room status



4.1 Filling the Mobile Magnet

4.1.1 General Information



CAUTION

Cold head failure

The cold head seals can fail and leak to the atmosphere.

- » Always ensure the cold head bypass valve is CLOSED during helium filling.

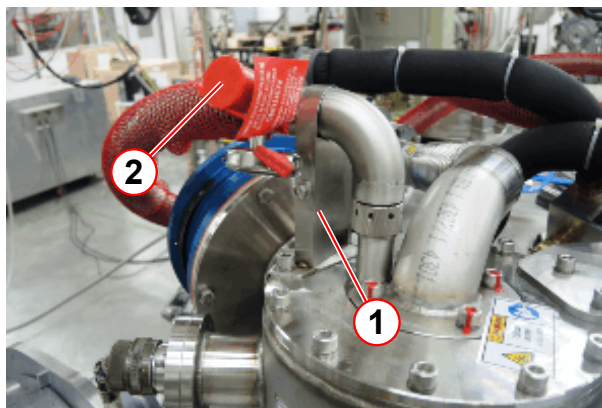
4.1.1.1 Mobile Systems Configuration Changes

The helium fill process is very similar to that of static systems.

Helium filling

- Mobile systems have an extended syphon permanently fitted in the turret.
- For access the coupling end of the syphon is located at the rear of the mobile trailer.

Fig. 166: Mobile syphon, turret transport bracket



- (1) Syphon transport bracket
- (2) Syphon vacuum valve

- The mobile syphon has an additional turret transport bracket (permanently fitted), to reduce vibration and movement.

This helium fill procedure can be performed on magnets at field and magnets at zero field.

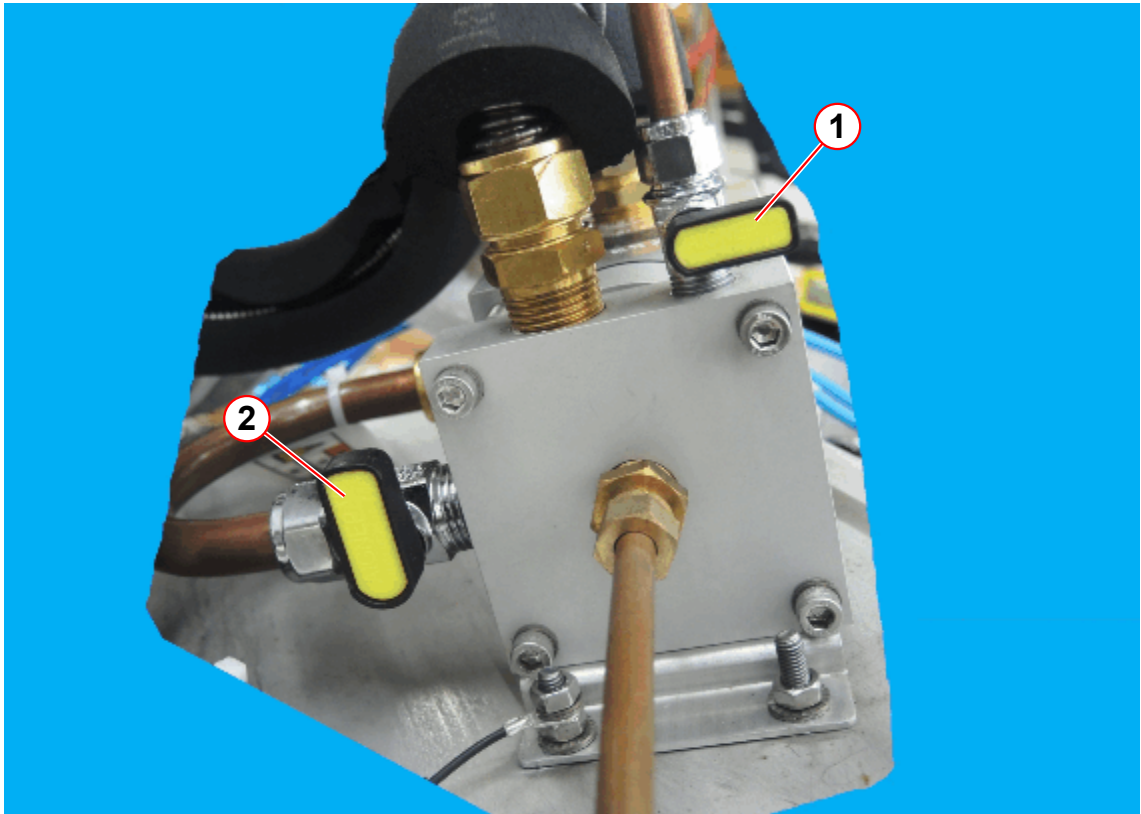


Only Siemens trained CSEs should perform this work.

Venting system

- The magnet bypass valve should remain CLOSED for the helium fill process.
- The quench valve drain screw is OPEN at all times.
- The acoustic de-coupler is fitted in the horizontal plane on a short quench line.

Fig. 167: OR98 vent valve assembly

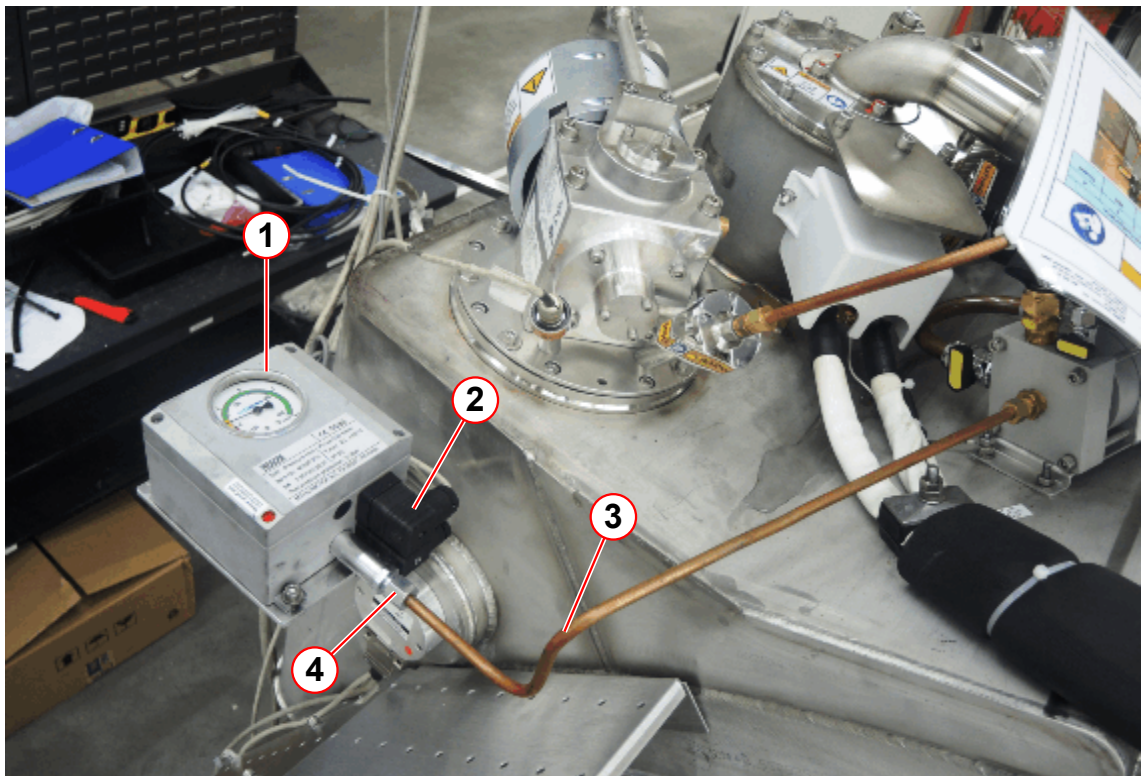


- (1) Cold head bypass valve
- (2) Magnet bypass valve

Pressure gauge

- For mobile systems, the pressure switch and gauge assembly has been turned past 90 degrees. This makes it easier for pressure observation during turret-related service.

Fig. 168: Pressure gauge of mobile system



- (1) Pressure gauge
- (2) Magnet pressure sensor cables (upper and lower)
- (3) Pressure pipe
- (4) Compression nut

4.1.2 Safety



CAUTION

Cryogenic liquids

Failure to observe the following safety warning may result in cold burns.

- » Protective clothing, cryogenic gloves and eye protection must be worn.



WARNING

Strong magnetic field

Failure to observe the following safety warning may result in physical injury and equipment damage.

- » Only non-magnetic equipment may be taken into the magnet trailer.



Ensure that the quench line has been correctly installed before helium filling.



WARNING

Risk of asphyxiation

Failure to observe the following safety warning may result in personnel experiencing dizziness or loss of consciousness while working with cryogen.

- » Do not vent helium gas into the magnet room during helium transfer.
- » Ensure that the turret venting kit is connected to the quench pipe.
- » Oxygen monitors must be activated at all times.

NOTICE

Mishandling of the dewar

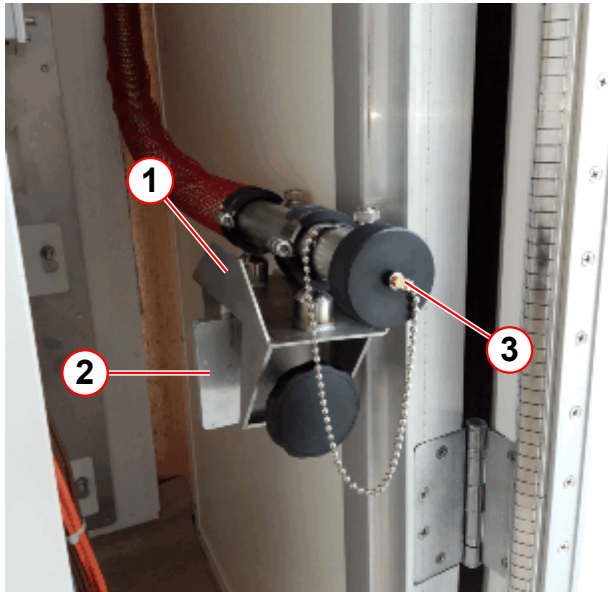
- » Observe the manufacturer's safety regulations for the dewar.

4.1.3 Required equipment

- Dewar syphon
- Dewar neck adaptor, clamps, seals, and hose
- Pressure service tool (PN: **03861221**)
- Heat gun and paper towels
- Helium gas supply
- Pressure regulator 200 bar/ 0 - 1.5 bar
- Protective clothing: gloves, safety glasses or face shield, lab coat, safety shoes
- O₂ Monitor - either room or personal
- Gas bottle wrench
- Leak detector

4.1.4 System Preparation

Fig. 169: Mobile syphon, transportation position (typical)



- (1) Syphon coupling clamp
- (2) Fixed wall bracket
- (3) Syphon end cap

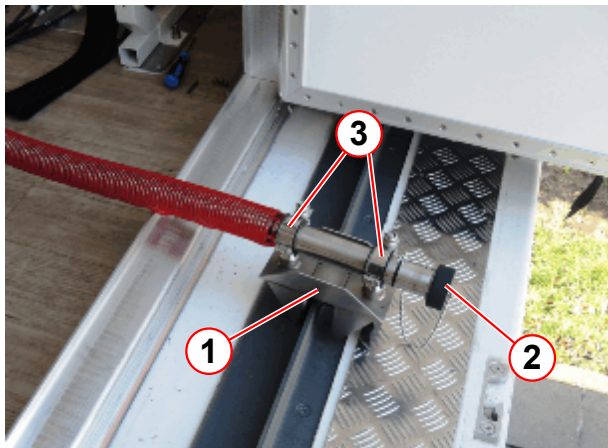
- The syphon is clamped to a wall bracket [1] positioned at the rear of the trailer.
- If necessary, remove the syphon clamp from the fixed wall bracket.

NOTE: Depending on how the extended syphon has been fixed in the trailer, it may not be necessary to remove the syphon from the wall bracket.



The positioning of the clamp will vary, depending on the trailer design.

Fig. 170: Mobile syphon, rear door fitment (typical)



- (1) Syphon clamped to rear door
- (2) Syphon end cap
- (3) Syphon bracket insulating bushes

- Secure the syphon clamp to the frame of the rear trailer door.
 - Avoid twisting the flexible syphon tube.
 - If required, loosen the syphon bracket insulating bushes to re-align the syphon.
- Ensure that there is open and clear access for helium dewar(s) at the rear of the trailer.
- Connect the laptop to the MSUP X6 via the RS232 cable, so that can continuous monitoring of the LHe levels during this procedure.

- Prepare the helium dewar.



All helium dewars must be vented of any excess pressure away from the trailer.

4.1.4.1 Performing a Helium Level Check

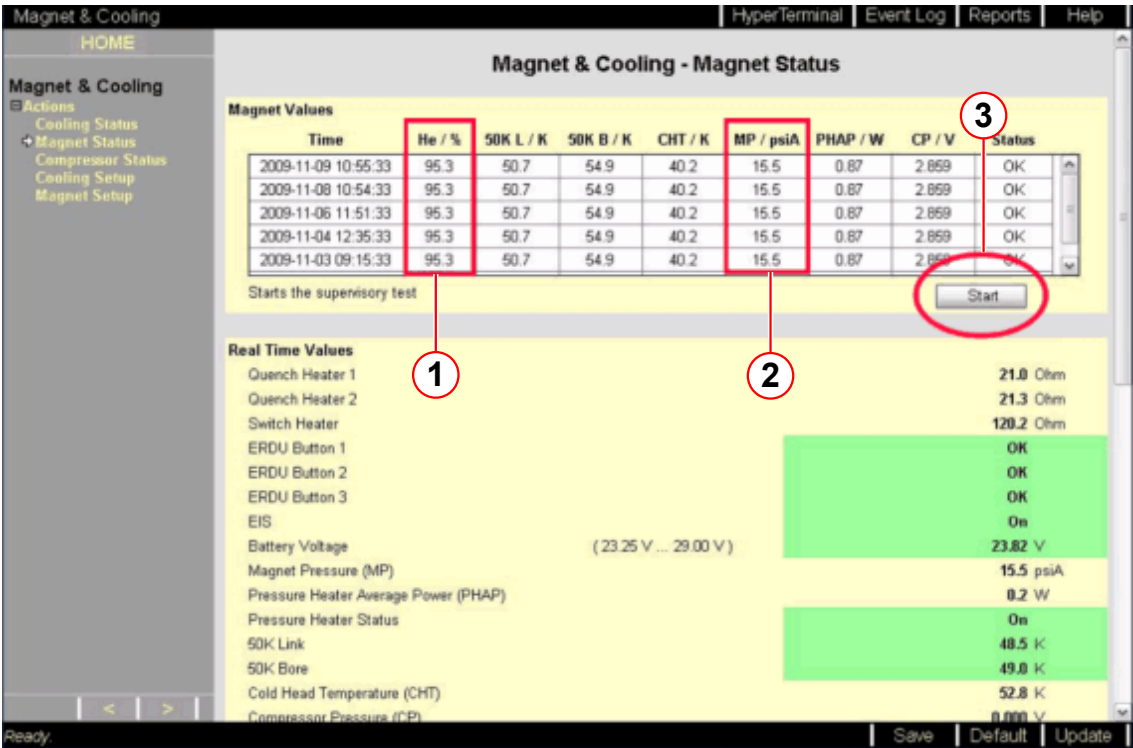


During installation, if the MRC is not available:
Take a helium sample using the MSUP.

Helium level check at the MRC

- Check the liquid helium level by using Service software on the MRC (MRAWP) host.
 - Check the liquid helium level using the MRC.
 - Access the Service software on the MRC.
 - Select the menu item Magnet & Cooling - Magnet Status.
 - Press the Start button (→ 3/ Fig. 171 Page 106) to initiate the helium measurement (→ 1/ Fig. 171 Page 106).

Fig. 171: Checking liquid helium and magnet pressure levels



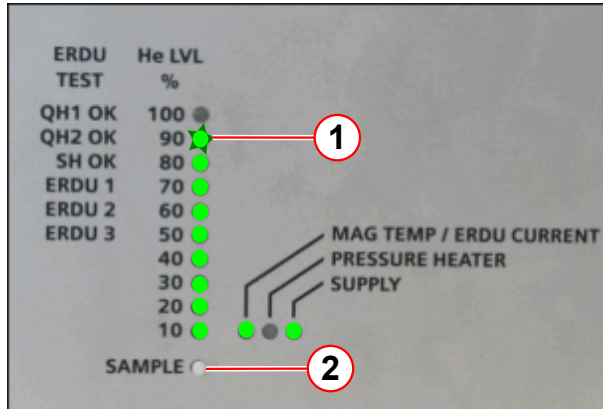
- (1) Liquid helium level
- (2) Magnet pressure
- (3) Start button



Data is only accurate from the last refresh of the page, or from when the page was opened last.

Helium level check at the MSUP

Fig. 172: MSUP - Helium level sample



- (1) Sample switch
- (2) Helium level

- Locate the SAMPLE switch on the MSUP lid.
- Depress the sample switch for 2 seconds and release.
- The green LEDs illuminate and display the actual liquid helium level percentage.

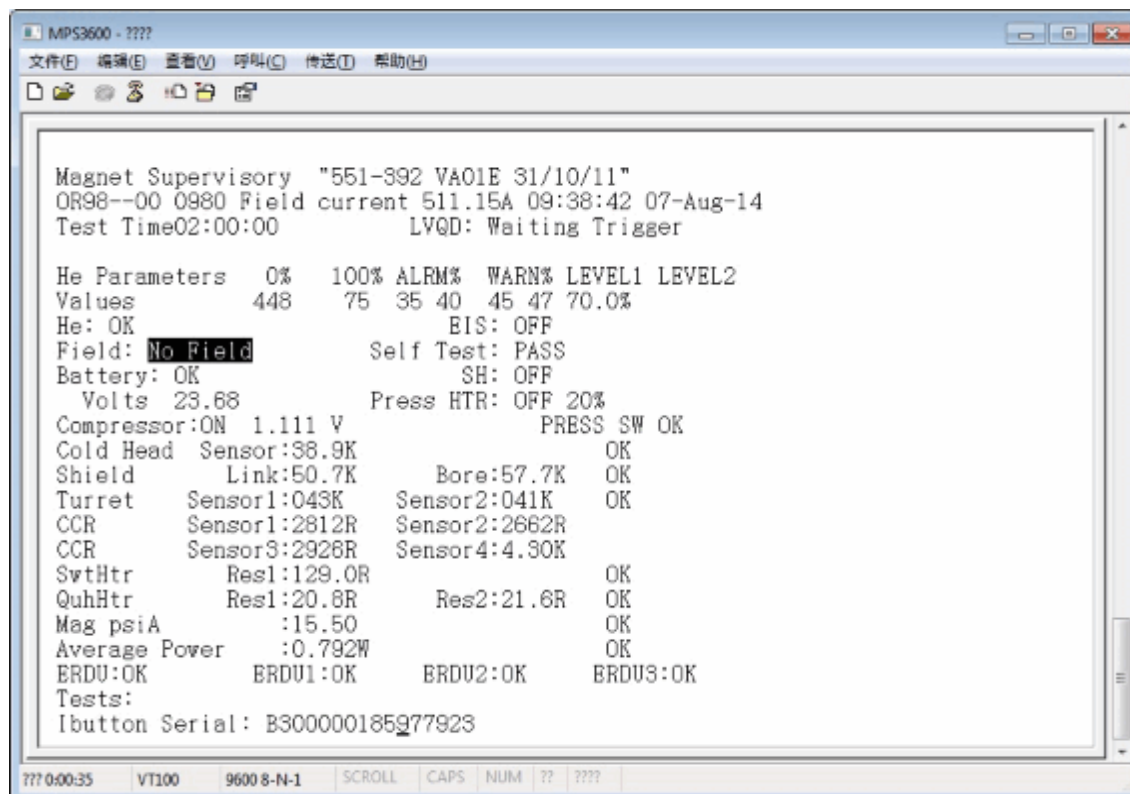
Helium level check with the laptop

The helium level could be check via the laptop if available.

- Turn on the laptop and open the MSUP Terminal Emulation software.
 - » Set bits per second to 9600.
 - » Set data bits to 8.
 - » Set parity to 1.
 - » Set flow control to none.
 - » Set emulation to VT100 (File: Properties:Settings)
- Connect the laptop to the MSUP X6 via the RS232 cable.
- Press **ESC**.
- Type **L0 6932 L0** <enter>.
- Type **R** <enter>.

- Press **ESC** to display the MSUP Process display screen.

Fig. 173: MSUP process display screen (example)



- Press **ESC** and type **HT** <enter>.
- Check the helium level every 5 minutes.

4.1.4.2 Verifying the Electronic Low-pressure Switch

The low pressure switch can be checked with any process involving depressurization of the helium vessel.



DO NOT turn off (fuse) F40 in the EPC. This shuts down the compressor and invalidates the low-pressure switch test.

Pressure control of the helium vessel is managed by the MSUP via the pressure heater control unit. The low-pressure switch must be checked whenever the magnet is depressurized.

When the pressure in the helium vessel drops to within 0.2 psi (G) above atmospheric pressure, the electrical pressure switch will open and turn off the compressor.

- » Check that the compressor turns OFF < 0.2 psi (G).
- » Check that the compressor turns ON > 0.2 psi (G).

As the helium vessel is pressurized, the switch closes and the compressor will restart.

4.1.5 Dewar Preparation

This procedure describes the use of **top fill** helium dewars used in conjunction with the Siemens Magnet Technology dewar syphon.

It covers all dewars that are externally pressurized with helium gas.



Refer to the dewar manufacturer's documentation when using side fill and/or self-pressurizing dewars.



Read the dewar manufacturer's safety guidelines and check the dewar valves before commencing the LHe fill.

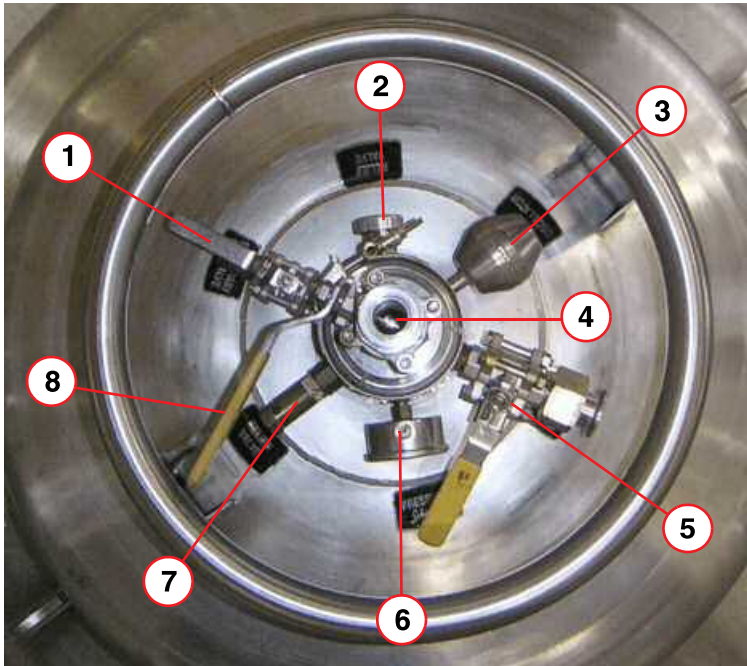
- If the dewar is highly pressurized, (greater than 0.3 bar (4.4 psi)), release the pressure by slowly opening the primary ball valve (→ 5/ Fig. 174 Page 110) to minimize the flash losses of helium.
 - This must be done in a **well-ventilated area outside of the trailer**.



Check for the cause of the high pressure, was the low pressure relief valve closed? Inform the supplier if the cause cannot be found.

- Connect the dewar siphon adaptor (→ 3/Fig. 177 Page 113) to the top of the dewar.

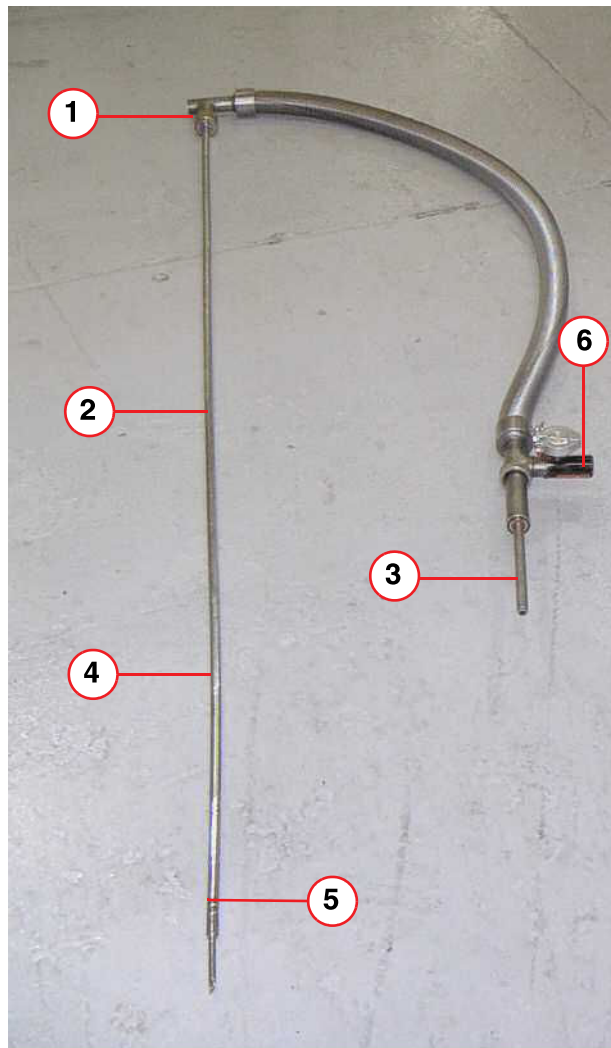
Fig. 174: Dewar Ports (typical)



- (1) Low pressure relief valve (shown open)
 - (2) Relief valve
 - (3) Anti-oscillator
 - (4) Syphon entry port (shown closed)
 - (5) Primary ball valve (shown closed)
 - (6) Pressure gauge
 - (7) Relief valve
 - (8) Syphon entry valve
- **Close** the primary ball valve (→ 5/Fig. 174 Page 110) and low-pressure relief valve (→ 1/Fig. 174 Page 110).

4.1.6 Preparation of dewar syphon

Fig. 175: Dewar syphon



- Attach extensions as necessary, including the sludge inhibitor, to the dewar syphon leg.

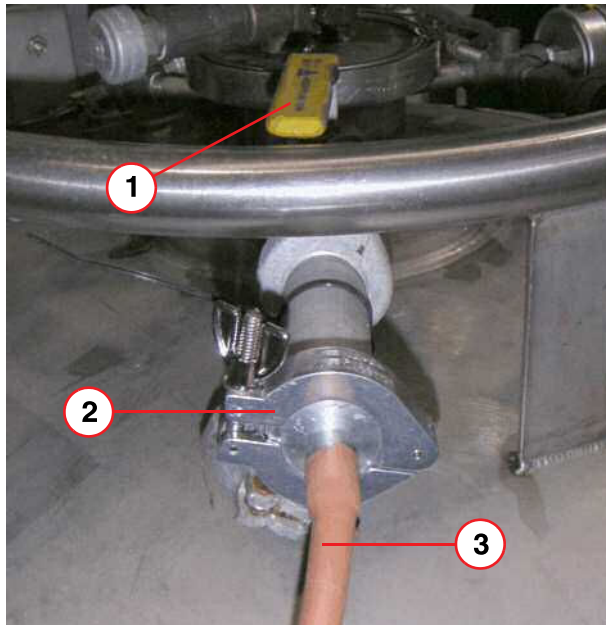
- (1) Syphon flow valve
- (2) Syphon leg
- (3) Bayonet connection (male)
- (4) Extension leg
- (5) Sludge inhibitor
- (6) Syphon vacuum valve



Do not open the syphon vacuum valve [6].

- Purge service and dewar syphons with helium gas to remove any debris.
- Connect the gas bottle to the dewar.

Fig. 176: Dewar gas connection

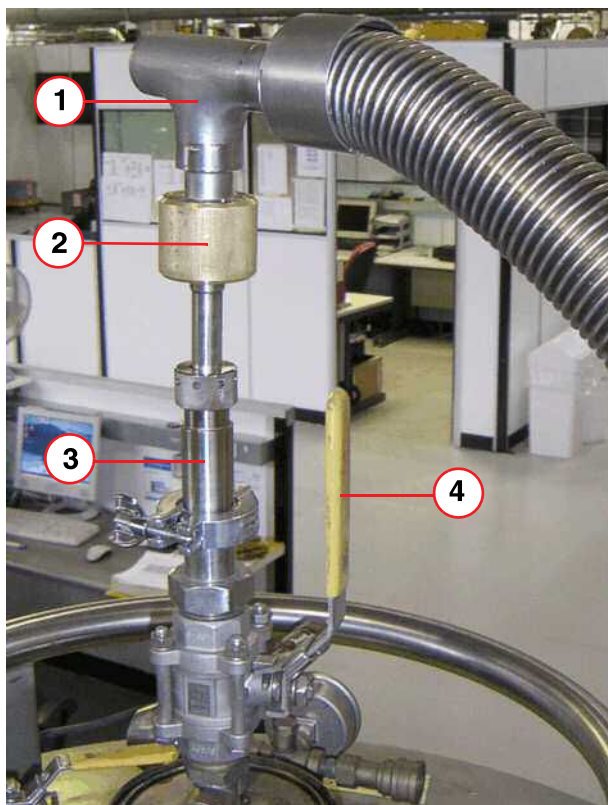


- (1) Primary ball valve
(2) NW25 clamp and seal
(3) Gas supply line

- Ensure that the **gas supply line** [3] is long enough to allow the dewar to be moved close to the end of the trailer, if the magnet is at field.
- Purge the helium **gas supply line** of air.
- Connect it to the **primary ball valve** [1] on the dewar, using an NW25 clamp and seal [2].

4.1.7 Pre-cool the dewar syphon.

Fig. 177: Dewar syphon fitment



- (1) Syphon
- (2) Syphon flow valve
- (3) Dewar syphon adaptor
- (4) Syphon entry valve

- Open the **syphon flow valve** [2] by two turns.
- Insert the syphon leg into the **dewar syphon adaptor** [3] and tighten the ring nut and seal to create a gas seal that the syphon can slide through.
(Do not fully tighten the ring nut at this stage).
- Open the **syphon entry valve** [4] and slowly slide the syphon into the dewar. Gas will begin to flow through the syphon.

- Push the syphon fully down into the dewar (very slowly to minimize pressure build-up and flash losses) until the sludge inhibitor touches the bottom.
- Fully tighten the ring nut and seal.
- Allow the syphon to cool until a dense white "cascading plume" appears. **Do not** allow high pressure to build-up in the dewar, only enough to cool the syphon.
- When the syphon is cold, close the syphon flow valve (→ 2/Fig. 177 Page 113), and move the dewar so that the syphon can be connected without excessive bending.
- **Open** the primary ball valve (→ 5/Fig. 174 Page 110) and pressurize the dewar to **0.27 bar (4 psi (G))**.

**WARNING**

Strong magnetic field

Failure to observe the following safety warning may result in physical injury and equipment damage.

- » Only non-magnetic equipment may be taken into the trailer.

4.1.8 Pre-cooling the Mobile Syphon

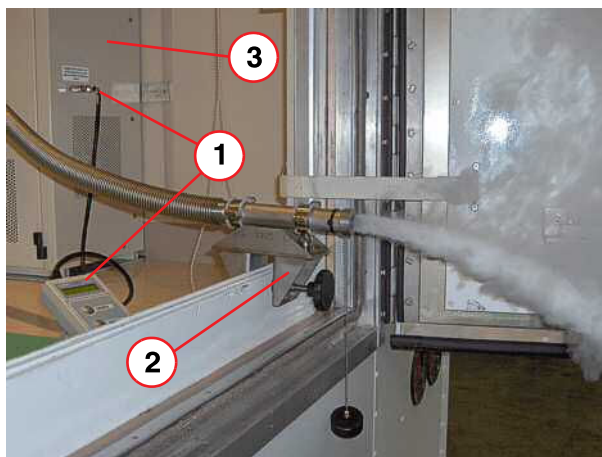


The helium dewar must be positioned close to the rear of the trailer with the dewar syphon attached, ready for use.

Cool down the mobile syphon as follows:

- Ensure the syphon end cap is fitted and the magnet bypass valve is **closed**.

Fig. 178: Mobile syphon venting (cascading plume)



- (1) Hand held connection to MSUP
- (2) Clamp position for venting (variable)
- (3) Electronics cabinet (E-box)

- **Remove** the mobile syphon **end cap** and allow the syphon to vent until it is fully cold and a dense white “cascading plume” appears.

**CAUTION**

Risk of cold burns.

The “cascading plume” of cold gas is very cold.

- » Stand clear when removing the cap.
- » Wear protective gloves and safety glasses.

4.1.9 Helium Pressures at Altitude

4.1.9.1 Helium Transfer with the Magnet Bypass Valve Closed

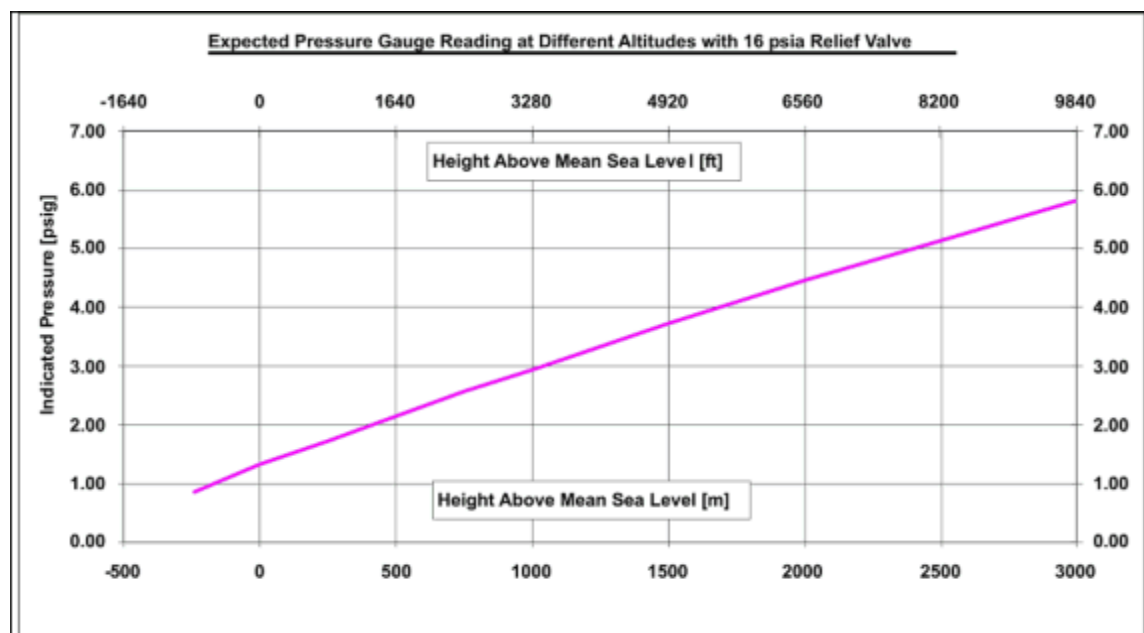
The graph(s) show the **gauge pressure** in the magnet as a function of altitude.

The optimum pressure difference between magnet and dewar for a helium transfer is 4 psi. To calculate this:

- Determine the **magnet's altitude**.
- Determine the **magnet gauge pressure** from the graph (→ Fig. 179 Page 115).

16 psi (A) valve

Fig. 179: Effect of a change in altitude on gauge pressure (16 psi (A))



Tab. 4 Dewar transfer pressures (typical)

Altitude (M)	0	1000	2000	3000
Dewar pressure psi (G)	5	7.0	8.5	9.5

- Add 4 psi to this number to calculate the **required transfer pressure**.
 - » **Example of 16 psi (A) vent valve assembly**
 Altitude (1000m) = 3.0 psi (G) + 4.0 psi = 7.0 psi (G)

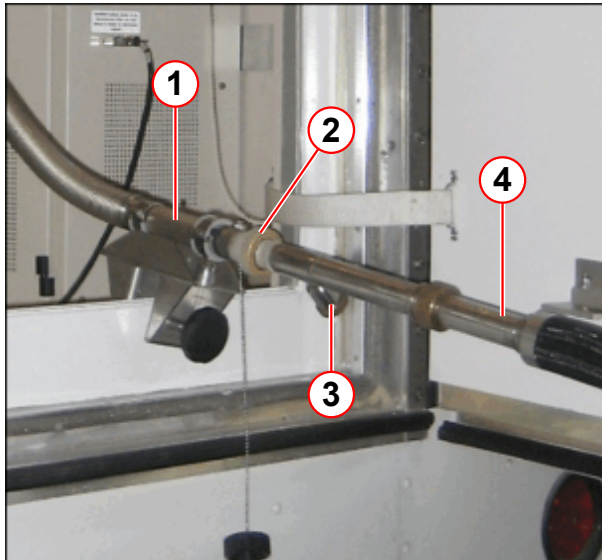


Maximum dewar pressure is 10.5 psi (G).

4.1.10 Liquid Helium Transfer, Mobile

- Check that the gas supply pressure is set to **0.27 bar (4 psi (G))**.
- As the **mobile syphon** nears the point of being fully cold, open the **dewar syphon flow valve** (→ 2/Fig. 177 Page 113) (two turns) and pre-cool the syphon again until a dense white “cascading plume” appears.

Fig. 180: Mobile - dewar and service syphon coupling



- (1) Service syphon
- (2) Lock nut
- (3) Dewar syphon vacuum valve
- (4) Dewar syphon

- When **both** syphons are giving a dense white “cascading plume” of helium gas, connect them together and **secure the lock nut** [2].

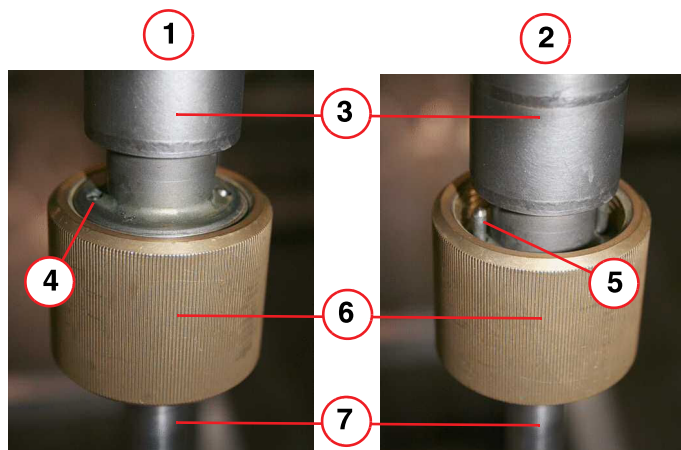
CAUTION

Risk of quench

Warm helium gas can cause a quench.

- » If the magnet is at field, it will quench if both siphons are not pre-cooled correctly.
- » Ensure syphons are correctly pre-cooled before joining together.
- » Frost or condensation on the body of either syphon is an indication of poor vacuum. Stop the transfer immediately; the syphon may need to be replaced.
- » Frost on the syphon coupling is acceptable.

Fig. 181: Dewar syphon flow valves - open and closed positions



- (1) FULLY OPEN position
- (2) FULLY CLOSED position
- (3) Stainless steel body
- (4) Guide pin flush with body
- (5) Guide pin
- (6) Brass nut
- (7) Syphon leg

- Fully open the syphon flow valve on the dewar syphon. (→ Fig. 181 Page 117)
 - » The flow valve is **fully open** when the two guide pins at the top of the brass nut are flush with the top of the stainless steel body.
 - » From fully closed to fully open is four turns.



Maintain a pressure of 0.27 bar (4 psi (G)) at the helium dewar.
Too much or too little pressure will result in inefficiency.

Fig. 182: Mobile - helium transfer



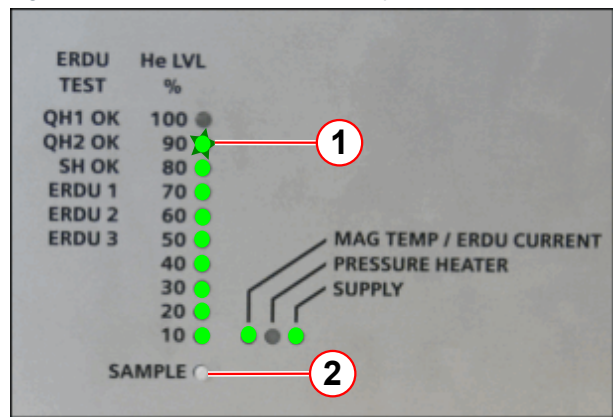
- At **4 psi (G)**, the transfer rate is approximately **10** liters per minute.
- **DO NOT LEAVE THE TRANSFER UN-ATTENDED.**
- A 250 liter dewar will empty in 25 to 30 minutes.

4.1.10.1 Helium Level Checks

- Repeat the helium measurement every 5 minutes. An increase in level should be seen within 20 minutes.
 - If not, stop the transfer and investigate the reason for poor transfer efficiency.

At 4 psi (G), the transfer rate is approximately 10 liters per minute. **Do not leave the transfer unattended.** A 250 liter dewar will empty in 25 to 30 minutes.

Fig. 183: MSUP - Helium level sample



- (1) Sample switch
- (2) Helium level

- Locate the SAMPLE switch on the MSUP lid.
- Depress the sample switch for 2 seconds and release.
- The green LEDs illuminate and display the actual liquid helium level percentage.
- If the 90% LED is flashing, STOP the transfer.

Tab. 5 MSUP helium display

MSUP LEDs	Helium level	Comments or actions
0 - 80% ON	As displayed	Continue to fill as required.
90% ON	As displayed	Stop the transfer.
*90% FLASHING 10 - 80% ON	95% - 97%	STOP the transfer IMMEDIATELY!
*100% FLASHING 10% - 90% ON	97% and over	Magnet is OVERFILLED.

i

* The LEDs return to ON after approximately 30 seconds.

i

During filling, do not exceed 95%.

4.1.10.2 Stopping the Helium Transfer

NOTICE

Risk of quench

Warm helium gas will cause a quench.

- » Stop the transfer immediately in case you hear whistling (a high-pitched sound) from the syphon, or the dewar pressure falls.

Fig. 184: Dewar manometer



- As soon as the dewar is empty, stop the transfer by closing the flow valve at the dewar syphon. **Do not fill above 95%.**
 - A whistling (high-pitched sound) of the syphon or a noticeable drop in dewar pressure indicates an empty dewar.
- The **dewar syphon flow valve** must be closed **immediately** if the dewar is empty - **if left open there is a risk of quenching.** (→ 6/Fig. 181 Page 117)
- Set the dewar gas pressure regulator to **zero**.
- Close the dewar **primary ball valve**. (→ 1/Fig. 176 Page 112)
- Refer to (→ Changing supply dewars during transfer / Page 119) if the magnet is not full.

4.1.11 Changing supply dewars during transfer

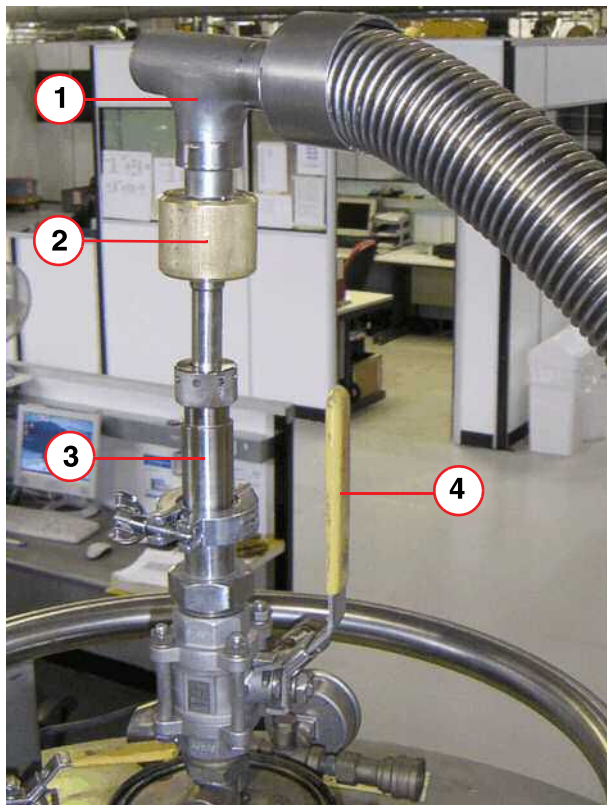


A whistling syphon (a high-pitched sound) or a noticeable drop in dewar pressure indicates an empty dewar.

If the supply dewar empties before the magnet is filled, it will be necessary to replace the supply dewar with a full one.

Ensure that a full dewar is available and depressurized, i.e. ready to receive the syphon.

Fig. 185: Dewar syphon fitment



- (1) Syphon
- (2) Syphon flow valve
- (3) Dewar syphon adaptor
- (4) Syphon entry valve

- When the dewar empties, stop the transfer **immediately** by closing the dewar syphon flow valve [2].



CAUTION

Risk of quench

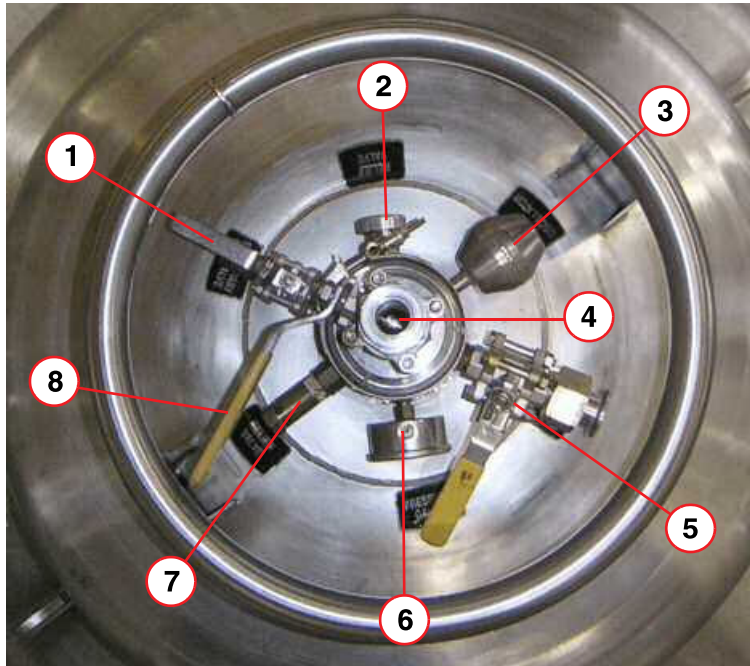
Failure to close the valve immediately, when the magnet is at field, will result in a quench.

- » Close the valve immediately.

- Turn **off** the dewar gas supply:
 - remove the **gas connection**.
 - vent the dewar down to atmospheric pressure.
 - **Ensure that the area is well ventilated.**

- Close the **primary ball valve**(→ 5/ Fig. 186 Page 121).

Fig. 186: Dewar Ports (typical)



- (1) Low pressure relief valve (shown open)
- (2) Relief valve
- (3) Anti-oscillator
- (4) Syphon entry port (shown closed)
- (5) Primary ball valve (shown closed)
- (6) Pressure gauge
- (7) Relief valve
- (8) Syphon entry valve
- Wait for the magnet pressure to drop to < 0.2 psi (G) before removing the dewar syphon.
- Using a heat gun, **warm** the syphon connecting joint to room temperature.
- **Disconnect** the syphons.



CAUTION

Risk of cold burns.

A "cascading plume" of cold gas will exhaust from the syphons when they are disconnected.

- » Stand clear of the syphon joint when disconnecting it.
- » Wear protective gloves and safety glasses.

- Remove any moisture from the end of the **service syphon** with paper toweling.

- Install the **siphon end cap**.



CAUTION

Danger of air ingress

Internal ice blockage could cause excessive pressure.

- » **Ensure the syphons are capped and plugged immediately.**

- Move the full dewar close to the empty dewar.
- Vent both storage dewars down to **atmospheric pressure**.
 - » **This must be done in a well-ventilated area outside the trailer.**
- **Open** the **low-pressure relief valve** on the **empty** dewar. (→ 1/Fig. 186 Page 121)
- Remove the dewar siphon and adapter from the empty dewar and **immediately**. Insert the siphon leg into the full dewar.
- **Close** the **siphon entry valve** (→ 4/Fig. 186 Page 121), of the **empty** dewar, and **open** the **low-pressure relief valve**. (→ 1/Fig. 186 Page 121)
- **Close** the **low-pressure relief valve** of the **full** dewar.
- Purge the gas supply line.
- **Connect** to the full dewar and **open** the dewar **primary ball valve**.
- Set the **gas pressure** to 0.27bar (4 psi (G)).
- **Pre-cool** the **dewar siphon**.
- Move the dewar next to the magnet system.
- Pre-cool the **service siphon**.
- When **both** siphons are giving a dense white "cascading plume" of helium gas, couple them together and secure the lock nut.
- **Continue** with the transfer.

4.1.12 Final Work Steps

After completing the transfer, check the following:

- The **magnet bypass valve** is **closed** and **locked**.
- The **pressure heater** is operating correctly.
- The **pressure heater LED** on the MSUP is illuminated.
- **No alarm signals** are evident on the Alarm Box.
- The **compressor** is switched **on** and running.
- **Manually** operate the pressure heater to re-pressurize the magnet.
 - Clear average power by manually turning off the pressure heater at **15.5 psi (A)**.
- The magnet pressure is stable at **15.5 psi (A)**.
- **Warm** the turret and venting system with the heat gun. Dry with paper toweling.

- Check the filling port and venting system for leaks.

4.2 Prerequisites for Ramping the Magnet

4.2.1 Quench Tube Check

- If the mobile trailer unit is parked in a public area there must be a defined barrier around the trailer for safety from:
 - » Gas cloud if a quench occurs.
 - » Stray field burst from the magnet if a quench occurs.
- Perform a visual inspection of the quench tube.
 - » Refer to the Safety-Related Tests Procedure of Aera system.
- Check whether the installation protocol is signed by the quench line manufacturer.



Ensure the quench line installation is complete before ramping the magnet. The Quench Line Acceptance Protocol must be checked before the first ramp. Check that the Quench Line Acceptance Protocol is signed off by the quench line manufacturer.

The acceptance protocol can be found under:

Product Information Home > Planning > Planning Guides > MR Systems > Quench Line Design and Calculation Tool

4.2.2 ERDU Button Test

- Mobile systems have three ERDU buttons on the alarm box, in the magnet room at the patient end, and on the right top of the RCR (→ ERDU and EPO Buttons / Page 36) behind the magnet.
- All the test procedures of the ERDU are identical to normal systems. Please refer to Safety Related Tests Procedure of Aera system.

4.2.3 Venting Leak Check

The leak check procedure of the venting system is the same as with static systems. Please refer to Start Up / Magnet of Aera system.

4.3 Ramping the Magnet



Only Siemens CSEs trained in ramping magnets are allowed to perform the magnet ramping procedure.



Before ramping check that the quench line is correctly installed from the magnet to the outlet.



Startup (1st) ramp helium level recommended at $\geq 60\%$.



Minimum helium level for ramping must be $> 40\%$.
Wait at least 2 hours after helium filling.



WARNING

Magnet safety checks

Failure to observe the following safety warning may result in physical injury and equipment damage.

- » Ramping is not allowed if the visual inspection of the quench line was not successful.
- » Before the magnet is ramped up initially, perform "Testing the ERDU circuit."
- » The procedure is described in the safety related tests.



Always inspect magnet room for the presence of unidentified objects. Before ramping, the nature of these objects must be clarified.

4.3.1 Moving MPSU 3600 to the Operator Room

- Due to the filter plate of mobile systems installed between the examination and operator rooms, the MPSU 3600 should be placed in the operator room when ramping.

- Moving the MPSU 3600 to the operator room using the patient lift.

Fig. 187: MPS on patient lift



- (1) MPS
(2) Patient lift

- Place the MPSU 3600 in the operator room and connect the ramp cable to the filter plate.

Fig. 188: MPS in operator room



4.3.2 OR98 Ramping

The ramping procedure is identical to the ramping of a static system. Please refer to Start Up / Magnet of Aera system.

4.3.3 Post-quench Procedure

The post-quench procedure is identical to that of a static system. Please refer to Start Up / Magnet of Aera system.

4.4 Shimming the Magnet

Magnet shimming is identical to the process used for the static system. Please refer to the Aera system online help.

- After shimming is done, please store the shim plates and ramp cables (other components also) in the trailer cabinet located near the SEP in the equipment room.

Fig. 189: Store the shim plates and ramp cables



- (1) Shim plates
- (2) Ramp cables

This is the initial version.

There are no Hazard IDs in this document.

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